Choosing the Dictionary and Penalty for IV-LASSO*

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Abstract

Estimating an instrumental variables (IV) first stage with the least absolute shrinkage and selection operator (LASSO) requires choosing a dictionary and penalty. Existing rules target first-stage objectives rather than precision of the structural parameter estimate. We derive a feasible approximate mean squared error (AMSE) criterion for full-sample IV-LASSO with one endogenous regressor and homoskedastic Gaussian errors. For a prespecified list of dictionary–penalty pairs, the criterion trades off approximation of the optimal instrument against a complexity cost that depends on the first two moments of the LASSO effective dimension and on the degree of endogeneity. The selected pair attains the smallest candidate-specific AMSE component up to vanishing relative error. In simulations calibrated to Paravisini et al. (2015), the criterion improves substantially on plug-in rules at low endogeneity and on cross-validation at high endogeneity, with nearly identical results under non-Gaussian errors. A supplementary appendix formally extends analysis to a class of non-Gaussian errors.

KEYWORDS: Instrumental Variables, LASSO, Model Selection, Second-Order Asymptotics

JEL CODES: C26, C36, C52

*An `R` package implementing IV-LASSO with the first-stage implementation guided by this paper can be found on GitHub. This package, along with the proofs of several technical lemmas in this paper, were developed with the help of Claude Fable. We take responsibility for all errors.

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1 Introduction

When empirical researchers have access to a large set of technical instruments, LASSO makes estimation of an IV first-stage feasible but leaves researchers to choose how heavily to penalize the fit and which dictionary of technical instruments to use.[1] Under standard sparse-first-stage conditions, Belloni et al. (2012) show that all sufficiently accurate LASSO first-fits yield IV estimators with the same first-order limiting distribution. Standard asymptotic theory therefore does not rank admissible implementations, even though their finite-sample performance can differ.

Figure 1 illustrates this issue in the eminent-domain application of Belloni et al. (2012). Across three nested first-stage dictionaries, the IV estimate changes materially as the penalty parameter varies. Moreover, different methods of selecting the first-stage penalty parameter can change whether the conventional pointwise 95% confidence interval for the resulting structural parameter estimate contains zero. For example, for the Case–Shiller home price index outcome, using cross-validation to select the penalty yields a significant result on the Restricted dictionary while using Stein's Unbiased Risk Estimate (SURE) does not. Given this variation, how should an applied researcher choose among possible first-stage implementations?

Existing first-stage tuning rules are not designed to target accuracy of the resulting structural parameter estimates. The calibrated penalty rule of Belloni et al. (2012) and the bootstrap-after-cross-validation method of Chetverikov and Sørensen (2025) are designed to control first-stage estimation error, while SURE and ordinary cross-validation target first-stage prediction (Tibshirani and Taylor, 2012; Chetverikov et al., 2021). Neither of these approaches necessarily yield high-quality structural parameter estimates. A LASSO fit with a lighter penalty or richer dictionary may better approximate the true first stage, but it can also fit more of the first-stage noise. When the first-stage noise is correlated with the structural error, this in turn introduces a bias into the resulting structural parameter estimate whose size grows with the degree of endogeneity. The optimal first-stage implementation therefore depends not only on goodness of first-stage fit and instrument strength, but also on the level of endogeneity.

[1] See Paravisini et al. (2015), Gilchrist and Sands (2016), Reimers and Waldfogel (2021), Bald et al. (2022), Derenoncourt (2022), Ryan and Sudarshan (2022), Agan et al. (2023), Bazzi et al. (2023), Beraja et al. (2023), Do and Jacoby (2025), Bazzi et al. (2026), and Cuddy and Currie (2026) for prominent applications of IV-LASSO and IV-post-LASSO methods in empirical economics.

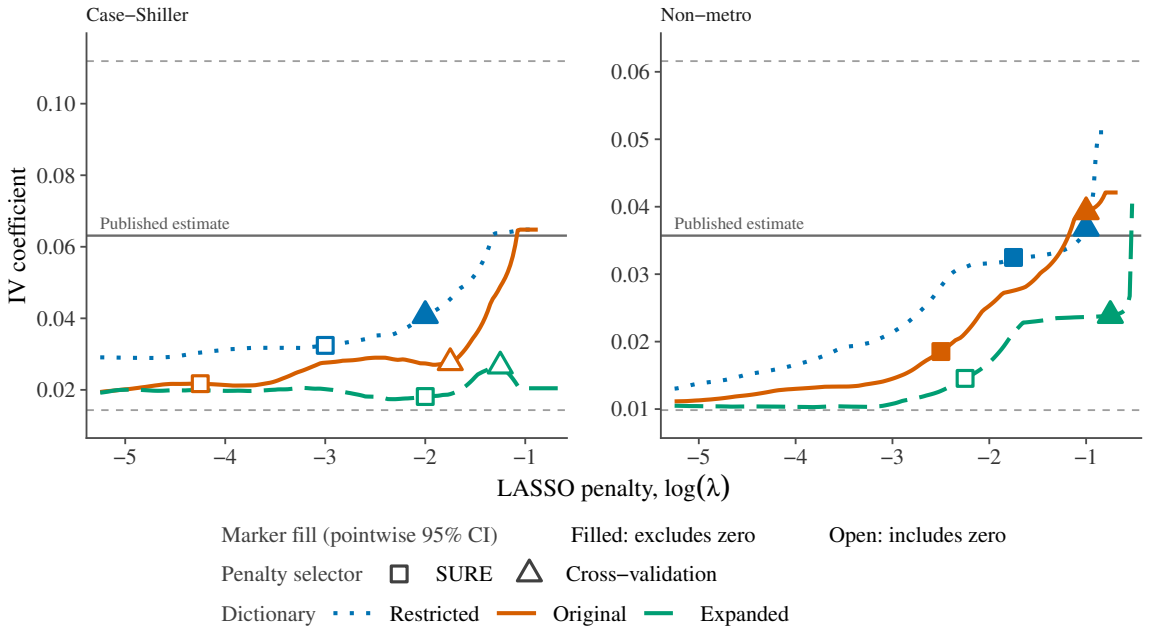


Figure 1: IV Estimates and Conventional Significance across First-Stage Implementations

Notes: Lines trace IV-LASSO estimates over LASSO penalties within three nested dictionaries. Original is the published dictionary; Restricted removes cross-products, and Expanded adds all non-degenerate second-order products. Filled markers denote pointwise 95% heteroskedasticity robust confidence intervals that exclude zero. The solid gray line and dashed bounds show the published IV-post-LASSO estimate and confidence interval. The panels report the two outcomes for which that interval excludes zero (183 Case–Shiller and 110 non-metro circuit-year observations).

This paper develops second-order asymptotic theory for the full-sample IV-LASSO estimator and uses it to provide a principled method for choosing the first-stage implementation. We focus on the leading case of a single endogenous regressor and work in a homoskedastic Gaussian model. As in Donald and Newey (2001), who build on the higher-order expansions of Nagar (1959) to choose among instrument sets, we take a finite, prespecified list of first-stage LASSO candidates, each consisting of a dictionary and a penalty, and rank them by the approximate mean squared error (AMSE) of the IV-LASSO estimators they produce. The candidate specific part of the AMSE can be decomposed into a first-stage approximation cost and a complexity cost, with complexity measured by a collinearity-robust analogue of the number of instruments selected by the LASSO. The size of this complexity cost rises with the squared covariance between the first-stage and structural errors, reflecting the discussion above.

This characterization leads to a feasible criterion that can be computed from each candidate's fitted first stage and its effective dimension, a measure of first-stage complexity closely related to the number of variables selected by the LASSO fit (Tibshirani and Taylor, 2012). Under our regularity conditions, the pair that minimizes this criterion has an AMSE as small as that of the best candidate in the list, up to vanishing relative error. This guarantee is stated relative to the second-order components that distinguish the candidates, rather than the first-order risk that all of them share. In a supplementary appendix, we extend these results to a class of non-Gaussian error distributions under some further regularity conditions, showing that the same feasible criterion minimizes AMSE in this more general model.

To derive the AMSE of the IV-LASSO estimator, we need to account for both the shrinkage and the variable selection performed by the first-stage LASSO. Although the LASSO fitted values are continuous in the endogenous variable, they need not be differentiable at points where the selected set of instruments changes. This lack of full differentiability complicates higher-order analysis, which typically relies on a smooth expansion of the first-stage estimator. Moreover, since our assumptions allow for coefficients near the selection threshold, we cannot avoid this difficulty by assuming that the selected model is eventually fixed. We address this by using the geometric results of Tibshirani and Taylor (2012) and the second-order Stein identities of Bellec and Zhang (2021) to characterize moments involving the LASSO fit and structural errors. The first-stage LASSO fit is Lipschitz continuous and thus differentiable almost everywhere, allowing these calculations despite its nonsmoothness.

Constructing a feasible criterion presents a further difficulty. Because the LASSO first-stages may converge slowly, estimation error in the feasible criterion can asymptotically dominate the candidate-specific AMSE terms. For our proposed feasible selection rule, we show that the leading part of this error is common to all candidates and therefore cancels when they are compared. After this common centering, the remaining estimation error is uniformly negligible relative to the candidate-specific AMSE components. Estimating the complexity cost requires fluctuations in the realized LASSO effective dimension to be small relative to its expectation. Using similar arguments to Bellec and Zhang (2021), we obtain this stability under a partial beta-min condition on a growing group of coefficients large enough for their instruments to be selected with probability approaching one. This group must account for a non-

vanishing share of the nonzero coefficients used to approximate the first stage. The condition also makes the AMSE approximation error small relative to the terms used to compare candidates, playing a similar role in our analysis to the growing basis condition of Donald and Newey (2001).

To evaluate the performance of the proposed criterion, we conduct a simulation study calibrated to the credit-supply and export data of Paravisini et al. (2015). We hold the original instruments fixed and vary the strength of identification and degree of endogeneity. At low levels of endogeneity, cross-validation and the proposed criterion perform comparably. As endogeneity rises, however, the criterion selects less complex first stages and achieves up to 50% lower mean squared error than cross-validation in designs with the highest levels of endogeneity. Compared to the cross-fit IV-LASSO estimator, the proposed criterion has substantially lower mean squared error when endogeneity is low and performs comparably when it is high. Interestingly, the relative performance of the criterion is nearly identical in the Gaussian designs and in the two non-Gaussian designs we consider, suggesting that the AMSE in more general models is well approximated by the AMSE in a Gaussian model.

Our results extend the literature on selecting and regularizing instruments according to the AMSE of the structural parameter estimator (Donald and Newey, 2001; Kuersteiner and Okui, 2010; Okui, 2011; Carrasco, 2012) to high-dimensional LASSO first stages, where each dictionary–penalty candidate itself selects instruments using the observed endogenous variable. Unlike a least-squares first stage on a prespecified instrument basis, fixing the dictionary and penalty does not fix the selected model. Our analysis accounts for this selection uncertainty as well as the shrinkage introduced by the penalty.

Our analysis also relates to a growing literature on the use of high-dimensional and machine-learning methods in econometrics (Belloni et al., 2014; Chernozhukov et al., 2018; Wager and Athey, 2018; Farrell et al., 2021) and an established literature studying higher-order asymptotic properties of econometric estimators (Sargan, 1976; Rothenberg, 1984; Newey and Smith, 2004; Hahn et al., 2004). Velez (2026) develops second-order approximations for double/debiased machine learning estimators to study the choice of the number of cross-fitting folds. His higher-order analysis assumes a stochastic linear expansion of the first-stage estimators. We instead account directly for LASSO shrinkage and variable selection to choose the dictionary and

penalty.

Section 2 presents the model and candidate first stages. Section 3 states the main results, and Section 4 reports the simulations. Proofs are in the main appendix, while a supplementary appendix extends our analysis to a class of non-Gaussian errors.

Notation. For $m \in \mathbb{N}$, let $[m] := \{1, \dots, m\}$. For an array $\{a_i\}_{i\in[n]}$, define

$$\mathbb{E}_n[a_i] := \frac{1}{n}\sum_{i\in[n]} a_i, \qquad \overline{\mathbb{E}}[a_i] := \mathbb{E}_n[\mathbb{E}[a_i]].$$

2 Model and Setup

To simplify exposition, we present the analysis assuming that any exogenous regressors have already been partialled out of both the outcome and the endogenous regressor. Our analysis assumes there is only a single endogenous variable.[2] The structural equation and first stage are given by

$$\begin{aligned} y_i &= \beta x_i + \varepsilon_i \\ x_i &= \Pi_i + v_i \end{aligned} \tag{2.1}$$

We assume the researcher observes n independent observations (y_i, x_i, z_i), where $y_i, x_i \in \mathbb{R}$ and $z_i \in \mathbb{R}^p$. In (2.1), we let $\Pi_i = \mathbb{E}[x_i \mid z_i]$ denote the conditional expectation of the endogenous variable given the instruments and assume that the structural error is conditional-mean independent of the instruments, so that $\mathbb{E}[\varepsilon_i \mid z_i] = \mathbb{E}[v_i \mid z_i] = 0$. Throughout the theoretical analysis, we condition on the realized instruments and write $x, \Pi, \varepsilon, v \in \mathbb{R}^n$ for the corresponding stacked vectors.[3] All expectations and probability statements below are therefore understood as conditional on this realization, i.e., $\Pi_i = \mathbb{E}[x_i]$.

Assumption 1 (Conditional Gaussian Sampling)**.** *Assume*

(i) The errors $\{(\varepsilon_i, v_i)\}_{i=1}^n$ are independent of the instruments and independently

[2] A formal treatment of the model with controls can be found in the arXiv version of this paper (Ma et al., 2026). An extension to multiple endogenous variables is available on request.

[3] That is, $x = (x_1, \dots, x_n) \in \mathbb{R}^n$.

and identically distributed according to

$$\begin{pmatrix} \varepsilon_i \\ v_i \end{pmatrix} \sim N(0, \Omega), \qquad \Omega := \begin{pmatrix} \sigma_\varepsilon^2 & \sigma_{\varepsilon v} \\ \sigma_{\varepsilon v} & \sigma_v^2 \end{pmatrix},$$

where Ω is a fixed nonsingular matrix.

(ii) For constants $0 < \underline{H} < \overline{H} < \infty$,

$$\underline{H} \leq H := \mathbb{E}_n[\Pi_i^2] \leq \overline{H}, \qquad \sigma_{\varepsilon v} \neq 0. \tag{2.2}$$

Assumption 1(i) presents the conditionally homoskedastic Gaussian model, while Assumption 1(ii) assumes the structural parameter is strongly identified and that there is non-negligible endogeneity problem, i.e., the structural parameter β cannot be recovered from ordinary least squares (OLS).

Under conditional homoskedasticity, the conditional mean Π_i is the optimal instrument: among IV estimators based on a nonrandom instrument, using Π_i itself minimizes the asymptotic variance of the resulting structural parameter estimate.[4] Since Π_i is unknown, feasible estimation replaces it with a first-stage estimate. Any IV estimator $\widehat{\beta}$ based on a sufficiently well-estimated first stage (Section 3 states sufficient conditions) admits the same expansion,

$$\sqrt{n}(\widehat{\beta} - \beta) = \frac{\sqrt{n}\,\mathbb{E}_n[\Pi_i \varepsilon_i]}{H} + o_p(1), \tag{2.3}$$

whose leading term does not involve the first-stage estimate and, conditional on the instruments, is distributed exactly as $N(0, \sigma_\varepsilon^2/H)$ at every sample size. Comparing first-stage estimators thus requires analyzing terms hidden in the remainder.

2.1 Candidate First Stages and the IV Estimator

We select from a prespecified list of J candidates, where J is fixed. Each $c \in [J]$ consists of a dictionary of technical instruments and a nonrandom penalty $\lambda_c > 0$. The dictionary maps z_i to $z_{c,i} := (z_{c,i1}, \ldots, z_{c,ip_c})' \in \mathbb{R}^{p_c}$, whose entries are fixed

[4] For nonrandom $g \in \mathbb{R}^n$ with $\mathbb{E}_n[g_i^2]$ bounded and $\mathbb{E}_n[g_i \Pi_i]$ bounded away from zero, the leading variance is $\sigma_\varepsilon^2 \mathbb{E}_n[g_i^2]/\mathbb{E}_n[g_i \Pi_i]^2 \geq \sigma_\varepsilon^2/H$, with equality when g is proportional to Π. The bound σ_ε^2/H is the fixed-design counterpart of the usual optimal-IV efficiency bound; see Chamberlain (1987) and Newey (1990).

functions of z_i. The dictionary dimension p_c may exceed n, and both the dictionary and the penalty may depend on n. Write

$$Z_c := (z_{c,1}, \ldots, z_{c,n})' \in \mathbb{R}^{n \times p_c}, \qquad Z_{c,B} := (Z_c)_{\cdot,B} \in \mathbb{R}^{n \times |B|} \quad \text{for } B \subseteq [p_c].$$

For each candidate implementation $c \in [J]$, we define the LASSO coefficient estimate and fitted first stage (Tibshirani, 1996) by

$$\widehat{\pi}_c \in \operatorname*{arg\,min}_{\pi \in \mathbb{R}^{p_c}} \left\{ \frac{1}{2} \mathbb{E}_n[(x_i - z_{c,i}'\pi)^2] + \lambda_c \|\pi\|_1 \right\}, \qquad \widehat{\Pi}_c := Z_c \widehat{\pi}_c. \tag{2.4}$$

When the columns of Z_c are linearly dependent, the coefficient minimizer need not be unique, but the fitted vector $\widehat{\Pi}_c$ remains unique (Tibshirani and Taylor, 2012; Tibshirani, 2013). Using the fitted values $\widehat{\Pi}_{c,i} := z_{c,i}'\widehat{\pi}_c$ as instruments, we obtain the IV-LASSO estimator

$$\widehat{\beta}_c := \begin{cases} \dfrac{\mathbb{E}_n[\widehat{\Pi}_{c,i} y_i]}{\mathbb{E}_n[\widehat{\Pi}_{c,i} x_i]}, & \mathbb{E}_n[\widehat{\Pi}_{c,i}^2] > 0, \\ 0, & \mathbb{E}_n[\widehat{\Pi}_{c,i}^2] = 0. \end{cases} \tag{2.5}$$

A natural criterion for comparing the candidate estimators $\widehat{\beta}_c$ is their scaled mean squared error (MSE), $\mathbb{E}[n(\widehat{\beta}_c - \beta)^2]$. The exact MSE of an IV estimator, however, need not exist in finite samples. To see this, consider the infeasible estimator that uses the optimal instrument Π_i in place of $\widehat{\Pi}_{c,i}$ in (2.5). This estimator satisfies

$$\widehat{\beta} - \beta = \frac{\mathbb{E}_n[\Pi_i \varepsilon_i]}{\mathbb{E}_n[\Pi_i x_i]}.$$

Under Assumption 1, the numerator and denominator on the right-hand side are jointly Gaussian. Because Ω is nonsingular, the numerator is not an exact linear function of the denominator, and their ratio does not have a finite second moment. More generally, even when the exact MSE exists, it may be neither tractable nor consistently estimable.

Given this restriction, we instead follow the approach of Donald and Newey (2001). Under suitable conditions, we show that the scaled squared error of each estimator

$\widehat{\beta}_c$ admits a decomposition of the form:

$$n(\widehat{\beta}_c - \beta)^2 = C_0 + Q_c + r_c, \tag{2.6}$$

where C_0 is common to all candidates and captures the component of squared error corresponding to the leading term in (2.3), Q_c captures the second-order contribution of candidate c's first-stage procedure, and r_c is asymptotically negligible in the sense that $\max_{c\in[J]} |r_c|/\mathbb{E}[Q_c] \to_p 0$.

We define $\mathbb{E}[C_0 + Q_c]$ as the AMSE of the candidate IV-LASSO estimator $\widehat{\beta}_c$. For the infeasible estimator based on the optimal instrument, the expectation of the candidate-specific component is zero. Hence, $\bar{Q}_c := \mathbb{E}[Q_c]$ measures the excess AMSE of candidate c relative to that benchmark. Because C_0 is common across candidates, ranking candidates by AMSE is equivalent to ranking them by $\bar{Q}_c$. In Section 3, we derive a closed-form asymptotic approximation to $\bar{Q}_c$, construct a consistent feasible counterpart, and use it to select among the first-order-equivalent candidate estimators.

2.2 The Effective Dimension

A key element of our analysis is the effective dimension of the LASSO fit, which will provide a measure of first-stage complexity. As discussed above, the number of selected instruments is not always well defined. With collinear dictionary columns, LASSO coefficient solutions can have different supports even though their fitted values are identical. The effective dimension therefore measures first-stage complexity using the rank of the equicorrelation set.

Letting $u_c := x - \widehat{\Pi}_c$ denote the LASSO residual from candidate implementation $c \in [J]$, define the equicorrelation set by

$$E_c := \left\{ j \in [p_c] : \left|\mathbb{E}_n[z_{c,ij} u_{c,i}]\right| = \lambda_c \right\}. \tag{2.7}$$

This is the set of dictionary columns for which the LASSO Karush–Kuhn–Tucker (KKT) inequality binds. Because the residual is unique, E_c does not depend on which coefficient solution is used, and it contains the support of every solution. We

define the effective dimension of the first-stage fit as

$$d_c := \operatorname{rank}(Z_{c,E_c}). \tag{2.8}$$

The quantity d_c is observable and invariant to the choice of coefficient solution. When the LASSO solution is unique, d_c agrees with the number of selected instruments almost surely under our sampling assumption.[5] As shown by Tibshirani and Taylor (2012), the degrees of freedom of the LASSO fit are given by $\mathbb{E}[d_c]$.

3 Main Results

After stating the conditions required for our analysis, we characterize the candidate-specific AMSE components (Theorem 1) and show that the selected candidate attains the smallest of them up to vanishing relative error (Theorem 2).

3.1 Conditions on the Candidate First Stages

Along with the Gaussian sampling and strong identification conditions in Assumption 1, we impose conditions for first-stage prediction consistency and a partial beta-min condition that is used to establish concentration of the LASSO effective dimension but does not require exact support recovery.

We use the approximately sparse representation of Belloni et al. (2012), in which for each candidate dictionary Z_c there is a nonrandom coefficient vector $\pi_c^0 \in \mathbb{R}^{p_c}$ and a prediction error $\xi_{c,i}$ such that

$$\Pi_i = z_{c,i}'\pi_c^0 + \xi_{c,i}, \qquad T_c := \operatorname{supp}(\pi_c^0), \qquad s_c := |T_c|, \tag{3.1}$$

where $\operatorname{supp}(b) := \{j : b_j \neq 0\}$, $s_c \geq 1$, and $\xi_c = (\xi_{c,1}, \ldots, \xi_{c,n})'$. The rate restrictions on s_c and ξ_c are stated below.

For a vector δ and an index set T, write $\delta_T := (\delta_j)_{j\in T}$ and $\|\delta\|_0 := |\operatorname{supp}(\delta)|$. Fix a constant $c_\lambda > 1$, common to all candidates, and set $L_\lambda := (c_\lambda + 1)/(c_\lambda - 1)$. Following Belloni et al. (2012), define the restricted eigenvalue (Bickel et al., 2009; van de Geer

[5] A sufficient condition for uniqueness is that the dictionary columns are in general position, which holds with probability one when the dictionary matrix has a joint continuous distribution (Tibshirani, 2013).

and Bühlmann, 2009)

$$\kappa_c := \min_{\substack{\delta \neq 0 \\ \|\delta_{T_c^c}\|_1 \leq L_\lambda \|\delta_{T_c}\|_1}} \frac{\sqrt{s_c \mathbb{E}_n[(z'_{c,i}\delta)^2]}}{\|\delta_{T_c}\|_1}, \tag{3.2}$$

and the upper and lower m-sparse eigenvalues

$$\phi_{\max,c}(m) := \max_{\substack{\delta \neq 0 \\ \|\delta\|_0 \leq m}} \frac{\mathbb{E}_n[(z'_{c,i}\delta)^2]}{\|\delta\|_2^2}, \qquad \phi_{\min,c}(m) := \min_{\substack{\delta \neq 0 \\ 1 \leq \|\delta\|_0 \leq m}} \frac{\mathbb{E}_n[(z'_{c,i}\delta)^2]}{\|\delta\|_2^2}. \tag{3.3}$$

These conditions control the design matrix only on sparse sets and the cone in (3.2), so they do not require the full dictionary to have linearly independent columns and, as such, do not rule out cases where $p_c \gg n$.

Assumption 2 (First-Stage Convergence)**.** *There are constants $0 \leq C_a < \infty$, $\alpha > 0$, $0 < \kappa_0 \leq 1$, $0 < \underline{\phi} \leq 1 \leq \overline{\phi} < \infty$, and $\vartheta > 4L_\lambda^2 \overline{\phi}/\kappa_0^2$, common to all candidates, such that the following hold for each $c \in [J]$:*

(i) The dictionary columns are normalized such that $\mathbb{E}_n[z_{c,ij}^2] = 1$ for each $j \in [p_c]$.

(ii) The approximation error satisfies

$$\mathbb{E}_n[\xi_{c,i}^2]^{1/2} \leq C_a \sqrt{\frac{s_c}{n}}.$$

(iii) The penalty satisfies

$$\lambda_c \geq c_\lambda \sqrt{2}\sigma_v \sqrt{\frac{\log(2p_c) + \alpha \log\log(p_c \vee n)}{n}}.$$

(iv) The design satisfies

$$\kappa_c \geq \kappa_0, \qquad \phi_{\max,c}(\lceil \vartheta s_c \rceil \wedge p_c) \leq \overline{\phi}, \qquad \phi_{\min,c}(\lceil (\vartheta + 1)s_c \rceil \wedge p_c) \geq \underline{\phi}.$$

(v) The sparsity index, s_c, and penalty λ_c satisfy

$$\frac{s_c}{\sqrt{n}} \to 0, \qquad \sqrt{s_c}\lambda_c \to 0.$$

Versions of each condition in Assumption 2 appear in Belloni et al. (2012), who also use them to establish convergence of first-stage LASSO estimates. Together with parts (i)–(iv) of Assumption 2, $\sqrt{s_c}\lambda_c \to 0$ yields first-stage prediction consistency, while $s_c/\sqrt{n} \to 0$ controls terms that arise in the higher-order expansion. Combined with Assumption 1, these conditions ensure that (2.3) holds for every candidate. Beyond guaranteeing suitable first-stage convergence of the candidate first-stage estimators, these conditions do little to restrict the candidate list, so dictionaries and penalties may vary across candidates (Bickel et al., 2009; van de Geer and Bühlmann, 2009; Belloni et al., 2012; Chetverikov and Sørensen, 2025).

The next condition ensures that the effective dimension grows and that its fluctuations become small relative to its mean. This allows us to estimate the complexity cost from the observed effective dimension, as described in Section 3.3, and to show that the terms omitted from the AMSE approximation are negligible relative to the candidate-specific component.

Assumption 3 (Partial Beta-Min)**.** *For each candidate, let*

$$\underline{b}_{n,c} := \frac{4}{\sqrt{\underline{\phi}}} \left\{ \frac{(1 + c_\lambda^{-1})\lambda_c \sqrt{s_c}}{\kappa_0} + C_a \sqrt{\frac{s_c}{n}} \right\},$$

where κ_0 and $\underline{\phi}$ are as in Assumption 2(iv). Define the set of strong coordinates by $A_c := \{j \in T_c : |\pi^0_{c,j}| \geq \underline{b}_{n,c}\}$ with $k_c := |A_c|$. There is a fixed constant $C_s \geq 1$ such that, for every $c \in [J]$,

$$s_c \leq C_s k_c, \qquad k_c \to \infty, \qquad \frac{\log(p_c/k_c)}{k_c} \to 0. \tag{3.4}$$

The strong coordinates A_c are selected with probability approaching one. Unlike full beta-min conditions for support recovery (Zhao and Yu, 2006; Wainwright, 2009), Assumption 3 does not impose a lower bound on the coefficients in $T_c \setminus A_c$, nor does it require recovery of the entire support. The requirement that $k_c \to \infty$ parallels the growing-basis condition in Donald and Newey (2001) and, together with the bound $s_c \leq C_s k_c$, underlies the relative concentration of d_c in Proposition 1. Since, under Assumption 2(v), $\underline{b}_{n,c} \to 0$, the coefficients in the strong core may shrink with n.

The partial beta-min condition may be particularly well motivated under strong iden-

tification. If all first-stage coefficients are close to zero, the instruments may not induce enough exogenous variation in the endogenous variable to consistently estimate the structural parameter.

3.2 The AMSE Criterion

This subsection formally establishes the decomposition in (2.6) and presents the infeasible criterion, a closed-form approximation to the expectation of the candidate-specific term. Exact expressions for the terms in (2.6) are given in the appendix but are not needed for the characterization below.

Two population quantities enter the infeasible criterion. The first is a measure of first-stage strength,

$$h_c := \overline{\mathbb{E}}[\widehat{\Pi}_{c,i}\Pi_i], \tag{3.5}$$

the population cross-moment between the fitted values and the optimal instrument. This is the population counterpart of the denominator of the IV estimator in (2.5). Under the conditions of Section 3.1, h_c converges to the second moment of the optimal instrument, H, and so is bounded away from zero in large samples.

The second quantity measures how well the fitted instrument approximates the optimal instrument. Because the IV estimator is unchanged when its instrument is multiplied by a nonzero constant, a fitted instrument proportional to the optimal instrument should register no error. We therefore measure the first-stage approximation error against the best rescaling of the optimal instrument,

$$\mathcal{A}_c := \min_{a\in\mathbb{R}} \overline{\mathbb{E}}\left[(\widehat{\Pi}_{c,i} - a\Pi_i)^2\right] = \overline{\mathbb{E}}[\widehat{\Pi}_{c,i}^2] - \frac{h_c^2}{H}. \tag{3.6}$$

The minimized value $\mathcal{A}_c$ is the mean squared residual from a population regression of the fitted instrument on Π. The ratio $\mathcal{A}_c/h_c^2$ appearing in the infeasible criterion below is invariant to rescaling of the fitted instrument.

The closed-form approximation to the expectation of the candidate-specific term combines these two quantities with certain error moments and moments of the effective dimension of the LASSO fit defined in Section 2.2,

$$S_c := \frac{\sigma_\varepsilon^2 \mathcal{A}_c}{h_c^2} + \frac{\sigma_{\varepsilon v}^2\, \mathbb{E}[d_c^2 + d_c]}{n h_c^2}. \tag{3.7}$$

We refer to S_c as the infeasible criterion. Each of its components is a population quantity, so the infeasible criterion cannot be computed directly from the data. Section 3.3 constructs a consistent feasible counterpart used to select a candidate.

Theorem 1 (Infeasible AMSE Criterion)**.** *Suppose Assumptions 1, 2, and 3 hold, with $J < \infty$ fixed. Then there exist random variables C_0, $\{Q_c\}_{c\in[J]}$, and $\{r_c\}_{c\in[J]}$ such that*

(i) For every candidate,

$$n(\widehat{\beta}_c - \beta)^2 = C_0 + Q_c + r_c, \tag{3.8}$$

where C_0 does not depend on the candidate, $\mathbb{E}[C_0] = \sigma_\varepsilon^2/H + O(n^{-1})$, and the remainder satisfies $\max_{c\in[J]} |r_c|/\mathbb{E}[Q_c] \longrightarrow_p 0$.

(ii) For every candidate, $\mathbb{E}[Q_c] > 0$ and $\mathbb{E}[Q_c] = S_c \left(1 + o(1)\right)$.

Theorem 1(i) establishes the validity of the decomposition in (2.6), while Theorem 1(ii) shows that S_c approximates the candidate-specific AMSE component $\bar{Q}_c = \mathbb{E}[Q_c]$ with vanishing relative error. Together, these results justify ranking candidates by S_c: minimizing S_c is asymptotically equivalent to minimizing $\bar{Q}_c$, and hence AMSE, over the candidate list. The remainder in part (i) is controlled in probability, so this result does not establish a ranking by exact finite-sample MSE.

The first term measures how well the candidate LASSO approximates the optimal instrument, while the second depends on the first two moments of the effective dimension. This term originates from the second moment of $\varepsilon'\widehat{\Pi}_c$, which we compute using results of Tibshirani and Taylor (2012) and Bellec and Zhang (2021).

When the first stage is least squares on a prespecified set of K linearly independent instruments ($\lambda = 0$), the effective dimension always equals K, and S_c reduces, up to lower-order terms, to the criterion of Donald and Newey (2001). With LASSO, however, the effective dimension depends on the realization of the sample, so the complexity cost depends on both its mean and its variance.[6]

A richer dictionary or a lighter penalty may reduce $\mathcal{A}_c$ while increasing the effective dimension. To examine this tradeoff, write $\rho = \mathrm{Corr}(\varepsilon_i, v_i)$, so that $\sigma_{\varepsilon v} = \rho\sigma_\varepsilon\sigma_v$, and

[6]Under the partial beta-min condition in Assumption 3, the effective dimension grows, so that $\mathbb{E}[d_c^2]$ asymptotically dominates $\mathbb{E}[d_c]$. However, both terms appear in the expression for $\mathbb{E}[(\varepsilon'\widehat{\Pi}_c)^2]$, and we keep them since they may both be relevant in finite samples.

rewrite (3.7) as

$$S_c = \frac{\sigma_\varepsilon^2}{h_c^2}\left\{\mathcal{A}_c + \rho^2\sigma_v^2\,\frac{\mathbb{E}[d_c^2 + d_c]}{n}\right\}. \tag{3.9}$$

In this form the second term is weighted by $\rho^2\sigma_v^2$, which is proportional to the squared covariance between the structural and first-stage errors. When the regressor is close to exogenous, the complexity term receives little weight.[7] When the regressor is highly endogenous, the same complexity produces a larger bias. This is the LASSO analogue of the many-instrument bias examined in Bekker (1994) and Donald and Newey (2001).

Cross-validation targets first-stage prediction error rather than the structural AMSE criterion in (3.7). Its rankings may therefore differ even when the complexity term receives little weight, because $\mathcal{A}_c$ allows rescaling and the criterion also depends on h_c. As endogeneity strengthens, the AMSE criterion prefers models with lower complexity, while cross-validation does not take this into consideration. The plug-in penalty rules of Belloni et al. (2012) and Chetverikov and Sørensen (2025) are designed to control first-stage noise as in Assumption 2(iii) and likewise do not account for endogeneity.

3.3 Feasible Candidate Selection

Since S_c depends on population quantities and is thus infeasible, we construct an observable criterion that estimates it up to a candidate-independent centering term and therefore preserves its ranking asymptotically. The only additional inputs not obtained from the candidate fits are estimates of the structural error moments σ_ε^2 and $\sigma_{\varepsilon v}$. We obtain these estimates using residuals from a pilot estimate $\check{\beta}$ of β.

For each candidate, the observed IV denominator

$$\widehat{h}_c := \mathbb{E}_n[\widehat{\Pi}_{c,i}x_i] \tag{3.10}$$

will serve as an estimator for the first-stage cross-moment h_c. The KKT conditions for the LASSO optimization problem in (2.4) yield $\widehat{h}_c = \mathbb{E}_n[\widehat{\Pi}_{c,i}^2] + \lambda_c\|\widehat{\pi}_c\|_1 \geq 0$, so $\widehat{h}_c = 0$ if and only if $\widehat{\pi}_c = 0$. The pilot $\check{\beta}$ can be any estimator with $\check{\beta} - \beta = O_p(n^{-1/2})$, such as the IV-LASSO estimate $\widehat{\beta}_{\check{c}}$ from a possibly data-dependent candidate $\check{c} \in [J]$.

[7] Exact exogeneity is excluded by Assumption 1(ii), and the case $\sigma_{\varepsilon v} = 0$ requires a different expansion.

Given the pilot, let

$$\widehat{\sigma}_{\varepsilon}^2 := \mathbb{E}_n[(y_i - \check{\beta} x_i)^2] \qquad \text{and} \qquad \widehat{\sigma}_{\varepsilon v} := \mathbb{E}_n[(y_i - \check{\beta} x_i) x_i]. \tag{3.11}$$

We put together these estimates to construct the feasible criterion

$$\widehat{S}_c := \frac{\widehat{\sigma}_{\varepsilon}^2 \mathbb{E}_n[\widehat{\Pi}_{c,i}^2] + n^{-1} \widehat{\sigma}_{\varepsilon v}^2 (d_c^2 + d_c)}{\widehat{h}_c^2}, \tag{3.12}$$

with the convention $\widehat{S}_c := +\infty$ when $\widehat{h}_c = 0$. The selected candidate minimizes the feasible criterion,

$$\widehat{c} := \min \left(\arg\min_{c \in [J]} \widehat{S}_c \right),$$

with ties broken by the smallest index.

The feasible criterion $\widehat{S}_c$ replaces the moment $\mathbb{E}[d_c^2 + d_c]$ in the infeasible S_c with the realized value $d_c^2 + d_c$. This replacement is justified by the following proposition.

Proposition 1 (Ratio Consistency of the Effective Dimension)**.** *Suppose Assumptions 1, 2, and 3 hold, with $J < \infty$ fixed. Then, for every candidate,*

$$\frac{d_c}{\mathbb{E}[d_c]} \longrightarrow_p 1, \tag{3.13}$$

$$\frac{d_c^2 + d_c}{\mathbb{E}[d_c^2 + d_c]} \longrightarrow_p 1. \tag{3.14}$$

The growth conditions on the strong core in Assumption 3, together with a variance bound derived from the second-order Stein identities of Bellec and Zhang (2021), yield relative concentration of d_c and hence of $d_c^2 + d_c$. This does not require the support of the LASSO solution to contain the support T_c in (3.1).

The numerator in $\widehat{S}_c$ contains the fitted second moment $\mathbb{E}_n[\widehat{\Pi}_{c,i}^2]$ rather than a direct estimate of the approximation error $\mathcal{A}_c$. Applying the decomposition in (3.6) to sample moments, with $\widehat{\mathcal{A}}_c := \min_{a \in \mathbb{R}} \mathbb{E}_n[(\widehat{\Pi}_{c,i} - a\Pi_i)^2]$, splits the first term of $\widehat{S}_c$ into two parts,

$$\frac{\widehat{\sigma}_{\varepsilon}^2 \mathbb{E}_n[\widehat{\Pi}_{c,i}^2]}{\widehat{h}_c^2} = \frac{\widehat{\sigma}_{\varepsilon}^2 \widehat{\mathcal{A}}_c}{\widehat{h}_c^2} + \frac{\widehat{\sigma}_{\varepsilon}^2}{H} \left(\frac{\mathbb{E}_n[\widehat{\Pi}_{c,i} \Pi_i]}{\widehat{h}_c} \right)^2.$$

The first part estimates $\sigma_\varepsilon^2 \mathcal{A}_c / h_c^2$, the first term of S_c. Because each candidate first stage is consistent, we can expand the second part around its value for the optimal instrument itself, $\widehat{\sigma}_\varepsilon^2 H / \mathbb{E}_n[\Pi_i x_i]^2$, where $\mathbb{E}_n[\Pi_i x_i] = H + \mathbb{E}_n[\Pi_i v_i]$ and $\mathbb{E}_n[\Pi_i v_i]$ is of order $n^{-1/2}$. This gives

$$B_n := \widehat{\sigma}_\varepsilon^2 \left(\frac{1}{H} - \frac{2\mathbb{E}_n[\Pi_i v_i]}{H^2} \right) \tag{3.15}$$

up to terms that are negligible relative to $\bar{Q}_c$. Theorem 2 below shows that

$$\widehat{S}_c = B_n + \bar{Q}_c(1 + o_p(1)).$$

Although B_n may dominate the candidate-specific AMSE components $\bar{Q}_c$, it is common to all candidates and therefore does not affect their feasible-criterion ranking. Minimizing $\widehat{S}_c$ is then asymptotically equivalent to minimizing $\bar{Q}_c$. This does not require B_n to be observed.

Theorem 2 (Feasible AMSE Selection)**.** *Suppose Assumptions 1, 2, and 3 hold, with $J < \infty$ fixed, and that the pilot satisfies $\check{\beta} - \beta = O_p(n^{-1/2})$.*

(i) Up to the common centering B_n, the feasible criterion consistently estimates the infeasible criterion,

$$\max_{c \in [J]} \frac{|\widehat{S}_c - B_n - S_c|}{S_c} \longrightarrow_p 0, \tag{3.16}$$

and thus the expectation of the candidate-specific term,

$$\max_{c \in [J]} \frac{|\widehat{S}_c - B_n - \bar{Q}_c|}{\bar{Q}_c} \longrightarrow_p 0. \tag{3.17}$$

(ii) The selected candidate $\widehat{c}$ satisfies

$$\frac{\bar{Q}_{\widehat{c}}}{\min_{c \in [J]} \bar{Q}_c} \longrightarrow_p 1. \tag{3.18}$$

The supplementary appendix establishes the same AMSE criterion and feasible selection result for a class of non-Gaussian errors. That extension requires the structural error to have a linear conditional mean and constant conditional variance given the first-stage error, along with additional regularity conditions. Under more general departures from Gaussianity or homoskedasticity, AMSE optimality is not established

here, although the results in Section 4.2 suggest that the AMSE in more general models may still be well approximated by the AMSE in the Gaussian model. If, under alternative conditions, all candidates continue to admit the common first-order expansion in (2.3), then the selected estimator retains that expansion and first-order inference remains valid.

4 Simulation Study

We compare the proposed criterion to first-stage selection via cross-validation and the plug-in penalty of Belloni et al. (2012). We also consider a cross-fitted IV estimator, with the dictionary and penalty chosen within each training sample (Chernozhukov et al., 2018). The simulations are calibrated to the data of Paravisini et al. (2015), who study the effect of credit supply on exports. We first consider Gaussian errors in Section 4.1 and then repeat the exercise in Section 4.2 with common-scale Laplace and smooth-transformation errors.

4.1 Calibrated Gaussian Designs

We hold the observed exposure instruments fixed and generate the outcome and endogenous variable from the IV model described in (2.1) with $\beta = 1$. The true first stage Π is proportional to the fitted values from a least-squares regression of credit growth on 17 instruments selected by a cross-validated LASSO from the original instruments and their 45 pairwise products. We also consider a dense alternative to this main first stage, proportional to the least-squares fitted values on all 55 of these instruments. We rescale these fitted values to vary identification strength, targeting expected first-stage F statistics of 2.7, 4.4, 7.7, 14.5, or 28.0 at the original sample size of $n = 14{,}208$ firm–product–destination export flows. Holding the instruments fixed and only rescaling the two first stages lets us examine how each rule's performance changes with identification strength and endogeneity in an empirically plausible setting.

The errors are generated homoskedastic Gaussian, with $\varepsilon_i = \rho v_i + \sqrt{1-\rho^2}\, e_i$ for $v_i, e_i \overset{\text{iid}}{\sim} N(0,1)$. The parameter ρ controls the degree of endogeneity. We consider a value calibrated to the data, $\widehat{\rho} \approx -0.22$, along with its multiples $\rho = c\,\widehat{\rho}$ for $c = 0, 0.25, \ldots, 2.75$ to examine how the performance of each rule changes with the level of endogeneity.[8] Each of the 120 designs is simulated 1,000 times. We report

[8] Assumption 1(ii) rules out exact exogeneity, so the designs with $\rho = 0$ lie outside our theory,

the average of $n(\widehat{\beta} - \beta)^2$, which we refer to as either the risk or the mean squared error.

Each candidate first stage implementation uses one of two nested dictionaries: either the ten original exposure instruments (Original) or a richer dictionary of 114 instruments including their pairwise products and the authors' alternative exposure measures (Expanded). For each dictionary, we consider 37 multiples of the plug-in penalty of Belloni et al. (2012) ("BCCH"), giving 74 candidates in total. This plug-in penalty level uses the known standard deviation of the first-stage error in place of an estimate. The proposed criterion selects among these candidate dictionary-penalty pairs by minimizing $\widehat{S}_c$ in (3.12), with estimated error moments coming from a pilot estimator of β. Cross-validation chooses from the same candidates using ten-fold first-stage prediction error, keeping observations from the same firm in the same fold. The cross-fitting estimator ("Cross-fit") makes this choice separately for each of ten folds formed by firm, using only the other nine folds, and the resulting predictions for each held-out fold are used as instruments in a single IV estimate. BCCH uses its plug-in penalty on the Original dictionary. When it selects no instruments, we use the pilot estimate instead.

Figure 2 shows how the rules respond to endogeneity. Consistent with the increasing complexity cost seen in (3.9), the criterion selects less complex first stages as endogeneity rises, while the other three rules do not account for the level of endogeneity. At both $F = 7.7$ and $F = 14.5$, the criterion's mean effective dimension falls by more than six-fold. Its risk is more than 50% below that of BCCH at the lowest levels of endogeneity and between 30% and 50% below that of cross-validation at the highest level, although cross-validation performs better in some designs with intermediate levels of endogeneity.

Table 1 shows that the criterion remains close to the best feasible rule across the design grid, but does not dominate every design. With the main first stage, cross-validation has lower average relative risk than the criterion at $F = 2.7$ and near the calibrated correlation. As identification strengthens, the criterion and cross-fitting perform more similarly. The criterion's largest relative risk is about 20% above that of the best rule. The large BCCH entries are driven by a few replications at $F = 2.7$ in which BCCH selects no instruments and the pilot estimate is used in its place.

and we include them only as a limiting case.

Table 1: Relative risk, Gaussian errors.

Design	Summary Measure	Feasible rules: Criterion	Cross-validation	Cross-fit	BCCH	Infeasible: Known Moments
Panel A: Main signal, by first-stage strength						
$F = 2.7$	Average	1.072	1.034	1.589	14.529	0.857
	Maximum	1.199	1.143	2.045	21.117	0.976
$F = 4.4$	Average	1.051	1.074	1.240	1.813	0.924
	Maximum	1.173	1.291	1.615	2.459	0.987
$F = 7.7$	Average	1.034	1.126	1.124	1.824	0.962
	Maximum	1.129	1.464	1.449	2.394	1.023
$F = 14.5$	Average	1.015	1.302	1.068	1.835	0.975
	Maximum	1.077	2.002	1.300	2.288	1.009
$F = 28.0$	Average	1.036	1.301	1.038	1.908	1.014
	Maximum	1.074	2.009	1.163	2.170	1.044
Panel B: Main signal, by degree of endogeneity						
$0 \leq \lvert\rho\rvert \leq 0.109$	Average	1.009	1.048	1.422	3.490	0.865
	Maximum	1.047	1.143	2.045	9.883	1.000
$0.164 \leq \lvert\rho\rvert \leq 0.273$	Average	1.105	1.014	1.229	4.105	0.967
	Maximum	1.199	1.107	1.725	14.566	1.007
$0.327 \leq \lvert\rho\rvert < 0.6$	Average	1.026	1.304	1.098	4.966	0.976
	Maximum	1.183	2.009	1.534	21.117	1.044
Panel C: Dense signal, all designs						
All F and ρ	Average	1.035	1.145	1.374	4.339	0.926
	Maximum	1.175	2.017	2.474	23.204	1.002

Notes: Risk is the mean of $n(\widehat{\beta} - \beta)^2$ over 1,000 replications. All panels present risk relative to the best of the four feasible rules in each design. Panels A and B use the main first stage. Panel A averages or maximizes over the twelve values of ρ at each F. Panel B averages or maximizes over the five values of F and the values of $|\rho|$ in each range. Panel C averages or maximizes over all 60 designs with the dense first stage. Known Moments evaluates the feasible criterion at population error moments and is not one of the four feasible rules, so its relative risk can fall below one.

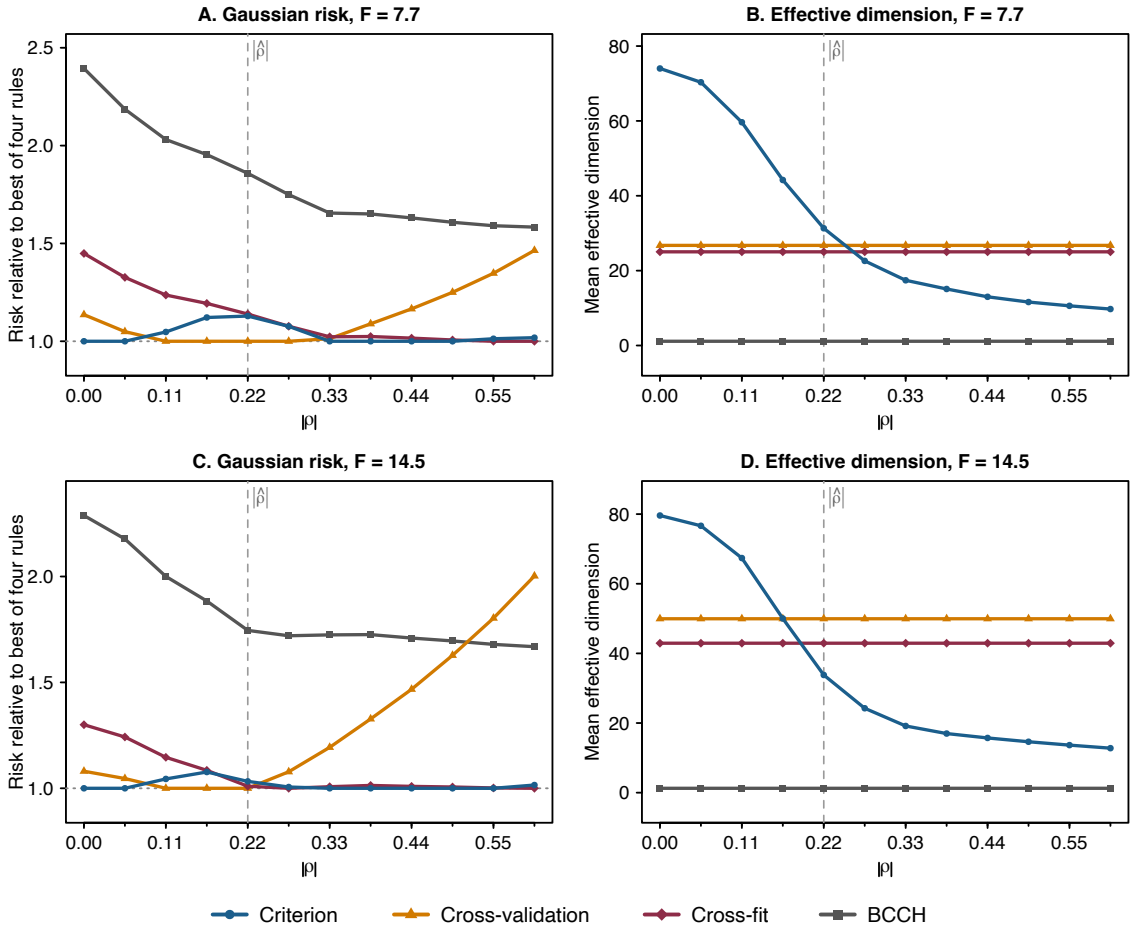


Figure 2: Relative risk and effective dimension under Gaussian errors.

Notes: Results use the main first stage, $n = 14{,}208$, and 1,000 replications. Panels A and C report risk relative to the lowest risk among the four rules; Panels B and D report mean effective dimension, averaged over folds for cross-fitting. Panels A–B use $F = 7.7$ and Panels C–D use $F = 14.5$. The horizontal axis is the absolute error correlation $|\rho|$, and the dashed line marks the calibrated correlation $|\widehat{\rho}| = 0.218$.

Setting the error moments to their population values, $\sigma_\varepsilon^2 = 1$ and $\sigma_{\varepsilon v} = \rho$, lowers the criterion's average relative risk for the main first stage from 1.042 to 0.946. The Known Moments benchmark still uses estimated first-stage quantities to rank candidates and need not attain the lowest risk in every design. We do not include it among the four rules used to normalize risk.

4.2 Robustness to Non-Gaussian Errors

We repeat the simulations of Section 4.1 under two departures from Gaussianity that differ in whether they preserve the constant conditional variance of the Gaussian

design. The simulation details are otherwise left completely unchanged.

For the common-scale Laplace design, we draw mutually independent $W_i \sim \text{Exp}(1)$ and $a_i, b_i \sim N(0,1)$, independently across observations, and set

$$v_i = \sqrt{W_i}\, a_i, \qquad \varepsilon_i = \sqrt{W_i}\left(\rho a_i + \sqrt{1-\rho^2}\, b_i\right).$$

The scale is shared by the two errors within each observation, and both errors have variance-one Laplace marginals.

For the smooth-transformation design, we normalize the following transformation to unit variance. Its unnormalized form appears in Chetverikov et al. (2020, Figure 1):

$$\psi(t) = \frac{t + t^3}{\nu(1+2t^2)}, \qquad \nu^2 = \mathbb{E}\left[\left(\frac{G+G^3}{1+2G^2}\right)^2\right], \quad G \sim N(0,1).$$

For independent standard Gaussian draws G_i, B_i, independent across observations, we set

$$v_i = \psi(G_i), \qquad \varepsilon_i = \rho v_i + \sqrt{1-\rho^2}\,\psi(B_i).$$

Both designs have mean-zero, unit-variance errors with correlation ρ. The smooth-transformation design also satisfies $\text{Var}(\varepsilon_i \mid v_i) = 1-\rho^2$, so it is covered by the non-Gaussian extension in the supplementary appendix. The common-scale Laplace design, in which this conditional variance depends on v_i, goes beyond the assumptions of that extension. As before, the designs with $\rho = 0$ serve as exogenous benchmarks.

Table 2 reports the criterion's relative risk under each error distribution, with Gaussian errors as the reference row. We use the same 60 designs for each first-stage signal and 1,000 replications per design. Average relative risks remain similar across error distributions, although the maximum for the dense signal reaches 1.221 under smooth-transformation errors. Under these errors, the criterion's unnormalized risk differs from its risk under Gaussian errors by only between -4.1% and 3.1%, and its mean effective dimension changes by at most 3.2%.

For the main signal at $F = 7.7$ under Laplace errors, the mean effective dimension of the first stage selected by the criterion still falls from 74.3 to 9.7 as $|\rho|$ increases. At the highest level of endogeneity, the criterion's risk is about 28% below that

Table 2: Relative risk of the proposed criterion across error distributions.

	Main signal		Dense signal	
Error distribution	Average	Maximum	Average	Maximum
Gaussian	1.042	1.199	1.035	1.175
Common-scale Laplace	1.041	1.174	1.035	1.193
Smooth transformation	1.045	1.188	1.041	1.221

Notes: Each signal comprises 60 designs (five values of F and twelve values of ρ) with 1,000 replications per design. Relative risk divides the criterion's risk by the lowest risk among the four feasible rules in the same design and under the same error distribution. Known Moments is not part of this comparison. The Gaussian design is described in Section 4.1, and the two non-Gaussian constructions are defined in Section 4.2.

of cross-validation but 5% above that of cross-fitting. Cross-validation retains an advantage in some designs with intermediate levels of endogeneity, and cross-fitting has lower average relative risk at $F = 28.0$. The largest BCCH risks under Laplace errors are again driven by a few replications in which BCCH selects no instruments. While AMSE optimality under the smooth-transformation design is established in the supplementary appendix, the results under Laplace errors suggest that the AMSE in more general models may be well approximated by the AMSE in the Gaussian model.

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A Notation and Preliminary Reductions

Here we collect the notation used throughout the appendix. We write C for a positive constant that does not depend on n or on the candidate and whose value may change from line to line. For positive a_n and b_n, we write $a_n \asymp b_n$ when a_n/b_n is bounded and bounded away from zero. Since J is fixed, a rate or limit established for each candidate holds uniformly over $c \in [J]$, and so we fix a candidate and suppress its subscript. For example, we write $\widehat{\Pi}$, h, d, and S for $\widehat{\Pi}_c$, h_c, d_c, and S_c. We only bring the subscript back where candidates are compared.

LASSO notation. To use the Stein identities, we need expressions for the derivatives of the LASSO fit with respect to its response. For $w \in \mathbb{R}^n$, let $\widehat{\pi}(w)$ be a minimizer of (2.4) with x replaced by w, and write

$$\mu(w) := Z\widehat{\pi}(w), \qquad u(w) := w - \mu(w), \qquad E(w) := \left\{ j \in [p] : |\mathbb{E}_n[z_{ij} u_i(w)]| = \lambda \right\}.$$

In this notation, $\mu(x) = \widehat{\Pi}$, $u(x) = u$, and $E(x) = E$.

Lemma D.1, below, recalls certain facts about the LASSO fit from Tibshirani and Taylor (2012). In particular, it notes that the LASSO fitted-value map $\mu : \mathbb{R}^n \to \mathbb{R}^n$ is one-Lipschitz, and hence Lebesgue-almost-everywhere differentiable by Rademacher's theorem. At a differentiability point $w \in \mathbb{R}^n$, write $D\mu(w)$ for the Jacobian of μ. Since $\Pi + v$ has a Lebesgue density under Assumption 1(i), the fit is almost surely differentiable at the observed $x = \Pi + v$, and we write $P := D\mu(\Pi + v)$ for the

Jacobian there. Then Lemma D.1 and (D.5) give, almost surely,

$$P = P' = P^2, \qquad \operatorname{tr}(P) = d, \qquad P\widehat{\Pi} = \widehat{\Pi}. \tag{A.1}$$

That is, P is the orthogonal projection onto the column space of $Z_{E(x)}$, the active columns indexed by the equicorrelation set in (2.7). Both P and $I - P$ are therefore contractions.

With d defined in (2.8), we will save notation later in the proofs and write

$$\bar{d} := \mathbb{E}[d], \qquad \text{so that} \qquad \mathbb{E}[d^2 + d] = \bar{d}^2 + \operatorname{Var}(d) + \bar{d}. \tag{A.2}$$

Population projection residual. The minimum in (3.6) is attained at $a = h/H$. We write

$$M := \widehat{\Pi} - \frac{h}{H}\Pi \tag{A.3}$$

for the residual from the population projection of $\widehat{\Pi}$ on Π, so that $\mathbb{E}\|M\|_2^2 = n\mathcal{A}$. This residual is the $m(v)$ of Lemma E.2 with $b = (h/H)\Pi$, and the identity (E.5) becomes

$$\mathbb{E}[(v'M - \sigma_v^2 d)^2] = \sigma_v^2 n\mathcal{A} + \sigma_v^4 \bar{d}. \tag{A.4}$$

Gaussian error decomposition. Under Assumption 1(i) set $e := \varepsilon - \frac{\sigma_{\varepsilon v}}{\sigma_v^2} v$. Then e is Gaussian with mean zero and $\operatorname{Cov}(e_i, v_i) = 0$, hence independent of v:

$$\varepsilon = \frac{\sigma_{\varepsilon v}}{\sigma_v^2} v + e, \qquad e \perp v, \qquad \operatorname{Var}(e_i) = \sigma_\varepsilon^2 - \frac{\sigma_{\varepsilon v}^2}{\sigma_v^2}. \tag{A.5}$$

Positive denominator event. By Lemma F.2 there is a fixed constant $\underline{h} > 0$ with $h \geq \underline{h}$ once n is large enough. We take n this large throughout, so that H, h, and their reciprocals are bounded, and we enlarge C where needed for the bounds below to hold for all such n. Every function of v to which we apply the Stein identities or the Poincaré inequality is locally Lipschitz, and by Lemma D.1 its value and gradient grow at most polynomially. As such, these results apply and all expectations below are finite. With $\widehat{h}$ the sample denominator in (3.10), define

$$\mathcal{H}_n := \left\{ \widehat{h} \geq \frac{h}{2} \right\}. \tag{A.6}$$

Since analyzing the properties of IV-LASSO is difficult when the denominator is close to zero, our proofs will often work on $\mathcal{H}_n$ to establish stochastic bounds. Because $\mathbb{P}(\mathcal{H}_n) \to 1$ by Lemma B.1, these bounds also hold without restricting to $\mathcal{H}_n$, while all expectations below are unconditional.

B Proof of Theorem 1

Proof. Begin by defining the numerator scaled by the population denominator, along with the denominator correction,

$$\Gamma := \frac{\sqrt{n}\,\mathbb{E}_n[\varepsilon_i \widehat{\Pi}_i]}{h}, \qquad \Delta := \frac{\widehat{h} - h}{h}. \tag{B.1}$$

On $\mathcal{H}_n$, Lemma B.1 shows that $\widehat{\beta}$ is the ratio in (2.5), and after substituting $y_i = \beta x_i + \varepsilon_i$ we can write

$$\sqrt{n}(\widehat{\beta} - \beta) = \frac{\sqrt{n}\,\mathbb{E}_n[\varepsilon_i \widehat{\Pi}_i]}{\widehat{h}} = \frac{\Gamma}{1+\Delta} = \Gamma(1-\Delta) + \frac{\Gamma\Delta^2}{1+\Delta}. \tag{B.2}$$

In the term $\Gamma\Delta$, we replace Γ by the leading term in (2.3),

$$L_n := \frac{\sqrt{n}\,\mathbb{E}_n[\varepsilon_i \Pi_i]}{H}, \tag{B.3}$$

and we set $q := \mathbb{E}_n[\Pi_i v_i]/H$ and define

$$\begin{aligned} \Lambda &:= \Gamma - \Delta L_n, & C_0 &:= (1-q)^2 L_n^2, \\ Q &:= \Lambda^2 - C_0, & r &:= n(\widehat{\beta} - \beta)^2 - \Lambda^2. \end{aligned} \tag{B.4}$$

These definitions make (3.8) an identity, and C_0 does not depend on the candidate. If we use the optimal instrument Π in place of $\widehat{\Pi}$, then h becomes H and $\widehat{h}$ becomes $\mathbb{E}_n[\Pi_i x_i] = H(1+q)$, so that $\Gamma = L_n$, $\Delta = q$, $\Lambda = (1-q)L_n$, and $Q = 0$.

Step 1: $\sqrt{n}(\widehat{\beta} - \beta) = \Lambda + o_p(S)$. Subtracting Λ from (B.2), we have

$$\sqrt{n}(\widehat{\beta} - \beta) - \Lambda = -\Delta(\Gamma - L_n) + \frac{\Gamma\Delta^2}{1+\Delta}$$

on $\mathcal{H}_n$, where $1 + \Delta = \widehat{h}/h$ is bounded away from zero. Apply Lemmas B.1 and B.2

to see that $\Delta = O_p(n^{-1/2})$ and $\Gamma - L_n = O_p(\sqrt{S})$, so that $\Gamma = O_p(1)$, and therefore

$$\sqrt{n}(\widehat{\beta} - \beta) - \Lambda = O_p\left(\sqrt{\frac{S}{n}} + \frac{1}{n}\right) = S \cdot O_p\left(\frac{1}{\sqrt{nS}} + \frac{1}{nS}\right),$$

which is $o_p(S)$ because $nS \to \infty$ by (F.14). The same bounds show that

$$\Lambda = O_p(1), \qquad \sqrt{n}(\widehat{\beta} - \beta) = O_p(1). \tag{B.5}$$

Step 2: $\mathbb{E}[\Gamma^2] = \sigma_\varepsilon^2/H + S$. From the second-moment identity (E.6) in Lemma E.3, we have

$$\mathbb{E}[\Gamma^2] = \frac{\mathbb{E}[(\varepsilon'\widehat{\Pi})^2]}{nh^2} = \frac{\sigma_\varepsilon^2 \mathbb{E}\|\widehat{\Pi}\|_2^2}{nh^2} + \frac{\sigma_{\varepsilon v}^2 \mathbb{E}[d^2 + d]}{nh^2}.$$

By (3.6), $\mathbb{E}\|\widehat{\Pi}\|_2^2/n = \mathcal{A} + h^2/H$, and substituting this into the first term yields

$$\mathbb{E}[\Gamma^2] = \frac{\sigma_\varepsilon^2}{H} + \underbrace{\frac{\sigma_\varepsilon^2 \mathcal{A}}{h^2} + \frac{\sigma_{\varepsilon v}^2 \mathbb{E}[d^2 + d]}{nh^2}}_{S}.$$

Step 3: $\mathbb{E}[Q] = S\{1 + o(1)\}$. Expand $\Lambda^2 = (\Gamma - \Delta L_n)^2$ and write $\Gamma = L_n + (\Gamma - L_n)$ in the cross term to obtain

$$\mathbb{E}[\Lambda^2] - \mathbb{E}[\Gamma^2] = -2\mathbb{E}[\Delta L_n^2] - 2\mathbb{E}[\Delta L_n(\Gamma - L_n)] + \mathbb{E}[\Delta^2 L_n^2].$$

By Lemma B.3, $|\mathbb{E}[\Delta L_n^2]| \le C\{\sqrt{S/n} + 1/n\}$ and $\mathbb{E}[\Delta^2 L_n^2] \le C/n$. For the middle term, Cauchy–Schwarz and Lemmas B.2 and B.3 give

$$|\mathbb{E}[\Delta L_n(\Gamma - L_n)]| \le \{\mathbb{E}[\Delta^2 L_n^2]\,\mathbb{E}[(\Gamma - L_n)^2]\}^{1/2} \le C\sqrt{\frac{S}{n}}.$$

To compute $\mathbb{E}[C_0]$, note that under Assumption 1 the pair $(\Pi'\varepsilon, \Pi' v)$ is centered jointly Gaussian with covariance matrix $\|\Pi\|_2^2 \Omega = nH\Omega$. Its odd moments vanish, and the product-moment formula of Isserlis (1918) states that $\mathbb{E}[X^2Y^2] = \mathbb{E}[X^2]\mathbb{E}[Y^2] + 2\{\mathbb{E}[XY]\}^2$ for a centered jointly Gaussian pair (X, Y). We therefore have

$$\mathbb{E}[L_n^2] = \frac{\sigma_\varepsilon^2}{H}, \qquad \mathbb{E}[qL_n^2] = 0, \qquad \mathbb{E}[q^2 L_n^2] = \frac{\sigma_\varepsilon^2 \sigma_v^2 + 2\sigma_{\varepsilon v}^2}{nH^2}.$$

Since $C_0 = (1-2q+q^2)L_n^2$, this gives $\mathbb{E}[C_0] = \sigma_\varepsilon^2/H + O(n^{-1})$, which is the bound on $\mathbb{E}[C_0]$ in part (i). Combining these bounds with Step 2 and $\mathbb{E}[Q] = \mathbb{E}[\Lambda^2] - \mathbb{E}[C_0]$, we can bound

$$|\mathbb{E}[Q] - S| = \left|\mathbb{E}[\Lambda^2] - \mathbb{E}[\Gamma^2] + \frac{\sigma_\varepsilon^2}{H} - \mathbb{E}[C_0]\right| \leq C\left\{\sqrt{\frac{S}{n}} + \frac{1}{n}\right\}.$$

After dividing by S, we use $nS \to \infty$ once more to conclude that $\mathbb{E}[Q]/S \to 1$, so that $\mathbb{E}[Q] > 0$ and part (ii) follows.

Step 4: $r/\mathbb{E}[Q] \to_p 0$. Factor the difference of squares in (B.4) to write

$$|r| = |\sqrt{n}(\widehat{\beta} - \beta) - \Lambda|\,|\sqrt{n}(\widehat{\beta} - \beta) + \Lambda|.$$

The first factor is $o_p(S)$ by Step 1, and the second is $O_p(1)$ by (B.5). Since $S \leq 2\mathbb{E}[Q]$ for all large n by Step 3, we have $r/\mathbb{E}[Q] = o_p(1)$, which proves part (i).

Finally, for the finitely many n at which $h_c < \underline{h}$ or $\mathbb{E}[Q_c] \leq 0$ for some candidate, set $C_0 := 0$, and set $Q_c := 1$ and $r_c := n(\widehat{\beta}_c - \beta)^2 - 1$ for every $c \in [J]$. This choice preserves the decomposition and keeps $\mathbb{E}[Q]$ positive, and it does not affect the asymptotic statements. □

B.1 Supporting Lemmas

Lemma B.1 (The Sample Denominator)**.** *(i) Under Assumption 1,*

$$\mathbb{E}[\widehat{h}] = h + \frac{\sigma_v^2 \bar{d}}{n}, \qquad \mathrm{Var}(\widehat{h}) \leq \frac{4\sigma_v^2(H + \sigma_v^2)}{n}. \tag{B.6}$$

(ii) If Assumptions 2 and 3 also hold, with $J < \infty$ fixed and Δ as in (B.1), then $\mathbb{E}[(\widehat{h} - h)^2] \leq C/n$ and

$$\widehat{h} - h = O_p(n^{-1/2}), \qquad \Delta = O_p(n^{-1/2}). \tag{B.7}$$

(iii) Under the hypotheses of (ii), $\mathbb{P}(\mathcal{H}_n) \to 1$, and on $\mathcal{H}_n$ we have $\mathbb{E}_n[\widehat{\Pi}_i^2] > 0$ and $\widehat{\beta} = \mathbb{E}_n[\widehat{\Pi}_i y_i]/\widehat{h}$.

Proof. Since $n\widehat{h} = x'\widehat{\Pi} = \Pi'\widehat{\Pi} + v'\widehat{\Pi}$, the first identity in (E.7) gives $\mathbb{E}[\widehat{h}] = h + \sigma_v^2\bar{d}/n$. For the variance, the gradient of $x'\widehat{\Pi} = x'\mu(x)$ with respect to v is $\widehat{\Pi} + Px$, whose

norm is at most $2\|x\|_2$ by (A.1) and (D.4). By the Gaussian Poincaré inequality (E.1) we can then bound

$$\mathrm{Var}(\widehat{h}) = \frac{\mathrm{Var}(x'\widehat{\Pi})}{n^2} \leq \frac{4\sigma_v^2 \mathbb{E}\|x\|_2^2}{n^2} = \frac{4\sigma_v^2(H+\sigma_v^2)}{n},$$

which proves (i). Under (ii), $\bar{d} = o(\sqrt{n})$ by (F.13), so that $\mathbb{E}[(\widehat{h}-h)^2] = \mathrm{Var}(\widehat{h}) + \sigma_v^4\bar{d}^2/n^2 \leq C/n$, and we obtain (B.7) from Markov's inequality and $h \geq \underline{h}$. Since $\mathcal{H}_n^c$ is contained in $\{|\widehat{h}-h| > \underline{h}/2\}$, it follows that $\mathbb{P}(\mathcal{H}_n) \to 1$. Finally, $\widehat{h} = 0$ whenever $\widehat{\Pi} = 0$, so on $\mathcal{H}_n$ the fit $\widehat{\Pi}$ is nonzero and the estimator is the ratio in (2.5). □

Lemma B.2. *Suppose Assumptions 1 and 2 hold, with $J < \infty$ fixed, and let Γ and L_n be as in (B.1) and (B.3). Then*

$$\mathbb{E}[(\Gamma - L_n)^2] \leq 2S. \tag{B.8}$$

Proof. With M the population projection residual in (A.3), we have $\Gamma - L_n = \varepsilon'\{\widehat{\Pi} - (h/H)\Pi\}/(\sqrt{n}\,h) = \varepsilon' M/(\sqrt{n}\,h)$. Using the Gaussian error decomposition (A.5), we can write

$$\varepsilon' M - \sigma_{\varepsilon v} d = \frac{\sigma_{\varepsilon v}}{\sigma_v^2}(v'M - \sigma_v^2 d) + e'M.$$

Conditional on v, the vector M and the rank d are fixed, while $e'M$ has mean zero and variance $(\sigma_\varepsilon^2 - \sigma_{\varepsilon v}^2/\sigma_v^2)\|M\|_2^2$. By (A.4) and $\mathbb{E}\|M\|_2^2 = n\mathcal{A}$, we therefore have

$$\mathbb{E}[(\varepsilon' M - \sigma_{\varepsilon v} d)^2] = \frac{\sigma_{\varepsilon v}^2}{\sigma_v^4}(\sigma_v^2 n\mathcal{A} + \sigma_v^4 \bar{d}) + \left(\sigma_\varepsilon^2 - \frac{\sigma_{\varepsilon v}^2}{\sigma_v^2}\right) n\mathcal{A} = \sigma_\varepsilon^2 n\mathcal{A} + \sigma_{\varepsilon v}^2 \bar{d}.$$

Writing $\varepsilon' M = (\varepsilon' M - \sigma_{\varepsilon v} d) + \sigma_{\varepsilon v} d$ and using $(a+b)^2 \leq 2a^2 + 2b^2$, we conclude that

$$\mathbb{E}[(\Gamma - L_n)^2] = \frac{\mathbb{E}[(\varepsilon' M)^2]}{nh^2} \leq \frac{2}{nh^2}\left\{\sigma_\varepsilon^2 n\mathcal{A} + \sigma_{\varepsilon v}^2 \mathbb{E}[d^2 + d]\right\} = 2S.$$

□

Lemma B.3 (Moments of the Denominator Correction)**.** *Suppose Assumptions 1, 2, and 3 hold, with $J < \infty$ fixed, and let Δ and L_n be as in (B.1) and (B.3). Then*

$$\mathbb{E}[\Delta^2 L_n^2] \leq \frac{C}{n}, \tag{B.9}$$

$$|\mathbb{E}[\Delta L_n^2]| \leq C\left\{\sqrt{\frac{S}{n}} + \frac{1}{n}\right\}. \tag{B.10}$$

Proof. We prove both bounds using the Stein identity

$$\mathbb{E}[(\Pi' v)g(v)] = \sigma_v^2 \mathbb{E}[\Pi' \nabla g(v)], \tag{B.11}$$

which follows from Stein's lemma (Stein, 1981, Lemma 2).

We will also need three facts. First, by the proof of Lemma B.1, the gradient of Δ with respect to v is

$$\nabla\Delta = \frac{\widehat{\Pi} + Px}{nh},$$

so that $\mathbb{E}\|\nabla\Delta\|_2^2 \leq 4\mathbb{E}\|x\|_2^2/(nh)^2 \leq C/n$. Second, Lemma B.1 gives $\mathbb{E}[\Delta] = \sigma_v^2 \bar{d}/(nh)$ and $\mathbb{E}[\Delta^2] \leq C/n$. Third, by (A.5),

$$L_n = \frac{\sigma_{\varepsilon v}}{\sigma_v^2 \sqrt{n}\, H}\Pi' v + \frac{\Pi' e}{\sqrt{n}\, H},$$

where the second term is independent of v and has mean zero and variance $(\sigma_\varepsilon^2 - \sigma_{\varepsilon v}^2/\sigma_v^2)/H$, so that

$$\mathbb{E}[L_n^2 \mid v] = \frac{\sigma_\varepsilon^2 - \sigma_{\varepsilon v}^2/\sigma_v^2}{H} + \frac{\sigma_{\varepsilon v}^2}{\sigma_v^4 n H^2}(\Pi' v)^2. \tag{B.12}$$

Step 1: $\mathbb{E}[\Delta^2 L_n^2] \leq C/n$. Since Δ depends only on v, multiplying (B.12) by Δ^2 and taking expectations yields $\mathbb{E}[\Delta^2 L_n^2] \leq C\mathbb{E}[\Delta^2] + (C/n)\mathbb{E}[(\Pi' v)^2 \Delta^2]$, and it thus suffices to show that $\mathbb{E}[(\Pi' v)^2 \Delta^2] \leq C$. To this end, apply (B.11) with $g(v) = (\Pi' v)\Delta^2$. Since $\nabla(\Pi' v) = \Pi$ and $\|\Pi\|_2^2 = nH$, the product rule gives

$$\begin{aligned}\mathbb{E}[(\Pi' v)^2 \Delta^2] &= \sigma_v^2 nH \mathbb{E}[\Delta^2] + 2\sigma_v^2 \mathbb{E}[(\Pi' v)\Delta \cdot \Pi' \nabla\Delta] \\ &\leq \sigma_v^2 nH \mathbb{E}[\Delta^2] + \tfrac{1}{2}\mathbb{E}[(\Pi' v)^2 \Delta^2] + 2\sigma_v^4 \mathbb{E}[(\Pi' \nabla\Delta)^2],\end{aligned}$$

where the inequality comes from Young's inequality.[9] By Cauchy–Schwarz, the last term is at most $2\sigma_v^4 nH \mathbb{E}\|\nabla\Delta\|_2^2 \leq C$, and the first is at most C because $\mathbb{E}[\Delta^2] \leq C/n$.

[9] Young's inequality states that $2ab \leq \eta a^2 + b^2/\eta$ for real a, b and any $\eta > 0$, which follows from expanding $(\sqrt{\eta}\, a - b/\sqrt{\eta})^2 \geq 0$. Here we apply it with $a = (\Pi' v)\Delta$, $b = \sigma_v^2 \Pi' \nabla\Delta$, and $\eta = 1/2$.

Moving $\frac{1}{2}\mathbb{E}[(\Pi'v)^2\Delta^2]$ to the left-hand side, we obtain $\mathbb{E}[(\Pi'v)^2\Delta^2] \leq C$.

Step 2: $|\mathbb{E}[\Delta L_n^2]| \leq C\{\sqrt{S/n} + 1/n\}$. This time we multiply (B.12) by Δ and take expectations. Apply (B.11) with $g(v) = (\Pi'v)\Delta$ to obtain $\mathbb{E}[(\Pi'v)^2\Delta] = \sigma_v^2 nH\mathbb{E}[\Delta] + \sigma_v^2\mathbb{E}[(\Pi'v)\Pi'\nabla\Delta]$. After this substitution, the two multiples of $\mathbb{E}[\Delta]$ add up to $(\sigma_\varepsilon^2 - \sigma_{\varepsilon v}^2/\sigma_v^2)/H + \sigma_{\varepsilon v}^2/(\sigma_v^2 H) = \sigma_\varepsilon^2/H$, so that

$$\mathbb{E}[\Delta L_n^2] = \frac{\sigma_\varepsilon^2}{H}\mathbb{E}[\Delta] + \frac{\sigma_{\varepsilon v}^2}{\sigma_v^2 nH^2}\mathbb{E}[(\Pi'v)\Pi'\nabla\Delta]. \tag{B.13}$$

The first term is at most $C\bar{d}/n$. For the second, notice that $(I-P)\Pi = -(H/h)(I-P)M$ because $P\widehat{\Pi} = \widehat{\Pi}$, and therefore

$$\mathbb{E}\|(I-P)\Pi\|_2^2 \leq \frac{H^2}{h^2}\mathbb{E}\|M\|_2^2 = \frac{H^2}{h^2}n\mathcal{A}.$$

Since P is symmetric, the formula for $\nabla\Delta$ above shows that $nh\,\Pi'\nabla\Delta = \Pi'\widehat{\Pi} + x'P\Pi$, and writing $x'P\Pi = x'\Pi - x'(I-P)\Pi$ with $x'\Pi = nH + \Pi'v$, we have

$$nh\,\mathbb{E}[(\Pi'v)\Pi'\nabla\Delta] = \underbrace{\mathbb{E}[(\Pi'v)\Pi'\widehat{\Pi}]}_{\text{I}} + \underbrace{\mathbb{E}[(\Pi'v)(nH+\Pi'v)]}_{\text{II}} - \underbrace{\mathbb{E}[(\Pi'v)x'(I-P)\Pi]}_{\text{III}}.$$

By (B.11) with $g(v) = \Pi'\widehat{\Pi}$, whose gradient is $P\Pi$,

$$0 \leq \text{I} = \sigma_v^2\mathbb{E}[\Pi'P\Pi] \leq \sigma_v^2 nH,$$

and $\text{II} = \mathbb{E}[(\Pi'v)^2] = \sigma_v^2 nH$ because $\mathbb{E}[\Pi'v] = 0$. For III, we have $\mathbb{E}[(\Pi'v)^4] + \mathbb{E}\|x\|_2^4 \leq Cn^2$ by Gaussian fourth moments, and Cauchy–Schwarz then bounds $\mathbb{E}[(\Pi'v)^2\|x\|_2^2]$ by Cn^2. With a second application we can bound

$$|\text{III}| \leq \left\{\mathbb{E}[(\Pi'v)^2\|x\|_2^2]\,\mathbb{E}\|(I-P)\Pi\|_2^2\right\}^{1/2} \leq Cn^{3/2}\sqrt{\mathcal{A}}.$$

Combining the bounds on I, II, and III and dividing by nh, we have $|\mathbb{E}[(\Pi'v)\Pi'\nabla\Delta]| \leq C(1+\sqrt{n\mathcal{A}})$, so that by (B.13),

$$|\mathbb{E}[\Delta L_n^2]| \leq C\left\{\frac{\bar{d}}{n} + \sqrt{\frac{\mathcal{A}}{n}} + \frac{1}{n}\right\}.$$

Finally, $\bar{d}/n + \sqrt{\mathcal{A}/n} \leq C\sqrt{S/n}$ by the first bound in (F.15), and (B.10) follows. □

Lemma B.4 (The First-Order Expansion)**.** *Suppose Assumptions 1 and 2 hold, with $J < \infty$ fixed. Then (2.3) holds for every candidate.*

Proof. By Lemma F.2, $h \geq \underline{h}$ and $\mathcal{A} \to 0$, while Lemma F.1 gives $d < \vartheta s + 1$ on $\mathcal{G}$. Since $\mathbb{P}(\mathcal{G}^c) \leq \eta_n \to 0$ by the proof of Lemma F.2 and $s/\sqrt{n} \to 0$ by Assumption 2(v), we have $d/\sqrt{n} = o_p(1)$ and $\bar{d}/n \leq (\vartheta s + 1)/n + \mathbb{P}(\mathcal{G}^c) \to 0$. With these rates, the calculation in the proof of Lemma B.2 shows that

$$\frac{\mathbb{E}[(\varepsilon' M - \sigma_{\varepsilon v} d)^2]}{nh^2} = \frac{\sigma_\varepsilon^2 \mathcal{A} + \sigma_{\varepsilon v}^2 \bar{d}/n}{h^2} \longrightarrow 0,$$

so that $\Gamma - L_n = (\varepsilon' M - \sigma_{\varepsilon v} d)/(\sqrt{n}\, h) + \sigma_{\varepsilon v} d/(\sqrt{n}\, h) = o_p(1)$. By Lemma B.1(i), $\mathbb{E}[\Delta] = \sigma_v^2 \bar{d}/(nh) \to 0$ and $\mathrm{Var}(\Delta) \leq C/n$, so that $\Delta = o_p(1)$ and $\mathbb{P}(\mathcal{H}_n) \to 1$. On $\mathcal{H}_n$, $\sqrt{n}(\widehat{\beta} - \beta) = \Gamma/(1 + \Delta)$ by (B.2), and therefore

$$\sqrt{n}(\widehat{\beta} - \beta) - L_n = \frac{\Gamma - L_n - \Delta L_n}{1 + \Delta} = o_p(1),$$

since $L_n = O_p(1)$. This is (2.3). □

C Proofs of Proposition 1 and Theorem 2

C.1 Proof of Proposition 1

Proof. The first conclusion (3.13) is part of (F.13) of Lemma F.2. For (3.14), write

$$\frac{d^2 + d}{\mathbb{E}[d^2 + d]} = \frac{(d/\bar{d})^2 + (d/\bar{d})/\bar{d}}{\mathbb{E}[d^2 + d]/\bar{d}^2}. \tag{C.1}$$

By (F.13), $d/\bar{d} \to_p 1$, $\bar{d} \to \infty$, and $\mathbb{E}[d^2 + d]/\bar{d}^2 \to 1$, so that the right-hand side of (C.1) converges to one in probability and (3.14) holds. □

C.2 Proof of Theorem 2

Proof. We work on $\mathcal{H}_n$, where $\widehat{S}$ is finite.

Step 1: $\widehat{\sigma}_\varepsilon^2 \mathbb{E}_n[\widehat{\Pi}_i^2]/\widehat{h}^2 = \sigma_\varepsilon^2 \mathcal{A}/h^2 + B_n + o_p(S)$. Applying the decomposition in (3.6) to sample moments, and recalling that $H = \mathbb{E}_n[\Pi_i^2]$, we have $\mathbb{E}_n[\widehat{\Pi}_i^2] = \widehat{\mathcal{A}} + \{\mathbb{E}_n[\Pi_i \widehat{\Pi}_i]\}^2/H$, with $\widehat{\mathcal{A}}$ as defined in Section 3.3. Since $x_i = \Pi_i + v_i$, we also have $\mathbb{E}_n[\Pi_i \widehat{\Pi}_i] = \widehat{h} - \mathbb{E}_n[v_i \widehat{\Pi}_i]$. With this in mind define $\widehat{q} := \mathbb{E}_n[v_i \widehat{\Pi}_i]/\widehat{h}$, so that

$$\frac{\widehat{\sigma}_\varepsilon^2 \mathbb{E}_n[\widehat{\Pi}_i^2]}{\widehat{h}^2} = \frac{\widehat{\sigma}_\varepsilon^2 \widehat{\mathcal{A}}}{\widehat{h}^2} + \frac{\widehat{\sigma}_\varepsilon^2}{H}(1-\widehat{q})^2. \tag{C.2}$$

For the first term, Lemmas B.1, C.1 and C.2 give $\widehat{\mathcal{A}} - \mathcal{A} = o_p(S)$ and $\widehat{h} - h, \widehat{\sigma}_\varepsilon^2 - \sigma_\varepsilon^2 = O_p(n^{-1/2})$. Using these rates and $\mathcal{A} \leq CS$ from (F.14), we can write

$$\frac{\widehat{\sigma}_\varepsilon^2 \widehat{\mathcal{A}}}{\widehat{h}^2} = \frac{\sigma_\varepsilon^2 \mathcal{A}}{h^2} + o_p(S) + O_p(\mathcal{A} n^{-1/2}) = \frac{\sigma_\varepsilon^2 \mathcal{A}}{h^2} + o_p(S).$$

For the second term, apply Lemma C.1 to see that $\widehat{q} = q + o_p(S)$, where $q = \mathbb{E}_n[\Pi_i v_i]/H = O_p(n^{-1/2})$ because $\mathbb{E}[q^2] = \sigma_v^2/(nH)$. We can then factor

$$(1-\widehat{q})^2 - (1-q)^2 = (q-\widehat{q})(2-q-\widehat{q}) = o_p(S),$$

while $(1-q)^2 = 1 - 2q + q^2$ with $q^2 = O_p(n^{-1}) = o_p(S)$, since $nS \to \infty$. As $\widehat{\sigma}_\varepsilon^2/H = O_p(1)$, the second term in (C.2) equals $\widehat{\sigma}_\varepsilon^2(1-2q)/H + o_p(S) = B_n + o_p(S)$, with B_n the common centering in (3.15). We keep $\widehat{\sigma}_\varepsilon^2$ inside B_n, since its estimation error is of order $n^{-1/2}$ and need not be $o_p(S)$. Adding the two terms proves Step 1.

Step 2: $\widehat{\sigma}_{\varepsilon v}^2(d^2+d)/(n\widehat{h}^2) = \sigma_{\varepsilon v}^2 \mathbb{E}[d^2+d]/(nh^2) + o_p(S)$. Write

$$\frac{\widehat{\sigma}_{\varepsilon v}^2(d^2+d)}{n\widehat{h}^2} = \frac{\sigma_{\varepsilon v}^2 \mathbb{E}[d^2+d]}{nh^2} \cdot \frac{\widehat{\sigma}_{\varepsilon v}^2}{\sigma_{\varepsilon v}^2} \cdot \frac{d^2+d}{\mathbb{E}[d^2+d]} \cdot \frac{h^2}{\widehat{h}^2}. \tag{C.3}$$

Of the last three factors, the first converges to one in probability by Lemma C.2 and $\sigma_{\varepsilon v} \neq 0$, the second by Proposition 1, and the third by Lemma B.1. Since the leading factor is at most S, the product equals $\sigma_{\varepsilon v}^2 \mathbb{E}[d^2+d]/(nh^2) + o_p(S)$.

Step 3: Part (i). Adding Steps 1 and 2 and using (3.12) and (3.7), we obtain $\widehat{S} =$

$B_n + S + o_p(S)$, which shows (3.16). By Theorem 1(ii), $\bar{Q}/S \to 1$, so that

$$\widehat{S} - B_n - \bar{Q} = (\widehat{S} - B_n - S) + (S - \bar{Q}) = o_p(S) = o_p(\bar{Q}),$$

which is (3.17).

Step 4: Part (ii). We now compare candidates, so we restore the subscript. By Theorem 1(ii), every $\bar{Q}_c$ is positive. Let

$$\omega_n := \max_{c \in [J]} \frac{|\widehat{S}_c - B_n - \bar{Q}_c|}{\bar{Q}_c}.$$

By Step 3, $\omega_n \to_p 0$. Since $\widehat{c}$ minimizes the feasible criterion, on $\{\omega_n < 1\}$ we have, for every $c \in [J]$,

$$B_n + (1 - \omega_n)\bar{Q}_{\widehat{c}} \le \widehat{S}_{\widehat{c}} \le \widehat{S}_c \le B_n + (1 + \omega_n)\bar{Q}_c.$$

After cancelling B_n and minimizing the right-hand side over c, we divide by $(1 - \omega_n)\min_c \bar{Q}_c$ to obtain

$$0 \le \frac{\bar{Q}_{\widehat{c}}}{\min_c \bar{Q}_c} - 1 \le \frac{2\omega_n}{1 - \omega_n}, \tag{C.4}$$

and the right-hand side converges to zero in probability. Since $\mathbb{P}(\omega_n < 1) \to 1$, (3.18) follows. □

C.3 Supporting Lemmas

Lemma C.1 (Sample Counterparts of $\mathcal{A}$ and q)**.** *Suppose Assumptions 1, 2, and 3 hold, with $J < \infty$ fixed, and let $q = \mathbb{E}_n[\Pi_i v_i]/H$ and $\widehat{q} = \mathbb{E}_n[v_i \widehat{\Pi}_i]/\widehat{h}$ be as in the proofs of Theorems 1 and 2. With $\widehat{\mathcal{A}} = \min_{a \in \mathbb{R}} \mathbb{E}_n[(\widehat{\Pi}_i - a\Pi_i)^2]$ as defined in Section 3.3, we have*

$$\widehat{\mathcal{A}} - \mathcal{A} = O_p\left(\sqrt{\frac{\mathcal{A}}{n}} + \frac{1}{n}\right) = o_p(S), \tag{C.5}$$

and, on $\mathcal{H}_n$,

$$\widehat{q} - q = O_p\left(\sqrt{\frac{\mathcal{A}}{n}} + \frac{\bar{d}}{n} + \frac{1}{n}\right) = o_p(S). \tag{C.6}$$

Proof. Step 1: $\widehat{\mathcal{A}} - \mathcal{A} = O_p(\sqrt{\mathcal{A}/n} + 1/n)$. Applying the decomposition in (3.6) to

sample moments as in the proof of Theorem 2, we can write

$$\widehat{\mathcal{A}} = \mathbb{E}_n[\widehat{\Pi}_i^2] - \frac{\{\mathbb{E}_n[\Pi_i\widehat{\Pi}_i]\}^2}{H} = \frac{1}{n}\left\|\left(I - \frac{\Pi\Pi'}{nH}\right)\widehat{\Pi}\right\|_2^2.$$

Since $\overline{\mathbb{E}}[\Pi_i\widehat{\Pi}_i] = h$, we have $\mathbb{E}[\{\mathbb{E}_n[\Pi_i\widehat{\Pi}_i]\}^2] = h^2 + \mathrm{Var}(\mathbb{E}_n[\Pi_i\widehat{\Pi}_i])$. Taking expectations and comparing with (3.6) then shows that $\mathcal{A} - \mathbb{E}[\widehat{\mathcal{A}}] = \mathrm{Var}(\mathbb{E}_n[\Pi_i\widehat{\Pi}_i])/H$. The gradient of $\mathbb{E}_n[\Pi_i\widehat{\Pi}_i]$ with respect to v is $P\Pi/n$, whose squared norm is at most $\|\Pi\|_2^2/n^2 = H/n$, so that by the Gaussian Poincaré inequality (E.1),

$$0 \le \mathcal{A} - \mathbb{E}[\widehat{\mathcal{A}}] \le \frac{\sigma_v^2}{n}. \tag{C.7}$$

Similarly, the gradient of $\widehat{\mathcal{A}}$ is $(2/n)P\{I - \Pi\Pi'/(nH)\}\widehat{\Pi}$, whose squared norm is at most $4\widehat{\mathcal{A}}/n$, so that

$$\mathrm{Var}(\widehat{\mathcal{A}}) \le \frac{4\sigma_v^2}{n}\mathbb{E}[\widehat{\mathcal{A}}] \le \frac{4\sigma_v^2\mathcal{A}}{n}.$$

The claim follows from Chebyshev's inequality and (C.7).

Step 2: $\widehat{q} - q = O_p(\sqrt{\mathcal{A}/n} + \bar{d}/n + 1/n)$ on $\mathcal{H}_n$. With M as in (A.3), we have $\mathbb{E}_n[v_i\widehat{\Pi}_i] = v'M/n + hq$, and therefore

$$\widehat{q} - q = \frac{v'M/n}{\widehat{h}} - q\frac{\widehat{h} - h}{\widehat{h}}. \tag{C.8}$$

By (A.4) and Markov's inequality, $v'M = \sigma_v^2 d + O_p(\sqrt{n\mathcal{A}} + \sqrt{\bar{d}})$, and $d = O_p(\bar{d})$ with $\bar{d} \to \infty$ by (F.13), so that $v'M/n = O_p(\sqrt{\mathcal{A}/n} + \bar{d}/n)$. Substituting this bound, $q = O_p(n^{-1/2})$, $\widehat{h} - h = O_p(n^{-1/2})$, and $\widehat{h} \ge \underline{h}/2$ on $\mathcal{H}_n$ into (C.8) yields the claim.

Step 3: Both rates are $o_p(S)$. By the second bound in (F.15), $\sqrt{\mathcal{A}/n} + \bar{d}/n \le CS/\bar{d}$, which is $o(S)$ because $\bar{d} \to \infty$ by (F.13). The remaining term $1/n$ is also $o(S)$, since $nS \to \infty$ by (F.14). □

Lemma C.2 (Estimating the Error Moments)**.** *Suppose Assumptions 1, 2, and 3 hold and $J < \infty$ is fixed. For any (possibly data-dependent) $[J]$-valued index $\check{c}$,*

$$\widehat{\beta}_{\check{c}} - \beta = O_p(n^{-1/2}). \tag{C.9}$$

If the pilot satisfies $\check{\beta} - \beta = O_p(n^{-1/2})$ and $\widehat{\sigma}^2_\varepsilon, \widehat{\sigma}_{\varepsilon v}$ are as in (3.11), then

$$\widehat{\sigma}^2_\varepsilon - \sigma^2_\varepsilon = O_p(n^{-1/2}), \qquad \widehat{\sigma}_{\varepsilon v} - \sigma_{\varepsilon v} = O_p(n^{-1/2}). \tag{C.10}$$

Proof. Since $\check{c} \in [J]$ and J is fixed, (B.5) in the proof of Theorem 1 gives (C.9).

By independence, Gaussian fourth moments, and Chebyshev's inequality,

$$\mathbb{E}_n[\varepsilon_i^2] = \sigma^2_\varepsilon + O_p(n^{-1/2}), \qquad \mathbb{E}_n[\varepsilon_i x_i] = \sigma_{\varepsilon v} + O_p(n^{-1/2}),$$

while $\mathbb{E}_n[x_i^2] = O_p(1)$ by Markov's inequality. Since $y_i - \check{\beta} x_i = \varepsilon_i - \delta_\beta x_i$ for $\delta_\beta := \check{\beta} - \beta$, we have

$$\widehat{\sigma}^2_\varepsilon = \mathbb{E}_n[\varepsilon_i^2] - 2\delta_\beta \mathbb{E}_n[\varepsilon_i x_i] + \delta_\beta^2 \mathbb{E}_n[x_i^2], \quad \widehat{\sigma}_{\varepsilon v} = \mathbb{E}_n[\varepsilon_i x_i] - \delta_\beta \mathbb{E}_n[x_i^2]. \tag{C.11}$$

We then obtain (C.10) by substituting these moment bounds and $\delta_\beta = O_p(n^{-1/2})$ into the identities (C.11). □

D LASSO Fitted-Value Geometry

We treat Z and $\lambda > 0$ as fixed and place no distributional restrictions on the response. As in Appendix A, let $u(w)$ and $E(w)$ denote the LASSO residual and equicorrelation set for response w, both of which are determined by the unique fitted vector $\mu(w)$. We also define the closed convex set containing zero

$$\mathcal{K} := \{a \in \mathbb{R}^n : \|Z'a/n\|_\infty \le \lambda\}.$$

In Lemma D.1, $\widehat{\pi}(w)$ is any minimizer of (2.4) with response w, while in Lemma D.2 we fix a Borel-measurable one.

Lemma D.1 (Geometry of the LASSO Fit)**.** *For every $w \in \mathbb{R}^n$ and every minimizer $\widehat{\pi}(w)$ of the problem in (2.4) with response w,*

$$|\mathbb{E}_n[z_{ij}u_i(w)]| \le \lambda, \qquad \widehat{\pi}_j(w)\mathbb{E}_n[z_{ij}u_i(w)] = \lambda|\widehat{\pi}_j(w)|, \quad j \in [p], \tag{D.1}$$

and hence $\operatorname{supp}\{\widehat{\pi}(w)\} \subseteq E(w)$ *and*

$$\mathbb{E}_n[\mu_i(w)w_i] = \mathbb{E}_n[\mu_i(w)^2] + \lambda\|\widehat{\pi}(w)\|_1. \tag{D.2}$$

Moreover,

$$u(w) = P_{\mathcal{K}}(w), \qquad \mu(w) = w - P_{\mathcal{K}}(w), \tag{D.3}$$

where $P_{\mathcal{K}}$ is Euclidean projection onto $\mathcal{K}$. The map μ is one-Lipschitz, $\mu(0) = 0$, and

$$\|\mu(w)\|_2 \le \|w\|_2. \tag{D.4}$$

For Lebesgue-almost every $w \in \mathbb{R}^n$,

$$D\mu(w) = P_{\operatorname{col}(Z_{E(w)})}, \qquad D\mu(w)' = D\mu(w) = D\mu(w)^2, \qquad D\mu(w)\mu(w) = \mu(w). \tag{D.5}$$

Proof. The KKT conditions imply (D.1), and summing the second equality over $j \in [p]$ gives (D.2). Multiplying the objective in (2.4) by n puts it in the standard form with penalty $n\lambda$, so that we can apply Lemmas 1 and 3 of Tibshirani and Taylor (2012) to obtain (D.3) and the one-Lipschitz property. Since $0 \in \mathcal{K}$, we have $\mu(0) = 0$ and hence (D.4). As in the proof of their Theorem 1, their Lemma 5 yields $D\mu(w) = Z_{E(w)}Z^{+}_{E(w)} = P_{\operatorname{col}(Z_{E(w)})}$ for Lebesgue-almost every w, with $+$ denoting the Moore–Penrose pseudoinverse. The remaining identities in (D.5) follow because this is an orthogonal projection and $\mu(w) \in \operatorname{col}(Z_{E(w)})$ by (D.1). □

Lemma D.2 (A Measurable LASSO Solution)**.** *There exists a Borel-measurable LASSO solution $w \mapsto \widehat{\pi}(w)$ such that, for every $w \in \mathbb{R}^n$,*

$$\operatorname{rank}\big(Z_{\operatorname{supp}\{\widehat{\pi}(w)\}}\big) = \big|\operatorname{supp}\{\widehat{\pi}(w)\}\big| \le \operatorname{rank}(Z) \le n, \tag{D.6}$$

and $Z\widehat{\pi}(w)$ equals the unique fitted vector $\mu(w)$.

Proof. By Lemma 14 of Tibshirani (2013) with penalty $n\lambda$, there is for every w a minimizer whose support indexes linearly independent columns. Order the finite set $\mathcal{B} = \{B \subseteq [p] : \operatorname{rank}(Z_B) = |B|\}$, including $B = \varnothing$. For nonempty $B \in \mathcal{B}$, set

$$b_B^{(B)}(w) = (Z_B' Z_B)^{-1} Z_B' \mu(w), \qquad b_{B^c}^{(B)}(w) = 0,$$

and set $b^{(\varnothing)}(w) = 0$. Let $\mathcal{R}_B$ be the set of responses for which $b^{(B)}(w)$ satisfies the KKT conditions (D.1) with residual $w - Zb^{(B)}(w)$. Since μ and $b^{(B)}$ are continuous by Lemma D.1, each $\mathcal{R}_B$ is the preimage of a closed set under a continuous map, and is thus closed. For any w, the minimizer above has support B, and since $\mu(w)$ is unique and Z_B has full column rank, it equals $b^{(B)}(w)$, including when $B = \varnothing$. The sets $\mathcal{R}_B$ therefore cover $\mathbb{R}^n$. Choose $b^{(B)}(w)$ for the first B with $w \in \mathcal{R}_B$. The regions on which each B is chosen are Borel, and the KKT conditions are sufficient for a minimum, so that this choice is a Borel-measurable minimizer. Its fitted vector is $\mu(w)$, and because its support lies in the independent set B, it satisfies (D.6). □

E Gaussian Identities and Rank Variance

By Rademacher's theorem, any locally Lipschitz $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable almost everywhere. Define $\nabla f(w)$ to be its gradient where it is differentiable and zero otherwise.

Lemma E.1 (Poincaré Inequality and Stein Identities)**.** *Let $W \sim N(0, \sigma^2 I_n)$ with $\sigma > 0$. If $g : \mathbb{R}^n \to \mathbb{R}$ is locally Lipschitz with $\mathbb{E}[g(W)^2] < \infty$ and $\mathbb{E}[\|\nabla g(W)\|_2^2] < \infty$, then*

$$\operatorname{Var}\{g(W)\} \le \sigma^2 \mathbb{E}[\|\nabla g(W)\|_2^2]. \tag{E.1}$$

Separately, let $m : \mathbb{R}^n \to \mathbb{R}^n$ be Lipschitz, let $Dm(w)$ be its Jacobian where m is differentiable and zero otherwise, and define $\operatorname{div} m(w) := \operatorname{tr}\{Dm(w)\}$. Then

$$\mathbb{E}[W' m(W)] = \sigma^2 \mathbb{E}[\operatorname{div} m(W)], \tag{E.2}$$

$$\mathbb{E}[\{W' m(W) - \sigma^2 \operatorname{div} m(W)\}^2] = \sigma^2 \mathbb{E}[\|m(W)\|_2^2] + \sigma^4 \mathbb{E}[\operatorname{tr}\{Dm(W)^2\}]. \tag{E.3}$$

For this m, if g is a locally Lipschitz scalar function with $|g(w)| + \|\nabla g(w)\|_2 \le C(1 + \|w\|_2^r)$ almost everywhere for some $C < \infty$ and $r \ge 0$, then

$$\mathbb{E}[\{W' m(W) - \sigma^2 \operatorname{div} m(W)\} g(W)] = \sigma^2 \mathbb{E}[m(W)' \nabla g(W)]. \tag{E.4}$$

Proof. Inequality (E.1) is the Gaussian Poincaré inequality, which is Theorem 2.2 of Bellec and Zhang (2021) with the vector field set to zero. By Lipschitz continuity, m has at most linear growth and a weak Jacobian that is bounded almost everywhere, so that its Gaussian moments are finite. We apply the first-order Stein formula of Stein (1981) componentwise to obtain (E.2). For (E.4), we instead apply it to $m_i g$, which is integrable by the growth condition on g, use the product rule, and sum over i. Theorem 2.1(ii) of Bellec and Zhang (2021) gives (E.3). The transpose convention in their Jacobian is immaterial here, since it leaves the divergence and $\mathrm{tr}\{(Dm)^2\}$ unchanged. □

Lemma E.2 (Moments of a Recentered LASSO Fit)**.** *Suppose Assumption 1(i) holds, and let $b \in \mathbb{R}^n$ be nonrandom. Define $m(a) := \mu(\Pi + a) - b$ for $a \in \mathbb{R}^n$. Let d be as in (2.8) and $\bar{d} := \mathbb{E}[d]$ as in (A.2). Then m is one-Lipschitz, and*

$$\mathbb{E}[v'm(v)] = \sigma_v^2 \bar{d}, \qquad \mathbb{E}[\{v'm(v) - \sigma_v^2 d\}^2] = \sigma_v^2 \mathbb{E}[\|m(v)\|_2^2] + \sigma_v^4 \bar{d}. \tag{E.5}$$

Proof. By Lemma D.1, m is one-Lipschitz and its weak Jacobian $Dm(a)$ is almost everywhere an orthogonal projection. Since $\operatorname{div} m(v) = \mathrm{tr}\{Dm(v)^2\} = d$ almost surely, (E.5) follows from (E.2) and (E.3). □

Lemma E.3 (Second Moment of the Numerator)**.** *Suppose Assumption 1(i) holds. Then*

$$\mathbb{E}[(\varepsilon'\widehat{\Pi})^2] = \sigma_\varepsilon^2 \mathbb{E}[\|\widehat{\Pi}\|_2^2] + \sigma_{\varepsilon v}^2 \mathbb{E}[d^2 + d]. \tag{E.6}$$

Define

$$\zeta := v'\widehat{\Pi} - \sigma_v^2 d.$$

Then

$$\mathbb{E}[v'\widehat{\Pi}] = \sigma_v^2 \mathbb{E}[d], \qquad \mathbb{E}[\zeta^2] = \sigma_v^2 \mathbb{E}[\|\widehat{\Pi}\|_2^2] + \sigma_v^4 \mathbb{E}[d], \qquad \mathbb{E}[d\zeta] = 0. \tag{E.7}$$

Proof. Apply Lemma E.2 at $b = 0$ to obtain the first two identities in (E.7). Write $m(a) := \mu(\Pi + a)$ and $P := Dm(v)$, where $P' = P = P^2$ and $P\widehat{\Pi} = \widehat{\Pi}$ almost surely by Lemma D.1. For $g(a) := a'm(a)$ we have $\nabla g(v) = \widehat{\Pi} + Pv$. By (E.4) together with

the first two identities in (E.7),

$$\begin{aligned}\sigma_v^2\mathbb{E}[d\zeta] &= \mathbb{E}[(v'\widehat{\Pi})\zeta] - \mathbb{E}[\zeta^2] \\ &= \sigma_v^2\mathbb{E}[\widehat{\Pi}'(\widehat{\Pi} + Pv)] - \mathbb{E}[\zeta^2] \\ &= \sigma_v^2\mathbb{E}[\|\widehat{\Pi}\|_2^2] + \sigma_v^4\mathbb{E}[d] - \mathbb{E}[\zeta^2] = 0.\end{aligned}$$

This proves the last identity in (E.7). Then

$$\mathbb{E}[(v'\widehat{\Pi})^2] = \mathbb{E}[(\zeta + \sigma_v^2 d)^2] = \sigma_v^2\mathbb{E}[\|\widehat{\Pi}\|_2^2] + \sigma_v^4\mathbb{E}[d^2 + d].$$

Meanwhile, by the Gaussian error decomposition (A.5) and the fact that $\widehat{\Pi}$ is $\sigma(v)$-measurable,

$$\mathbb{E}[(\varepsilon'\widehat{\Pi})^2] = \frac{\sigma_{\varepsilon v}^2}{\sigma_v^4}\mathbb{E}[(v'\widehat{\Pi})^2] + \left(\sigma_\varepsilon^2 - \frac{\sigma_{\varepsilon v}^2}{\sigma_v^2}\right)\mathbb{E}[\|\widehat{\Pi}\|_2^2].$$

Combining the last two displays proves (E.6). □

Bellec and Zhang (2021, Theorem 3.11) give a bound of this form for the size of the LASSO support when the dictionary is in general position, so that the LASSO solution is unique. Our proof follows theirs, although we do not need this condition.

Lemma E.4 (Variance of the Effective Dimension)**.** *Fix $\Pi \in \mathbb{R}^n$, $Z \in \mathbb{R}^{n\times p}$, and $\lambda > 0$, and let μ denote the LASSO fitted-value map associated with (2.4). For $v \sim N(0, \sigma_v^2 I_n)$ with $\sigma_v^2 > 0$, define*

$$d := \operatorname{rank}(Z_{E(\Pi+v)}), \qquad \bar{d} := \mathbb{E}[d].$$

Then $d = \operatorname{div}\mu(\Pi + v)$ $\mathbb{P}$-almost surely. There is a universal constant $C < \infty$ such that

$$\operatorname{Var}(d) \le C\left[1 + \bar{d}\log\left\{\frac{ep}{\bar{d}\vee 1}\right\}\right]. \tag{E.8}$$

Proof. Begin by defining

$$G := v/\sigma_v, \qquad F(w) := \mu(\Pi + \sigma_v w)/\sigma_v.$$

Then $G \sim N(0, I_n)$. By Lemma D.1, $DF(G) = P = P_{\operatorname{col}(Z_{E(\Pi+v)})}$ is an orthogonal projection with $\operatorname{tr}(P) = \operatorname{tr}(P^2) = d$ almost surely. The map F is one-Lipschitz,

while $w'F(w)$ and its gradient are square integrable under $N(0, I_n)$, so Theorem 2.2 of Bellec and Zhang (2021) gives

$$\mathrm{Var}(d) \le \bar{d} + \mathbb{E}[\|PG\|_2^2]. \tag{E.9}$$

For $1 \le r \le \mathrm{rank}(Z)$, let

$$\mathscr{P}_r := \big\{ P_{\mathrm{col}(Z_B)} : B \subseteq [p],\ |B| = \mathrm{rank}(Z_B) = r \big\}.$$

Since Z is fixed, this family is deterministic. On $\{d = r\}$, an independent subset of $E(\Pi + v)$ spans $\mathrm{col}(Z_{E(\Pi+v)})$, so

$$|\mathscr{P}_r| \le \binom{p}{r} \le (ep/r)^r, \qquad P \in \mathscr{P}_r \quad \text{on } \{d = r\}. \tag{E.10}$$

For each fixed $P_0 \in \mathscr{P}_r$ we have $\|P_0 G\|_2^2 \sim \chi^2_r$, so we can apply Laurent and Massart (2000, Lemma 1) with $u = 2r\log(ep/r) + r + t$ and take a union bound. The threshold $r + 2\sqrt{ru} + 2u$ is then at most $11\{r\log(ep/r) + t\}$, and the resulting bound for each r is $|\mathscr{P}_r| e^{-u} \le e^{-r-t}$. Since $P = 0$ on $\{d = 0\}$, summing over $r \ge 1$ gives

$$\mathbb{P}\big\{ \|PG\|_2^2 > 11[d\log\{ep/(d \vee 1)\} + t] \big\} \le \frac{e^{-t}}{e-1}, \qquad t \ge 0. \tag{E.11}$$

Tail integration yields

$$\mathbb{E}[\|PG\|_2^2] \le 11\mathbb{E}[d\log\{ep/(d \vee 1)\}] + \frac{11}{e-1}. \tag{E.12}$$

The function $x \mapsto x\log\{ep/(x \vee 1)\}$ is concave and dominates x on $[0, p]$. By Jensen's inequality and (E.9)–(E.12), we have $\mathrm{Var}(d) \le 12[1 + \bar{d}\log\{ep/(\bar{d} \vee 1)\}]$, which is (E.8). □

F Sparse First-Stage Bounds

Following Belloni et al. (2012), define the score event

$$\mathcal{G} := \left\{ \big\|\mathbb{E}_n[z_i v_i]\big\|_\infty \le \frac{\lambda}{c_\lambda} \right\}. \tag{F.1}$$

Lemma F.1 (LASSO Rates and Recovery of the Strong Coordinates)**.** *Under Assumption 2, for all sufficiently large n, the following hold on the event $\mathcal{G}$ defined by (F.1). First, the prediction norm satisfies*

$$\mathbb{E}_n[\{z_i'(\widehat{\pi} - \pi^0)\}^2]^{1/2} \leq 2\mathbb{E}_n[\xi_i^2]^{1/2} + \frac{2(1 + c_\lambda^{-1})\lambda\sqrt{s}}{\kappa_0}. \tag{F.2}$$

Next, for the measurable solution $\widehat{\pi}$ of Lemma D.2,

$$\|\widehat{\pi} - \pi^0\|_2 \leq \frac{2}{\sqrt{\underline{\phi}}} \left\{ \mathbb{E}_n[\xi_i^2]^{1/2} + \frac{(1 + c_\lambda^{-1})\lambda\sqrt{s}}{\kappa_0} \right\}. \tag{F.3}$$

The effective dimension d defined in (2.8) satisfies

$$d < \vartheta s + 1. \tag{F.4}$$

If Assumption 3 holds as well, then, again on $\mathcal{G}$,

$$k \leq d, \tag{F.5}$$

and, for the measurable solution $\widehat{\pi}$ above,

$$A \subseteq \operatorname{supp}(\widehat{\pi}), \qquad \operatorname{sign}(\widehat{\pi}_j) = \operatorname{sign}(\pi_j^0), \quad j \in A. \tag{F.6}$$

Proof. Define $\delta := \widehat{\pi} - \pi^0$, $a := \mathbb{E}_n[\xi_i^2]^{1/2}$, $t := \mathbb{E}_n[(z_i'\delta)^2]^{1/2}$, and $\theta := c_\lambda^{-1}$. By Assumption 2(ii)–(iii),

$$\frac{a}{\lambda\sqrt{s}} \leq \frac{C_a}{c_\lambda\sqrt{2}\sigma_v\{\log(2p) + \alpha\log\log(p \vee n)\}^{1/2}} \longrightarrow 0, \tag{F.7}$$

where the denominator diverges because $p \vee n \geq n$.

Prediction bound. We follow the basic inequality argument of Belloni et al. (2012, Lemma 6), whose penalty is $2n\lambda$ with unit loadings under our normalization, and compare the objective at $\widehat{\pi}$ and π^0. On $\mathcal{G}$, Hölder's inequality, Cauchy–Schwarz, and $\pi_{T^c}^0 = 0$ give

$$\frac{1}{2}t^2 + (1 - \theta)\lambda\|\delta_{T^c}\|_1 \leq (1 + \theta)\lambda\|\delta_T\|_1 + at. \tag{F.8}$$

If $t \le 2a$, (F.2) is immediate. Otherwise (F.8) gives the cone condition $\|\delta_{T^c}\|_1 \le L_\lambda \|\delta_T\|_1$, so that $\|\delta_T\|_1 \le \sqrt{s}\, t/\kappa_0$ by Assumption 2(iv). Substituting this into (F.8) yields

$$t \le 2a + \frac{2(1+\theta)\lambda\sqrt{s}}{\kappa_0},$$

which proves (F.2).

Rank bound. Let $E := E(x)$, so that $d = \operatorname{rank}(Z_E)$. We proceed by contradiction, assuming that $d \ge \lceil \vartheta s \rceil$. Choose $B \subseteq E$ with $|B| = \lceil \vartheta s \rceil$ and linearly independent columns. Since $\lceil \vartheta s \rceil \le d \le p$, the upper sparse-eigenvalue bound in Assumption 2(iv) applies at $\lceil \vartheta s \rceil$. The residual is $u(x) = \xi + v - Z\delta$, so that by the KKT equalities on B, the score event, and this upper sparse-eigenvalue bound we have

$$\begin{aligned}\lambda\sqrt{\lceil \vartheta s \rceil} &= \|(\mathbb{E}_n[z_i u_i(x)])_B\|_2 \le \|(\mathbb{E}_n[z_i v_i])_B\|_2 + \sup_{\|\gamma\|_2=1,\ \operatorname{supp}\gamma \subseteq B} \mathbb{E}_n[(z_i'\gamma)(\xi_i - z_i'\delta)] \\ &\le \theta\lambda\sqrt{\lceil \vartheta s \rceil} + \sqrt{\overline{\phi}}\,(a+t).\end{aligned}$$

By (F.2), $a + t \le 3a + 2(1+\theta)\lambda\sqrt{s}/\kappa_0$, and therefore

$$\frac{\lceil \vartheta s \rceil}{s} \le \frac{\overline{\phi}}{(1-\theta)^2}\left\{\frac{3a}{\lambda\sqrt{s}} + \frac{2(1+\theta)}{\kappa_0}\right\}^2 < \vartheta$$

for large n, since $a/(\lambda\sqrt{s}) \to 0$ by (F.7) and $\vartheta > 4L_\lambda^2\overline{\phi}/\kappa_0^2$. This contradicts $\lceil \vartheta s \rceil \ge \vartheta s$, so $d < \lceil \vartheta s \rceil < \vartheta s + 1$ and (F.4) follows.

Coefficient bound and strong core. We now use the measurable solution of Lemma D.2 and let $\widehat{T} := \operatorname{supp}(\widehat{\pi})$. By Lemmas D.1 and D.2,

$$|\widehat{T}| \le d, \qquad |\operatorname{supp}(\delta)| \le (s+d) \wedge p \le \lceil (\vartheta+1)s \rceil \wedge p.$$

Using the lower sparse-eigenvalue bound in Assumption 2(iv) and (F.2) we can therefore write

$$\sqrt{\underline{\phi}}\,\|\delta\|_2 \le t \le 2a + \frac{2(1+\theta)\lambda\sqrt{s}}{\kappa_0},$$

which also holds when $\delta = 0$ and proves (F.3). If Assumption 3 holds, then by Assumption 2(ii) and the beta-min threshold of Assumption 3,

$$|\delta_j| \le \|\delta\|_2 \le \underline{b}_n/2 \le |\pi_j^0|/2, \qquad j \in A.$$

Every $j \in A$ is therefore selected with the correct sign, and $k \le |\widehat{T}| \le d$. This proves (F.6) and (F.5). □

Lemma F.2 (Consequences of the First-Stage Conditions)**.** *Suppose Assumptions 1 and 2 hold, with $J < \infty$ fixed. Let Φ denote the standard normal distribution function and define*

$$\eta_n := 2p\Phi\left(-\frac{\sqrt{n}\lambda}{c_\lambda \sigma_v}\right)$$

and

$$a_n := \mathbb{E}_n[\xi_i^2] + s\lambda^2 + \sqrt{\eta_n}.$$

Then

$$\eta_n \to 0, \qquad a_n \to 0,$$

and

$$\overline{\mathbb{E}}[(\widehat{\Pi}_i - \Pi_i)^2] = O(a_n), \tag{F.9}$$

$$h - H = O(\sqrt{a_n}), \qquad \overline{\mathbb{E}}[\widehat{\Pi}_i^2] - H = O(\sqrt{a_n}), \quad \mathcal{A} = O(a_n). \tag{F.10}$$

There are constants $\underline{h} > 0$ and $C < \infty$, independent of n and of the candidate, such that, for all sufficiently large n,

$$0 < \underline{h} \le h, \qquad h + \overline{\mathbb{E}}[\widehat{\Pi}_i^2] \le C. \tag{F.11}$$

If Assumption 3 also holds, then, for all sufficiently large n,

$$k(1 - \eta_n) \le \bar{d} \le 2(\vartheta s + 1), \tag{F.12}$$

and

$$\begin{gathered} \bar{d} \asymp k, \qquad \bar{d} \to \infty, \qquad \frac{\mathrm{Var}(d)}{\bar{d}^2} \to 0, \\ \frac{d}{\bar{d}} \to_p 1, \qquad \frac{\mathbb{E}[d^2 + d]}{\bar{d}^2} \to 1, \qquad \frac{\bar{d}}{\sqrt{n}} \to 0. \end{gathered} \tag{F.13}$$

Under the same additional Assumption 3,

$$S \asymp \mathcal{A} + \frac{\mathbb{E}[d^2 + d]}{n} \asymp \mathcal{A} + \frac{\bar{d}^2}{n}, \qquad S \to 0, \qquad nS \to \infty. \tag{F.14}$$

Moreover,

$$\sqrt{\frac{\mathcal{A}}{n}} + \frac{\bar{d}}{n} \le C\sqrt{\frac{S}{n}}, \quad \sqrt{\frac{\mathcal{A}}{n}} + \frac{\bar{d}}{n} \le \frac{CS}{\bar{d}}. \tag{F.15}$$

Proof. We use column normalization, the Gaussian bound $\Phi(-u) \le \exp(-u^2/2)$ for $u \ge 0$, and Assumption 2(iii) to obtain

$$\mathbb{P}(\mathcal{G}^c) \le 2p\Phi\left(-\frac{\sqrt{n}\lambda}{c_\lambda \sigma_v}\right) = \eta_n \le \{\log(p \vee n)\}^{-\alpha} \to 0.$$

Then $a_n \to 0$ by parts (ii) and (v) of Assumption 2.

Prediction moments. Write $U(v) := \|\mu(\Pi + v) - \Pi\|_2/\sqrt{n}$, so that $\mathbb{E}[U^2] = \overline{\mathbb{E}}[(\widehat{\Pi}_i - \Pi_i)^2]$. By Lemma D.1, U is $n^{-1/2}$-Lipschitz, and thus the Gaussian Poincaré inequality (E.1) gives $\mathrm{Var}(U) \le \sigma_v^2/n$. On $\mathcal{G}$, Lemma F.1 and the triangle inequality imply

$$U \le \overline{U} := 3\mathbb{E}_n[\xi_i^2]^{1/2} + \frac{2(1 + c_\lambda^{-1})\lambda\sqrt{s}}{\kappa_0}. \tag{F.16}$$

Since $\mathbb{P}(\mathcal{G}) \ge 1/2$ eventually and on $\mathcal{G}$ we have $|U - \mathbb{E}U| \ge \mathbb{E}U - \overline{U}$ whenever $\mathbb{E}U > \overline{U}$,

$$\mathbb{P}(\mathcal{G})(\mathbb{E}U - \overline{U})_+^2 \le \mathrm{Var}(U).$$

Then $\mathbb{E}U \le \overline{U} + \sqrt{2\,\mathrm{Var}(U)}$, so that

$$\mathbb{E}[U^2] = \mathrm{Var}(U) + (\mathbb{E}U)^2 \le 2\overline{U}^2 + 5\sigma_v^2/n = O(a_n).$$

Here $n^{-1} = O(s\lambda^2)$ follows from Assumption 2(iii) and $s \ge 1$. This proves (F.9).

Fitted moments. By (3.5), Cauchy–Schwarz, and (F.9),

$$|h - H|^2 \le H\mathbb{E}[U^2] = O(a_n), \qquad 0 \le \mathcal{A} \le \mathbb{E}[U^2] = O(a_n),$$

where the bound on $\mathcal{A}$ follows by evaluating the objective in (3.6) at coefficient one. Moreover,

$$\overline{\mathbb{E}}[\widehat{\Pi}_i^2] - H = 2(h - H) + \mathbb{E}[U^2] = O(\sqrt{a_n}).$$

Since H is bounded and bounded away from zero by (2.2), these bounds prove (F.10) and (F.11).

Rank moments and ratios. Suppose Assumption 3 holds. By (F.5), $\bar{d} \geq k\mathbb{P}(\mathcal{G}) \geq k(1-\eta_n) \geq k/2$ eventually. Combining this lower bound with Lemma E.4 and (3.4), we obtain

$$V := \frac{\mathrm{Var}(d)}{\bar{d}^2} \leq C\left\{\frac{1}{k^2} + \frac{\log(2ep/k)}{k}\right\} \to 0.$$

This bound uses the fact that $d \leq p$ implies $\bar{d} \leq p$, and that $\log(ep/\bar{d}) \leq \log(2ep/k)$ because $\bar{d} \geq k/2$. Since $d < \vartheta s + 1$ on $\mathcal{G}$,

$$(1-\eta_n)\left(1 - \frac{\vartheta s + 1}{\bar{d}}\right)_+^2 \leq V.$$

If $\bar{d} > 2(\vartheta s + 1)$ along a subsequence, the left side is bounded away from zero, which is impossible. Therefore $\bar{d} \leq 2(\vartheta s + 1)$ eventually, which proves (F.12), and since $s \leq C_s k$ we also have $\bar{d} \asymp k \to \infty$. Chebyshev's inequality yields $d/\bar{d} \to_p 1$. Finally,

$$\frac{\mathbb{E}[d^2 + d]}{\bar{d}^2} = 1 + V + \frac{1}{\bar{d}} \to 1, \qquad \frac{\bar{d}}{\sqrt{n}} \leq \frac{2(\vartheta s + 1)}{\sqrt{n}} \to 0$$

by (A.2) and Assumption 2(v), which completes the proof of (F.13).

Criterion rates. With σ_ε^2 and $|\sigma_{\varepsilon v}|$ fixed and positive, the closed form (3.7) and the bounds (F.11) give

$$S \asymp \mathcal{A} + \mathbb{E}[d^2 + d]/n \asymp \mathcal{A} + \bar{d}^2/n.$$

Here the first comparison uses $\sigma_\varepsilon^2 > 0$ by Assumption 1(i) and $\sigma_{\varepsilon v} \neq 0$ by (2.2), and the second uses $\mathbb{E}[d^2 + d]/\bar{d}^2 \to 1$ from (F.13). Since $\mathcal{A} \to 0$, $\bar{d}/\sqrt{n} \to 0$, and $\bar{d} \to \infty$ by the preceding parts, these comparisons prove (F.14).

For (F.15), notice that $\mathcal{A} + \bar{d}^2/n \leq CS$ by the same comparison, and

$$2\bar{d}\sqrt{\mathcal{A}/n} \leq \mathcal{A} + \bar{d}^2/n.$$

Taking square roots gives the first bound in (F.15), and dividing by $\bar{d}$ gives the second. □

G Included Exogenous Controls

We now allow a fixed number of exogenous controls in the structural equation and every first stage. After partialling out the controls and expressing the residuals in an

orthonormal basis, we recover the model in Section 2, with sample size equal to the residual degrees of freedom, and under conditional Gaussian sampling the transformed errors remain independent and identically distributed. If the residualized design satisfies the identification and first-stage conditions, the AMSE criterion applies with complexity measured only in the remaining excluded-instrument directions.

G.1 Model and First-Stage Estimation

Let the original sample size be N, and let $W \in \mathbb{R}^{N\times q}$ have rank q, where q is fixed as $N \to \infty$. The columns of W enter both the structural equation and every first stage, and they may include an intercept. Write $\mathsf{Z}_c \in \mathbb{R}^{N\times p_c}$ for candidate c's excluded-instrument dictionary and $\mathscr{Z} := (W, \mathsf{Z}_1, \ldots, \mathsf{Z}_J)$. As in the main text, we condition on the realized design $\mathscr{Z}$, and take the candidate list and penalties to be fixed under this conditioning. Consider

$$y^\circ = \beta x^\circ + W\gamma + \varepsilon^\circ, \qquad x^\circ = m + v^\circ, \qquad m := \mathbb{E}[x^\circ \mid \mathscr{Z}]. \tag{G.1}$$

Conditional on $\mathscr{Z}$, suppose the pairs $\{(\varepsilon_i^\circ, v_i^\circ)\}_{i\in[N]}$ are independent and identically distributed $N(0,\Omega)$, with Ω as in Assumption 1(i). We impose no linearity restriction on m.

Define

$$P_W := W(W'W)^{-1}W', \qquad M_W := I_N - P_W, \qquad n := N - q.$$

Choose a $\mathscr{Z}$-measurable matrix $U \in \mathbb{R}^{N\times n}$ whose columns form an orthonormal basis for $\mathrm{col}(M_W)$. Then

$$U'U = I_n, \qquad UU' = M_W, \qquad U'W = 0, \qquad (U'a)'(U'b) = a'M_W b. \tag{G.2}$$

Set

$$\begin{aligned} y &:= U'y^\circ, \qquad & x &:= U'x^\circ, \qquad & \Pi &:= U'm, \\ \varepsilon &:= U'\varepsilon^\circ, \qquad & v &:= U'v^\circ, \qquad & Z_c &:= U'\mathsf{Z}_c. \end{aligned} \tag{G.3}$$

These variables satisfy

$$y = \beta x + \varepsilon, \qquad x = \Pi + v. \tag{G.4}$$

Although the residual vectors $M_W\varepsilon^\circ$ and $M_W v^\circ$ generally have dependent coordi-

nates, in the orthonormal coordinates (G.3) the error pairs are again independent and identically distributed, as the proof below shows.

We drop any dictionary column annihilated by M_W, since its contribution can be absorbed by the controls. We scale each remaining column so that $\mathsf{Z}'_{c,\cdot j} M_W \mathsf{Z}_{c,\cdot j}/n = 1$, and assign $\lambda_c > 0$ after this scaling. The transformed dictionary then satisfies Assumption 2(i). We estimate the first stage with the controls left unpenalized:

$$(\widehat{\alpha}_c, \widehat{\pi}_c^W) \in \underset{\alpha \in \mathbb{R}^q,\, \pi \in \mathbb{R}^{p_c}}{\arg\min} \left\{ \frac{\|x^\circ - W\alpha - \mathsf{Z}_c \pi\|_2^2}{2n} + \lambda_c \|\pi\|_1 \right\}. \tag{G.5}$$

Profiling out α leaves the LASSO problem in (2.4) with response x and dictionary Z_c, so that the residualized fitted instrument is uniquely defined by

$$\widehat{g}_c := M_W \mathsf{Z}_c \widehat{\pi}_c^W = U \widehat{\Pi}_c, \qquad \widehat{\Pi}_c := \mu_c(x). \tag{G.6}$$

If the loss is instead normalized by $1/(2N)$ with penalty λ_c° and the same scaled dictionary columns, the equivalent penalty in (G.5) is $\lambda_c = (N/n)\lambda_c^\circ$.

G.2 Estimator and Feasible Criterion

Let $\widehat{\beta}_c^W$ be the coefficient on x° in the IV regression of y° on (x°, W), using $(\widehat{g}_c, W)$ as instruments. The estimator is unchanged if we use the full first-stage fit $P_W x^\circ + \widehat{g}_c$ in place of $\widehat{g}_c$, because W is itself among the instruments. Partialling out W yields

$$\widehat{\beta}_c^W = \frac{\widehat{g}_c' M_W y^\circ}{\widehat{g}_c' M_W x^\circ} = \frac{\widehat{\Pi}_c' y}{\widehat{\Pi}_c' x} = \widehat{\beta}_c \tag{G.7}$$

when the fit is nonzero. Set $\widehat{\beta}_c^W = 0$ when $\widehat{g}_c = 0$, as in (2.5).

Define the equicorrelation set and effective dimension in the original coordinates by

$$E_c^W(x^\circ) := \left\{ j \in [p_c] : \left| \frac{\mathsf{Z}'_{c,\cdot j} M_W (x^\circ - \mathsf{Z}_c \widehat{\pi}_c^W)}{n} \right| = \lambda_c \right\},$$
$$d_c^W := \operatorname{rank}(M_W \mathsf{Z}_{c, E_c^W(x^\circ)}) = d_c. \tag{G.8}$$

The effective dimension counts only the excluded-instrument directions that remain after partialling out W. By comparison, the derivative of the full first-stage fitted-

value map has trace $q + d_c^W$ almost surely, and the degrees of freedom of the full fit are $q + \mathbb{E}[d_c^W]$. An implementation that reports the realized divergence of the full fit should subtract q to obtain d_c^W. As in Section 2.2, a count of nonzero coefficients need not equal the rank in (G.8) when coefficient solutions are nonunique.

Write $\widehat{h}_c^W := \widehat{g}_c' M_W x^\circ / n$. We choose the same pilot candidate $\check{c} \in [J]$ as in the transformed model and set $\check{\beta}^W := \widehat{\beta}_{\check{c}}^W$. The index may be data dependent. For instance, maximizing $\widehat{h}_c^W$ with ties broken by the smallest index selects the same pilot in either set of coordinates. Define

$$\begin{aligned} \widehat{\sigma}_{\varepsilon,W}^2 &:= \frac{(y^\circ - \check{\beta}^W x^\circ)' M_W (y^\circ - \check{\beta}^W x^\circ)}{n}, \\ \widehat{\sigma}_{\varepsilon v,W} &:= \frac{(y^\circ - \check{\beta}^W x^\circ)' M_W x^\circ}{n}, \end{aligned} \tag{G.9}$$

and

$$\widehat{S}_c^W := \frac{\widehat{\sigma}_{\varepsilon,W}^2 \|\widehat{g}_c\|_2^2 / n + \widehat{\sigma}_{\varepsilon v,W}^2 \{(d_c^W)^2 + d_c^W\}/n}{(\widehat{h}_c^W)^2}. \tag{G.10}$$

Set $\widehat{S}_c^W := +\infty$ if $\widehat{h}_c^W = 0$, and select

$$\widehat{c}^W := \min \left(\arg\min_{c \in [J]} \widehat{S}_c^W \right). \tag{G.11}$$

If every fit is zero, $\widehat{c}^W = 1$. The quantities in (G.9)–(G.11) can be computed by residualizing the original data, without constructing U. We require the same control set for every candidate, so that the candidates share the same residual space and optimal instrument.

G.3 AMSE and Candidate Selection

To state the result in the original coordinates, define

$$\begin{aligned} m_W &:= M_W m, \qquad & H_W &:= \|m_W\|_2^2 / n, \\ h_c^W &:= \mathbb{E}[\widehat{g}_c' m_W]/n, \qquad & \mathcal{A}_c^W &:= \min_{a \in \mathbb{R}} \mathbb{E}[\|\widehat{g}_c - a m_W\|_2^2]/n. \end{aligned} \tag{G.12}$$

These equal H, h_c, and $\mathcal{A}_c$ in the transformed model, whose infeasible criterion is therefore

$$S_c^W := \frac{\sigma_\varepsilon^2 \mathcal{A}_c^W}{(h_c^W)^2} + \frac{\sigma_{\varepsilon v}^2 \mathbb{E}[(d_c^W)^2 + d_c^W]}{n (h_c^W)^2}. \tag{G.13}$$

Corollary G.1 (Fixed-Control Equivalence)**.** *Suppose q and J are fixed, W has rank q, and (G.1) satisfies the conditional Gaussian sampling condition stated above. Suppose H_W and $\sigma_{\varepsilon v}$ satisfy Assumption 1(ii), and the transformed systems (Π, Z_c, λ_c) satisfy Assumptions 2 and 3 with $n = N-q$. Then Theorems 1 and 2 and proposition 1 hold for the controlled procedure. In particular, there are random variables C_0^W, Q_c^W, and r_c^W such that*

$$n(\widehat{\beta}_c^W - \beta)^2 = C_0^W + Q_c^W + r_c^W, \qquad \bar{Q}_c^W := \mathbb{E}[Q_c^W] = S_c^W\{1 + o(1)\}, \tag{G.14}$$

where C_0^W is common to all candidates and

$$\mathbb{E}[C_0^W] = \sigma_\varepsilon^2/H_W + O(n^{-1}), \qquad \max_c |r_c^W|/\bar{Q}_c^W \to_p 0.$$

The selected candidate satisfies

$$\frac{\bar{Q}^W_{\widehat{c}^W}}{\min_{c\in[J]} \bar{Q}_c^W} \longrightarrow_p 1. \tag{G.15}$$

The decomposition in (G.14) also holds with $N(\widehat{\beta}_c^W - \beta)^2$ on the left and a remainder satisfying the same bound. The first-order expansion holds with either $\sqrt{n}$ or $\sqrt{N}$ scaling.

Replacing n by N uses the growing effective dimension in Assumption 3, while all population quantities and criteria above remain normalized by n.

Proof. Step 1: The transformed model. Because U is fixed conditional on $\mathscr{Z}$, the vector $(\varepsilon', v')'$ is Gaussian with mean zero and covariance

$$\begin{pmatrix} \sigma_\varepsilon^2 U'U & \sigma_{\varepsilon v} U'U \\ \sigma_{\varepsilon v} U'U & \sigma_v^2 U'U \end{pmatrix} = \begin{pmatrix} \sigma_\varepsilon^2 I_n & \sigma_{\varepsilon v} I_n \\ \sigma_{\varepsilon v} I_n & \sigma_v^2 I_n \end{pmatrix}.$$

The coordinate pairs are therefore independent and identically distributed $N(0, \Omega)$. Since $U'W = 0$, premultiplying (G.1) by U' gives (G.4), and $H = H_W$ by (G.2). Assumption 1 then holds for the transformed model. The first-stage conditions do not depend on the choice of U, since the dictionary Gram matrices and prediction norms are determined by M_W, while the coefficient vectors and penalties are unchanged by an orthogonal change of basis.

Step 2: First-stage fits and effective dimension. Fix a candidate and suppress its subscript. For each π, least squares over α gives

$$\min_{\alpha}\|x^{\circ} - W\alpha - \mathsf{Z}\pi\|_2^2 = \|M_W(x^{\circ} - \mathsf{Z}\pi)\|_2^2 = \|x - Z\pi\|_2^2.$$

The last equality follows from $M_W = UU'$ and $U'U = I_n$. The profiled coefficient minimizers thus coincide with those in (2.4). By uniqueness of the LASSO fitted vector we have $Z\widehat{\pi}^W = \widehat{\Pi}$, and $M_W\mathsf{Z} = UZ$ implies $\widehat{g} = U\widehat{\Pi}$. For each dictionary column,

$$\frac{\mathsf{Z}'_{\cdot j}M_W(x^{\circ} - \mathsf{Z}\widehat{\pi}^W)}{n} = \frac{Z'_{\cdot j}(x - Z\widehat{\pi}^W)}{n}.$$

The equicorrelation sets therefore agree. Since U is injective,

$$\operatorname{rank}(M_W\mathsf{Z}_E) = \operatorname{rank}(UZ_E) = \operatorname{rank}(Z_E) = d,$$

which proves (G.8).

The full first-stage fitted-value map is $w^{\circ} \mapsto P_W w^{\circ} + U\mu(U'w^{\circ})$. At the observed response its derivative is $P_W + UPU'$ almost surely, where $P = D\mu(x)$ as in (A.1). Its trace is

$$\operatorname{tr}(P_W + UPU') = q + \operatorname{tr}(PU'U) = q + d.$$

Step 3: Equality of the IV estimators and criteria. The moment conditions of the controlled IV regression imply $\widehat{g}'M_W(y^{\circ} - \widehat{\beta}^W x^{\circ}) = 0$. Using $\widehat{g} = U\widehat{\Pi}$ and $U'M_W = U'$ gives (G.7) whenever the denominator is nonzero. By the LASSO KKT conditions,

$$\widehat{h}^W = \frac{\widehat{\Pi}'x}{n} = \frac{\|\widehat{\Pi}\|_2^2}{n} + \lambda\|\widehat{\pi}^W\|_1.$$

The denominator is thus positive when the fit is nonzero, and both estimators are zero by convention otherwise. In either case, $\widehat{\beta}_c^W = \widehat{\beta}_c$ for every candidate, so the pilot estimates also agree.

The identities $\widehat{g}_c = U\widehat{\Pi}_c$, $m_W = U\Pi$, and $U'U = I_n$ yield

$$H_W = H, \qquad h_c^W = h_c, \qquad \mathcal{A}_c^W = \mathcal{A}_c, \qquad \widehat{h}_c^W = \widehat{h}_c, \qquad \|\widehat{g}_c\|_2^2/n = \mathbb{E}_n[\widehat{\Pi}_{c,i}^2].$$

Similarly, $U'(y^\circ - \check{\beta}^W x^\circ) = y - \check{\beta} x$, so that by (G.9),

$$\widehat{\sigma}^2_{\varepsilon,W} = \widehat{\sigma}^2_{\varepsilon}, \qquad \widehat{\sigma}_{\varepsilon v,W} = \widehat{\sigma}_{\varepsilon v}.$$

Combining these equalities with $d_c^W = d_c$, we obtain

$$S_c^W = S_c, \qquad \widehat{S}_c^W = \widehat{S}_c, \qquad \widehat{c}^W = \widehat{c}. \tag{G.16}$$

These equalities also hold under the conventions for a zero fit. Since J is fixed, we have $\check{\beta} - \beta = O_p(n^{-1/2})$ by Lemma B.4 even when the pilot index is data dependent. Applying Theorems 1 and 2 and proposition 1 to the transformed model and taking $C_0^W = C_0$, $Q_c^W = Q_c$, and $r_c^W = r_c$ gives (G.14), and (G.15) follows from (G.16).

Step 4: Scaling by the original sample size. Since q is fixed, $N/n - 1 = q/n = O(n^{-1})$ and $\sqrt{N/n} - 1 = O(n^{-1})$. By Lemma B.4, $\max_c |\sqrt{n}(\widehat{\beta}_c^W - \beta)| = O_p(1)$, so that

$$N(\widehat{\beta}_c^W - \beta)^2 - n(\widehat{\beta}_c^W - \beta)^2 = \frac{q}{n}\, n(\widehat{\beta}_c^W - \beta)^2 = O_p(n^{-1})$$

for every candidate. By (F.14) and Theorem 1(ii), $\min_c n\bar{Q}_c^W \to \infty$. The added term is therefore $o_p(\bar{Q}_c^W)$, and the stated squared-error expansion follows. The difference between the $\sqrt{N}$- and $\sqrt{n}$-scaled estimators is likewise $O_p(n^{-1})$, so the first-order expansion is unchanged. □

H Simulation Implementation and Additional Error Design

H.1 Implementation Details

We use the credit and export data for the intensive-margin sample of Paravisini et al. (2015), with 14,208 firm–product–destination export flows from 2,091 firms. We partial out product–destination fixed effects by demeaning credit growth and every instrument within the 5,996 product–destination groups. The Original dictionary contains the ten exposure measures used for LASSO by Paravisini et al. (2015). These are the share of a firm's 2006 credit from banks with more than 10% foreign funding, an indicator that this share is at least one half, and credit-weighted averages of the first eight powers of its lenders' foreign-funding shares. The Expanded dictionary adds pairwise products, alternative exposure measures, threshold and bin indicators, cubic B-splines, and interactions. We remove two collinear or nearly collinear columns

using a screen on the instruments alone, which leaves a dictionary with 114 columns and numerical rank 109. We center each column and scale it to have unit empirical root mean square.

The main first-stage signal is a least-squares refit on 17 variables selected by LASSO from the Original instruments and their pairwise products (55 variables in all). The LASSO penalty minimizes ten-fold cross-validation error averaged over five fold assignments, each of which keeps observations from the same firm together. Meanwhile, the dense signal is the least-squares fit on all 55 variables, rescaled to match the main signal's H. Neither signal uses the export outcome. We divide both signals by the residual standard deviation of the 55-variable fit, 1.975, since the simulated errors have unit variance. We use $H \in \{0.00605, 0.0121, 0.0242, 0.0484, 0.0968\}$, where 0.0484 is the main signal's value in the data. Both signals lie in the span of the Original instruments and their pairwise products, which has numerical rank 51. Under iid errors, the expected F statistic $1 + nH/51$ takes the reported values 2.7, 4.4, 7.7, 14.5, and 28.0. This expected F statistic differs from the firm-clustered first-stage F statistic in the data, which is smaller. We set $\rho = c\widehat{\rho}$, where $\widehat{\rho} = -0.218$ is the correlation between the residuals from the main signal's fit and those from the just-identified IV regression of log export volume on credit growth with exposed-bank share as the instrument. Section 4 reports $c = 0, 0.25, \ldots, 2.75$, while the additional $c = 3$ cases remain in the design-level outputs. Each design uses 1,000 replications, and we pair the underlying error draws across designs.

For each dictionary with p_c columns, the first-stage BCCH plug-in penalty and its grid of 37 multipliers are

$$\lambda_c = \frac{1.1\,\varkappa_c}{\sqrt{n}}\Phi^{-1}\left(1 - \frac{\gamma_c}{2p_c}\right), \qquad \gamma_c = \frac{0.1}{\log(\max\{p_c, n\})},$$
$$\varkappa_c = 2^a, \quad a = -15, -14.5, \ldots, 3.$$

Here Φ is the standard normal distribution function. The standard deviation of the first-stage error is known to be one in these simulations, but we estimate the structural-error moments in the feasible criterion using the pilot below. A rule's simulated risk is the average of $n(\widehat{\beta} - \beta)^2$ over replications, while its relative risk is this risk divided by the lowest risk in the same design among the four feasible rules, namely the criterion, cross-validation, cross-fitting, and BCCH. We exclude Known

Moments from this denominator, so that its relative risk can fall below one.

The cross-validation rule uses one prespecified assignment of firms to ten folds. In each training fit we keep the full-sample penalty level and rescale the instruments to unit root mean square within the training sample. We then select the candidate with the smallest average prediction error over held-out observations and refit it on the full sample. The cross-fitting rule uses a separate prespecified assignment of firms to ten folds. For each outer fold, it selects the dictionary and penalty by ten-fold cross-validation using only the other nine folds, again splitting by firm. The plug-in penalties are based on the size of that outer training sample, on which the selected candidate is then fit. The held-out predictions serve as instruments, and the IV numerator and denominator are pooled across folds. The selected candidate may differ across outer folds. For cross-fitting we report the average of the fold-specific effective dimensions, with weights proportional to held-out fold size.

For the pilot we take the cross-fitting rule's folds, fit the first stage in each fold on the Original dictionary with the BCCH penalty, and pool the IV numerator and denominator. If all ten BCCH fits are empty, we instead use in every fold the candidate that minimizes held-out prediction error pooled over the ten folds. The pooled choice can then depend on each fold's held-out observations, whereas the cross-fitting rule tunes on its training folds alone. When the full-sample BCCH rule selects no instruments, we report the pilot estimate in place of its IV estimate. These fallbacks occur only rarely in the most weakly identified designs, and the design-level outputs record how often the full-sample BCCH rule falls back.

We compute the effective dimension as the rank of the instruments with nonzero LASSO coefficients, which in our checks equals the rank of the equicorrelation set in (2.8). A candidate fit whose Karush–Kuhn–Tucker violation exceeds $10^{-3}\lambda_c$ is unavailable to every rule. When full-sample BCCH selects no instruments, we record its effective dimension as zero even though its IV estimate is replaced by the pilot.

The common-scale Laplace errors are defined in Section 4.2. The Gaussian designs use the same underlying draws a_i and b_i, and all other implementation choices are shared. Known Moments evaluates the same feasible criterion with the structural-error moments set to their population values, $\sigma_\varepsilon^2 = 1$ and $\sigma_{\varepsilon v} = \rho$, and leaves its other estimated inputs unchanged. Design-level simulation outputs and the code used to

generate the reported summaries are provided in the replication materials.

H.2 Smooth-Transformation Errors

For the smooth-transformation errors defined in Section 4.2, we leave the first-stage signals and the other simulation details of Section 4.1 and appendix H.1 unchanged. We apply ψ to the same Gaussian draws used in the Gaussian designs and use 1,000 replications for each of the 120 designs. As under Gaussian errors, we retain the designs with $\rho = 0$ as exogenous benchmarks.

Section 4.2 and Table 2 report the risk and complexity of the criterion. Cross-validation retains an advantage near the calibrated correlation at $F = 7.7$ and $F = 14.5$, while cross-fitting remains competitive under stronger identification. The unnormalized risks of these two rules change by at most 6% relative to their Gaussian counterparts. By comparison, BCCH shows larger differences at $F = 2.7$, where it selects nothing in a few replications and falls back on the pilot.

Supplemental Appendix: Non-Gaussian Errors

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S1 Non-Gaussian Extension of the Main Results

We retain the model in (2.1) from the main text, along with the candidate LASSO fits and IV estimators in (2.4)–(2.5) and the effective dimension in (2.8). For the proofs below, we also consider fitting the LASSO to an arbitrary response $w \in \mathbb{R}^n$, and write $\mu_c(w)$ for the fitted values from (2.4) with x replaced by w. Let $E_c(w)$ denote the corresponding equicorrelation set in (2.7) and define $d_c(w) := \operatorname{rank}(Z_{c,E_c(w)})$, so that $\mu_c(x) = \widehat{\Pi}_c$, $E_c(x) = E_c$, and $d_c(x) = d_c$. As in the main text, there is one endogenous regressor and all statements condition on the realized instruments. We do not treat exogenous controls separately in this supplementary appendix. The expectations and probabilities below are taken under the non-Gaussian error distribution specified here, and the maintained conditional-mean restrictions give $\mathbb{E}[\varepsilon_i] = \mathbb{E}[v_i] = 0$ and $\Pi_i = \mathbb{E}[x_i]$.

S1.1 Non-Gaussian Sampling Conditions

The following conditions replace the main-text Assumption 1. Using the noise representation of Chetverikov et al. (2020), we allow the first-stage error to be a smooth transformation of a standard Gaussian variable. Many of the ideas behind the extended Stein identities and Poincaré inequality in Appendix S4 are also taken from that paper.

Assumption S1 (Non-Gaussian Sampling)**.** *Conditional on the instruments, the pairs (ε_i, v_i) are independent and identically distributed with a fixed distribution, and:*

(i) For fixed $a \neq 0$, $\sigma_u^2 > 0$, and $C_u < \infty$,

$$\varepsilon_i = a v_i + u_i, \qquad \mathbb{E}[u_i \mid v_i] = 0, \qquad \mathbb{E}[u_i^2 \mid v_i] = \sigma_u^2, \qquad \mathbb{E}[|u_i|^4] \leq C_u. \quad \text{(S1)}$$

(ii) There is a twice continuously differentiable function $\psi : \mathbb{R} \to \mathbb{R}$ *such that*

$$v_i = \psi(e_i), \qquad e_i \sim N(0,1), \tag{S2}$$

with $\mathbb{E}[\psi(e_i)] = 0$. *For fixed* $0 < c_\psi \le C_\psi < \infty$,

$$c_\psi \le \psi'(t) \le C_\psi \quad (t \in \mathbb{R}). \tag{S3}$$

(iii) The strong-identification bound on $H = \mathbb{E}_n[\Pi_i^2]$ *in the main-text* (2.2) *holds, and for a fixed* $K_\Pi < \infty$,

$$\max_{i \in [n]} |\Pi_i| \le K_\Pi. \tag{S4}$$

Part (i) specifies a linear conditional mean and a constant conditional variance for the structural error given v_i, but does not require u_i and v_i to be independent. We use the fourth-moment bound when estimating the moments of the structural error. Write $\sigma_v^2 = \mathbb{E}[v_i^2]$, $\sigma_{\varepsilon v} = a\sigma_v^2$, and $\sigma_\varepsilon^2 = a^2\sigma_v^2 + \sigma_u^2$. The error covariance is therefore nonsingular and $\sigma_{\varepsilon v} \neq 0$, as in the main text. Part (ii) permits asymmetric, non-Gaussian first-stage errors, and includes the Gaussian specification as the case $\psi(t) = \sigma_v t$. In Lemma S2 we derive the tail, Stein-kernel, and Poincaré bounds used below from the bounds on ψ', and since $\psi \in C^2$ the density and Stein kernel are continuously differentiable, as Lemma S3 requires. The additional bound in part (iii) controls projections weighted by the Stein kernel and the third-moment term in $\mathbb{E}[C_0]$.

The smooth-transformation design in Section 4.2 provides an example satisfying Assumption S1(i)–(ii) when $0 < |\rho| < 1$. In that design we have $a = \rho$, $u_i = \sqrt{1-\rho^2}\,\psi(B_i)$, and $\sigma_u^2 = 1 - \rho^2$, while the transformation satisfies $7/(16\nu) \le \psi'(t) \le 1/\nu$ and has bounded second and third derivatives.

S1.2 Changes to the First-Stage Conditions

As in the main text, we use the approximately sparse representation in (3.1), the restricted eigenvalue in (3.2), the upper and lower sparse eigenvalues defined immediately after it, and the constants of Assumption 2. Below we retain some parts of that assumption and replace the rest.

Assumption S2 (Non-Gaussian First-Stage Conditions)**.** *For every $c \in [J]$, with constants common to all candidates:*

(i) The column normalization in the main-text Assumption 2(i) holds, and, for a fixed $K_Z < \infty$, $\max_{i,j} |z_{c,ij}| \le K_Z$.

(ii) The approximate-sparsity bound in the main-text Assumption 2(ii) holds.

(iii) In place of the main-text Assumption 2(iii), the penalty satisfies

$$\lambda_c \ge c_\lambda C_\psi \sqrt{\frac{2\{\log(2p_c) + 20\log n\}}{n}}. \tag{S5}$$

(iv) The restricted and sparse eigenvalue bounds in the main-text Assumption 2(iv) hold.

(v) In place of the main-text Assumption 2(v), the sparsity indices, dimensions, and penalties satisfy

$$\sqrt{s_c}\lambda_c \to 0, \qquad \frac{\log(p_c n)}{s_c} \to 0, \qquad \frac{s_c^2 \log(p_c n)}{n} \to 0. \tag{S6}$$

The bound on the dictionary entries in part (i) controls the weighted Gram matrices in the Stein identities. We strengthen the lower bound in part (iii) so that the penalty dominates the empirical score with probability at least $1 - n^{-20}$, which lets us integrate the first-stage bounds over the exceptional event. Part (v) controls terms from the nonconstant Stein kernel and implies $s_c/\sqrt{n} \to 0$, as required in the main text. We also impose Assumption 3 as stated in the main text, using the same constants $C_a, \kappa_0, \underline{\phi}$ and the same threshold, set of strong coordinates A_c, and size k_c. Our arguments do not require exact support recovery.

S1.3 Extension of the Main Results

Define h_c, $\mathcal{A}_c$, and S_c by (3.5), (3.6), and (3.7), respectively, with moments taken under Assumption S1, and write $\bar{Q}_c = \mathbb{E}[Q_c]$ for the random variables Q_c below. We use the residual moment estimators, the feasible criterion, the selected candidate, and the common centering B_n in (3.11), (3.12), and (3.15). Both criteria are defined as in the main text, but with population moments computed under non-Gaussian errors. Since our results concern the AMSE, they do not require the estimator's exact mean

squared error to exist.

Theorem S1 (Infeasible AMSE Criterion). *Suppose Assumptions S1 and S2 and the main-text Assumption 3 hold, with $J < \infty$ fixed. Then there exist random variables C_0, $\{Q_c\}_{c\in[J]}$, and $\{r_c\}_{c\in[J]}$ such that*

(i) For every candidate,

$$n(\widehat{\beta}_c - \beta)^2 = C_0 + Q_c + r_c, \tag{S7}$$

where C_0 does not depend on the candidate, $\mathbb{E}[C_0] = \sigma_\varepsilon^2/H + O(n^{-1})$, and the remainder satisfies $\max_{c\in[J]} |r_c|/\bar{Q}_c \to_p 0$.

(ii) For every candidate, $\bar{Q}_c > 0$ and $\bar{Q}_c = S_c\{1 + o(1)\}$.

Theorem S2 (Feasible AMSE Selection). *Suppose the hypotheses of Theorem S1 hold and that the pilot satisfies $\check{\beta} - \beta = O_p(n^{-1/2})$.*

(i) Up to the common centering B_n, the feasible criterion consistently estimates the infeasible criterion,

$$\max_{c\in[J]} \frac{|\widehat{S}_c - B_n - S_c|}{S_c} \to_p 0, \tag{S8}$$

and thus the expectation of the candidate-specific term,

$$\max_{c\in[J]} \frac{|\widehat{S}_c - B_n - \bar{Q}_c|}{\bar{Q}_c} \to_p 0. \tag{S9}$$

(ii) The selected candidate $\widehat{c}$ satisfies

$$\frac{\bar{Q}_{\widehat{c}}}{\min_{c\in[J]} \bar{Q}_c} \to_p 1. \tag{S10}$$

By Lemma S12, the pilot may be the IV-LASSO estimate from a possibly data-dependent candidate in $[J]$. The proofs of both theorems are given in Appendix S3.

S2 Notation and Preliminary Reductions

We write C for a constant that depends only on the fixed constants in the assumptions and whose value may change between displays. Since J is fixed, a rate or limit established for each candidate holds uniformly over $c \in [J]$. As in the main appendix,

we therefore fix a candidate and suppress its subscript. We write T^0 for the support T_c in (3.1), since we use T for the Stein-kernel matrix.

Squared-error decomposition. We use Γ, Δ, L_n, q, Λ, C_0, Q, and r as defined in (B.1)–(B.4) in the proof of the main-text Theorem 1, except that expectations are now taken under Assumption S1. We keep its convention for small n, with $\underline{H}/2$ in place of $\underline{h}$.

Positive denominator event. Since $h \to H$ by (S30) and $\underline{H} \le H \le \overline{H}$ by the main-text bound (2.2), we have $\underline{H}/2 \le h \le 2\overline{H}$ once n is large enough. We take n this large throughout, so that H, h, and their reciprocals are bounded, and enlarge C where needed for the bounds below to hold for all such n. Every function of v to which we apply the weighted Stein identities or the weighted Poincaré inequality is locally Lipschitz, with value and gradient of at most polynomial growth by Lemma D.1. The identities and the inequality therefore apply, and all expectations below are finite by (S18). In place of the event used in the main appendix, we work on

$$\mathcal{H}_n := \{\widehat{h} \ge \underline{H}/4\},$$

on which the sample denominator is bounded away from zero. Since $\widehat{h} = 0$ whenever $\widehat{\Pi} = 0$, $\widehat{\beta}$ is the ratio in the main-text (2.5) on $\mathcal{H}_n$. Lemma S8 shows that $\mathbb{P}(\mathcal{H}_n) \to 1$.

Population projection residual. With P as in (S15), write

$$m = \mu(\Pi + v) - \frac{h}{H}\Pi \tag{S11}$$

for the residual from the population projection in the main-text identity (3.6). This is the residual M of the main appendix, written in the notation of Lemma S3. By the definition of h, that identity, and (S15), we have

$$\mathbb{E}[\|m\|_2^2] = n\mathcal{A}, \qquad Dm = P \quad \text{almost everywhere.} \tag{S12}$$

Let T and $D_\tau = \operatorname{tr}(TP)$ be the diagonal Stein-kernel matrix and the kernel-weighted rank in (S16). Taking $b = (h/H)\Pi$ in (S26) gives the equality in

$$\mathbb{E}[(v'm - D_\tau)^2] = \mathbb{E}[m'Tm + \operatorname{tr}(TPTP)] \leq C(n\mathcal{A} + s), \tag{S13}$$

and we bound its two terms using (S17) and $P^2 = P$ together with (S30) and (S12).

S3 Proofs of the Main Results

S3.1 Proof of Theorem S1

Proof. We follow the four steps of the proof of the main-text Theorem 1, with Lemma S7 in place of (E.6) and Lemma B.2. Similarly, Lemmas S8 and S9 replace Lemmas B.1 and B.3, while Lemma S10 replaces the Gaussian moments of L_n and C_0.

Step 1: $\sqrt{n}(\widehat{\beta} - \beta) = \Lambda + o_p(S)$. Step 1 of the Gaussian proof uses $\Delta = O_p(n^{-1/2})$, $\mathbb{E}[(\Gamma - L_n)^2] \leq CS$, $\mathbb{E}[L_n^2] = \sigma_\varepsilon^2/H$, $S \to 0$, $nS \to \infty$, and the fact that $1 + \Delta = \widehat{h}/h$ is bounded away from zero on $\mathcal{H}_n$. The first five hold by Lemmas S7, S8 and S10, and the last because $\widehat{h} \geq \underline{H}/4$ on $\mathcal{H}_n$ and $h \leq 2\overline{H}$. We can therefore repeat the argument of that step, which gives the claim as well as $\Lambda = O_p(1)$ and $\sqrt{n}(\widehat{\beta} - \beta) = O_p(1)$.

Step 2: $\mathbb{E}[\Gamma^2] = \sigma_\varepsilon^2/H + S + o(S)$. We divide (S39) by nh^2 and substitute $\mathbb{E}\|\widehat{\Pi}\|_2^2/n = \mathcal{A} + h^2/H$ from the main-text identity (3.6) to obtain

$$\mathbb{E}[\Gamma^2] = \frac{\sigma_\varepsilon^2}{H} + \frac{\sigma_\varepsilon^2 \mathcal{A}}{h^2} + \frac{\sigma_{\varepsilon v}^2 \mathbb{E}[d^2 + d]}{nh^2} + \frac{o(n\mathcal{A} + s^2)}{nh^2} = \frac{\sigma_\varepsilon^2}{H} + S + o(S),$$

where the last equality uses $S \asymp \mathcal{A} + s^2/n$ from Lemma S7.

Step 3: $\bar{Q} = S\{1 + o(1)\}$. Expanding Λ^2 as in Step 3 of the proof of the main-text Theorem 1, we have

$$\mathbb{E}[\Lambda^2] - \mathbb{E}[\Gamma^2] = -2\mathbb{E}[\Delta L_n^2] - 2\mathbb{E}[\Delta L_n(\Gamma - L_n)] + \mathbb{E}[\Delta^2 L_n^2].$$

By Lemma S9, $|\mathbb{E}[\Delta L_n^2]| \leq C\{\sqrt{S/n} + 1/n\}$ and $\mathbb{E}[\Delta^2 L_n^2] \leq C/n$. We can then bound the middle term by $C\sqrt{S/n}$ using Cauchy–Schwarz and Lemma S7. Meanwhile,

$\mathbb{E}[C_0] = \sigma_\varepsilon^2/H + O(n^{-1})$ by Lemma S10. Combining these bounds with Step 2 gives

$$|\bar{Q} - S| \le \left|\mathbb{E}[\Lambda^2] - \mathbb{E}[\Gamma^2]\right| + \left|\mathbb{E}[\Gamma^2] - \frac{\sigma_\varepsilon^2}{H} - S\right| + \left|\frac{\sigma_\varepsilon^2}{H} - \mathbb{E}[C_0]\right|$$
$$\le C\left\{\sqrt{\frac{S}{n}} + \frac{1}{n}\right\} + o(S),$$

which is $o(S)$ because $nS \to \infty$. Dividing by S shows that $\bar{Q}/S \to 1$, so that $\bar{Q} > 0$ and part (ii) follows.

Step 4: $r/\bar{Q} \to_p 0$. This follows from Steps 1 and 3 by the argument in Step 4 of the proof of the main-text Theorem 1, and it proves part (i). □

S3.2 Proof of Theorem S2

Proof. The proof follows the four steps of the proof of the main-text Theorem 2, with Lemmas S11, S12, S6, and S8 in place of the main-text Lemmas C.1 and C.2, Proposition 1, and Lemma B.1, respectively. We work on $\mathcal{H}_n$, whose probability tends to one by Lemma S8.

Step 1: $\widehat{\sigma}_\varepsilon^2 \mathbb{E}_n[\widehat{\Pi}_i^2]/\widehat{h}^2 = \sigma_\varepsilon^2 \mathcal{A}/h^2 + B_n + o_p(S)$. The decomposition (C.2) is an algebraic identity, and the rest of Step 1 of the Gaussian proof uses only the following facts. Lemma S11 shows that $\widehat{\mathcal{A}} - \mathcal{A}$ and $\widehat{q} - q$ are $o_p(S)$, while $\widehat{h} - h$ and $\widehat{\sigma}_\varepsilon^2 - \sigma_\varepsilon^2$ are $O_p(n^{-1/2})$ by Lemmas S8 and S12. By independence, $\mathbb{E}[q^2] = \sigma_v^2/(nH)$. Finally, $\mathcal{A} \le CS$ and $nS \to \infty$ by Lemma S7. The claim then follows as in the Gaussian proof.

Step 2: $\widehat{\sigma}_{\varepsilon v}^2 (d^2 + d)/(n\widehat{h}^2) = \sigma_{\varepsilon v}^2 \mathbb{E}[d^2 + d]/(nh^2) + o_p(S)$. Of the last three factors in (C.3), the first converges to one in probability by Lemma S12 and $\sigma_{\varepsilon v} = a\sigma_v^2 \neq 0$, the second by Lemma S6, and the third by Lemma S8. Since the leading factor is at most S, the claim follows.

Step 3: Part (i). Adding Steps 1 and 2, we obtain $\widehat{S} = B_n + S + o_p(S)$, which is (S8). Since $\bar{Q}/S \to 1$ by Theorem S1, adding and subtracting S in $\widehat{S} - B_n - \bar{Q}$ yields (S9).

Step 4: Part (ii). By Theorem S1, every $\bar{Q}_c$ is positive. Define ω_n as in the proof of the main-text Theorem 2, with $\bar{Q}_c$ now taken under Assumption S1. Then $\omega_n \to_p 0$

by Step 3, and on $\{\omega_n < 1\}$ the comparison leading to (C.4) gives

$$0 \le \frac{\bar{Q}_{\widehat{c}}}{\min_c \bar{Q}_c} - 1 \le \frac{2\omega_n}{1-\omega_n}.$$

Since $\mathbb{P}(\omega_n < 1) \to 1$, this bound implies (S10). □

S4 Conditional Moments, Geometry, and Stein Identities

Lemma S1 (Conditional Moments of the Structural Error)**.** *Under Assumption S1, for any measurable vectors $b(v), g(v)$ for which the expectations exist,*

$$\mathbb{E}[\varepsilon' b \mid v] = a v' b, \quad \mathbb{E}[(\varepsilon' b)(\varepsilon' g) \mid v] = a^2 (v' b)(v' g) + \sigma_u^2 b' g. \tag{S14}$$

Proof. Independence of the pairs and (S1) imply $\mathbb{E}[u \mid v] = 0$ and $\mathbb{E}[uu' \mid v] = \sigma_u^2 I_n$, so that $\mathbb{E}[\varepsilon \mid v] = av$ and $\mathbb{E}[\varepsilon\varepsilon' \mid v] = a^2 vv' + \sigma_u^2 I_n$. Multiplying these by the v-measurable vectors b and g yields (S14). □

LASSO geometry. Throughout we rely on the deterministic properties of the fit in Lemma D.1 and on the Borel choice of coefficients in Lemma D.2. Define $P(w) := P_{\mathrm{col}(Z_{E(w)})}$ for every w. This map is Borel because the continuous residual determines $E(w)$ among finitely many subsets. Write $P := P(\Pi + v)$, $d := d(\Pi + v)$, and $\mu := \mu(\Pi + v)$ when no argument is shown. Under Assumption S1, $\Pi + v$ has a Lebesgue density, so that by (D.5), almost surely,

$$D\mu(\Pi + v) = P, \qquad P = P' = P^2, \qquad P\mu = \mu, \qquad \operatorname{tr} P = d. \tag{S15}$$

When we need coefficients, we take $\widehat{\pi}(w)$ to be the Borel solution in Lemma D.2. Its support is contained in $E(w)$ by (D.1), and its selected columns are linearly independent by that lemma, so its support size is at most $\operatorname{rank}(Z_{E(w)}) = d(w)$.

Let p_v denote the density of v_i, which exists and is positive because $\psi' > 0$. Define the canonical Stein kernel and its diagonal matrix by

$$\tau(t) = \frac{\int_t^\infty z p_v(z)\, dz}{p_v(t)}, \quad T = \operatorname{diag}\{\tau(v_i)\}, \quad W = T - \sigma_v^2 I, \quad D_\tau = \operatorname{tr}(TP). \tag{S16}$$

Lemma S2 (Properties of the First-Stage Errors)**.** *Under Assumption S1, p_v is positive and continuously differentiable, and τ is continuously differentiable. The kernel satisfies*

$$c_\psi^2 \le \tau(t) \le C_\psi^2, \quad \mathbb{E}[\tau(v_i)] = \sigma_v^2, \quad (\tau p_v)'(t) = -t\, p_v(t). \tag{S17}$$

For every $b \in \mathbb{R}^n$ and $r > 0$,

$$\mathbb{E}e^{b'v} \le e^{C_\psi^2 \|b\|_2^2/2}, \quad \mathbb{P}(|b'v| > r) \le 2e^{-r^2/(2C_\psi^2\|b\|_2^2)} \quad (b \ne 0). \tag{S18}$$

For locally Lipschitz f with square-integrable weak gradient,

$$\mathrm{Var}(f(v)) \le \mathbb{E}[\nabla f(v)' T \nabla f(v)] \le C_\psi^2 \mathbb{E}[\|\nabla f(v)\|_2^2]. \tag{S19}$$

Finally, if $f : \mathbb{R} \to \mathbb{R}$ is locally absolutely continuous and $\mathbb{E}|v_i f(v_i)| + \mathbb{E}|\tau(v_i) f'(v_i)| < \infty$, then

$$\mathbb{E}[v_i f(v_i)] = \mathbb{E}[\tau(v_i) f'(v_i)]. \tag{S20}$$

Proof. Step 1: Density and kernel bounds.

Let φ denote the standard normal density. Since $c_\psi \le \psi' \le C_\psi$,

$$c_\psi |z| \le |\psi(z) - \psi(0)| \le C_\psi |z|, \qquad \psi(\mathbb{R}) = \mathbb{R}.$$

By the change-of-variable formula,

$$p_v(\psi(z)) = \frac{\varphi(z)}{\psi'(z)}. \tag{S21}$$

Since $\psi \in C^2$ and $\psi' > 0$, p_v is positive and continuously differentiable, and

$$\frac{e^{-(t-\psi(0))^2/(2c_\psi^2)}}{C_\psi\sqrt{2\pi}} \le p_v(t) \le \frac{e^{-(t-\psi(0))^2/(2C_\psi^2)}}{c_\psi\sqrt{2\pi}}. \tag{S22}$$

In particular, $\int (1+|t|^j) e^{\gamma t} p_v(t)\, dt < \infty$ for all fixed $j \ge 0$ and $\gamma \in \mathbb{R}$.

For independent standard Gaussian $e, \widetilde{e}$ and nondecreasing F, G for which the following expectations are finite,

$$\mathrm{Cov}(F(e), G(e)) = \tfrac{1}{2}\mathbb{E}[(F(e) - F(\widetilde{e}))(G(e) - G(\widetilde{e}))] \ge 0.$$

We apply this identity with $G(e) = \mathbb{1}_{\{e>z\}}$, first taking $F(e) = \psi(e) - c_\psi e$ and then $F(e) = C_\psi e - \psi(e)$. Both choices of F are centered and nondecreasing. Since $\mathbb{E}[e\mathbb{1}_{\{e>z\}}] = \varphi(z)$, we have for every $z \in \mathbb{R}$

$$c_\psi \varphi(z) \le \int_z^\infty \psi(t)\varphi(t)\,dt \le C_\psi \varphi(z), \quad \tau(\psi(z)) = \frac{\psi'(z)}{\varphi(z)} \int_z^\infty \psi(t)\varphi(t)\,dt.$$

From these bounds we obtain $c_\psi^2 \le \tau(\psi(z)) \le C_\psi^2$ for all z, as required. The definition of τ implies $(\tau p_v)'(t) = -t\,p_v(t)$, and since p_v is positive and continuously differentiable, τ is continuously differentiable as well.

Step 2: First-order identity and tails.

For compactly supported, locally absolutely continuous f,

$$\mathbb{E}[v_i f(v_i)] = -\int_{\mathbb{R}} f(t)(\tau p_v)'(t)\,dt = \int_{\mathbb{R}} f'(t)\tau(t)p_v(t)\,dt.$$

To extend this to general f satisfying the stated integrability conditions, we choose smooth χ_R equal to one on $[-R, R]$, zero outside $[-2R, 2R]$, and with $|\chi_R'| \le C/R$. Applying the preceding identity to $f\chi_R$ gives

$$\mathbb{E}[v_i f(v_i)\chi_R(v_i)] = \mathbb{E}[\tau(v_i)f'(v_i)\chi_R(v_i)] + \mathbb{E}[\tau(v_i)f(v_i)\chi_R'(v_i)].$$

We can bound the last term as

$$|\mathbb{E}[\tau(v_i)f(v_i)\chi_R'(v_i)]| \le \frac{C}{R^2}\mathbb{E}[|v_i f(v_i)|\mathbb{1}_{\{|v_i|\ge R\}}] \longrightarrow 0.$$

Dominated convergence proves (S20), which at $f(t) = t$ reduces to $\mathbb{E}\tau(v_i) = \sigma_v^2$. By the density envelope (S22), all exponential moments are finite. Since $\mathbb{E}[v_i] = 0$ and $\|\tau\|_\infty \le C_\psi^2$, we can apply the Laplace-transform bound in the proof of Theorem 14 of Saumard (2018) to the one-Lipschitz functions $t \mapsto t$ and $t \mapsto -t$ to obtain

$$\mathbb{E}e^{tv_i} \le e^{C_\psi^2 t^2/2}, \qquad t \in \mathbb{R}.$$

By independence, $\mathbb{E}e^{b'v} = \prod_i \mathbb{E}e^{b_i v_i} \le e^{C_\psi^2\|b\|_2^2/2}$. The two-sided Chernoff bound gives (S18) for $b \ne 0$, while for $b = 0$ the tail probability is zero.

Step 3: Weighted Poincaré inequality.

Since the density is positive on $\mathbb{R}$ and $\mathbb{E}|v_i| < \infty$, Theorem 6 of Saumard (2018) implies

$$\mathrm{Var}(f(v_i)) \le \mathbb{E}[\tau(v_i) f'(v_i)^2], \qquad f \in L^2(p_v) \text{ absolutely continuous.}$$

For a smooth compactly supported function f of n variables, let $v_{-1} = (v_2, \dots, v_n)$. Via the conditional variance decomposition, induction, and Jensen's inequality, we can bound

$$\begin{aligned}
\mathrm{Var}(f(v)) &= \mathbb{E}[\mathrm{Var}(f(v) \mid v_{-1})] + \mathrm{Var}\{\mathbb{E}[f(v) \mid v_{-1}]\} \\
&\le \mathbb{E}[\tau(v_1)(\partial_1 f)^2] + \sum_{j=2}^{n} \mathbb{E}[\tau(v_j)\{\mathbb{E}[\partial_j f \mid v_{-1}]\}^2] \\
&\le \sum_{j=1}^{n} \mathbb{E}[\tau(v_j)(\partial_j f)^2] \le C_\psi^2 \mathbb{E}[\|\nabla f\|_2^2].
\end{aligned}$$

For locally Lipschitz f with square-integrable weak gradient, define the truncations $f_N = (-N) \vee (f \wedge N)$. Using spatial cutoffs and local convolution as in Step 3 of the proof of Lemma S3 below, we extend the inequality to each bounded f_N, with $\|\nabla f_N\|_2 \le \|\nabla f\|_2$ almost everywhere, so that $\sup_N \mathrm{Var}(f_N) \le C_\psi^2 \mathbb{E}\|\nabla f\|_2^2$. On a fixed ball $\mathcal{U}$ with positive probability we have $|f_N| \le \sup_{\mathcal{U}} |f|$, so

$$\mathbb{P}(v \in \mathcal{U})(|\mathbb{E} f_N| - \sup_{\mathcal{U}} |f|)_+^2 \le \mathrm{Var}(f_N).$$

It follows that $\sup_N \mathbb{E}[f_N^2] < \infty$, and Fatou's lemma gives $f \in L^2$. Passing to the limit in the truncations, we arrive at (S19). □

For a locally Lipschitz vector field m, set

$$\mathfrak{D}(m) = v'm - \sum_i \tau(v_i) \partial_i m_i. \tag{S23}$$

Derivatives are understood as weak derivatives, which are defined almost everywhere. The identities below extend Stein's lemma to the first-stage errors of Assumption S1, following Chetverikov et al. (2020).

Lemma S3 (Weighted Stein Identities)**.** *Under Assumption S1, let m, g and their first derivatives have polynomial growth, with $m : \mathbb{R}^n \to \mathbb{R}^n$ and $g : \mathbb{R}^n \to \mathbb{R}$ locally Lipschitz. Then*

$$\mathbb{E}[\mathfrak{D}(m)] = 0,$$
$$\mathbb{E}[\mathfrak{D}(m)g] = \mathbb{E}[m'T\nabla g], \tag{S24}$$
$$\mathbb{E}[\mathfrak{D}(m)^2] = \mathbb{E}\{m'Tm + \mathrm{tr}(TDm\,TDm)\}. \tag{S25}$$

In particular, for $m(v) = \mu(\Pi + v) - b$ and nonrandom b,

$$\mathbb{E}[v'm] = \mathbb{E}[D_\tau], \quad \mathbb{E}[(v'm - D_\tau)^2] = \mathbb{E}[m'Tm + \mathrm{tr}(TPTP)]. \tag{S26}$$

Proof. Step 1: Coordinate and covariance identities.

Apply (S20) conditionally on the other coordinates to see that

$$\mathbb{E}[v_i f(v)] = \mathbb{E}[\tau(v_i)\partial_i f(v)].$$

By Fubini's theorem, the same identity holds for locally Lipschitz f whenever both sides are absolutely integrable. For smooth compactly supported m and continuously differentiable compactly supported g, the product rule gives

$$\begin{aligned}\mathbb{E}[\mathfrak{D}(m)g] &= \sum_i \mathbb{E}[v_i m_i g - \tau(v_i)(\partial_i m_i)g] \\ &= \sum_i \mathbb{E}[\tau(v_i)\partial_i(m_i g) - \tau(v_i)(\partial_i m_i)g] \\ &= \sum_i \mathbb{E}[\tau(v_i)m_i\partial_i g] = \mathbb{E}[m'T\nabla g].\end{aligned}$$

The coordinate identity applied to each m_i also shows that $\mathbb{E}\mathfrak{D}(m) = 0$.

Step 2: Second moments for smooth fields.

We first take m smooth and compactly supported. Since τ is continuously differentiable, all terms involving τ' are integrable on this support. Moreover, $\mathfrak{D}(m)$ is continuously differentiable and compactly supported, so Step 1 applies with $g = \mathfrak{D}(m)$.

We save notation by writing $m_{j,i} = \partial_i m_j$, $\tau_i = \tau(v_i)$, and $\tau_i' = \tau'(v_i)$. By Step 1,

$$\begin{aligned}\mathbb{E}[\mathfrak{D}(m)^2] &= \mathbb{E}\sum_i \tau_i m_i \partial_i \left[\sum_j \{v_j m_j - \tau_j m_{j,j}\}\right] \\ &= \mathbb{E}\sum_i \tau_i m_i^2 + \mathbb{E}\sum_{i,j} \tau_i m_i v_j m_{j,i} \\ &\quad - \mathbb{E}\sum_i \tau_i \tau_i' m_i m_{i,i} - \mathbb{E}\sum_{i,j} \tau_i \tau_j m_i m_{j,ji}.\end{aligned}$$

Integration by parts in coordinate j gives

$$\begin{aligned}\mathbb{E}\sum_{i,j} \tau_i m_i v_j m_{j,i} &= \mathbb{E}\sum_{i,j} \tau_j \partial_j(\tau_i m_i m_{j,i}) \\ &= \mathbb{E}\sum_{i,j} \tau_i \tau_j m_{i,j} m_{j,i} + \mathbb{E}\sum_i \tau_i \tau_i' m_i m_{i,i} \\ &\quad + \mathbb{E}\sum_{i,j} \tau_i \tau_j m_i m_{j,ij}.\end{aligned}$$

Since $m_{j,ij} = m_{j,ji}$, substitution yields

$$\mathbb{E}[\mathfrak{D}(m)^2] = \mathbb{E}\sum_i \tau_i m_i^2 + \mathbb{E}\sum_{i,j} \tau_i \tau_j m_{i,j} m_{j,i} = \mathbb{E}\{m' T m + \operatorname{tr}(T D m\, T D m)\}.$$

The terms involving τ' cancel on each compact support, and the approximation below requires only boundedness of τ.

Step 3: Extension to locally Lipschitz fields.

Fix n and put

$$w_0(v) = 1 + \|v\|_2^2.$$

By (S18), for every fixed $j \geq 0$,

$$\mathbb{E}[w_0(v)(1 + \|v\|_2^j)] < \infty. \tag{S27}$$

Choose smooth cutoffs χ_R equal to one on $\{\|v\|_2 \leq R\}$, zero on $\{\|v\|_2 \geq 2R\}$, and with $\|\nabla \chi_R\|_2 \leq C/R$. For a smooth compactly supported approximate identity ω_δ, set

$$m_R = \chi_R m, \qquad m_{R,\delta} = \omega_\delta * m_R,$$

and define $g_R, g_{R,\delta}$ in the same way. On compact sets, w_0 is bounded and the density $\prod_i p_v(v_i)$ of v is bounded above and below by positive constants, so that for fixed R

$$\mathbb{E}\big[w_0\|m_{R,\delta} - m_R\|_2^2 + \|Dm_{R,\delta} - Dm_R\|_F^2\big] \longrightarrow 0 \text{ as } \delta \downarrow 0.$$

Polynomial growth and (S27) allow us to apply dominated convergence to each term, including $m(\nabla\chi_R)'$, so that

$$\mathbb{E}\big[w_0\|m_R - m\|_2^2 + \|Dm_R - Dm\|_F^2\big] \longrightarrow 0 \text{ as } R \to \infty.$$

The same limits hold for g and its gradient. Choose a diagonal sequence m_ℓ, g_ℓ of these smooth approximations. Write $\|\cdot\|_{L^2(w_0)}$ for the norm $\mathbb{E}[w_0\|\cdot\|^2]^{1/2}$, with Frobenius norms for matrices, and $\|\cdot\|_{L^2}$ when w_0 is omitted. Then

$$\|\mathfrak{D}(m_\ell) - \mathfrak{D}(m)\|_{L^2} \le \|m_\ell - m\|_{L^2(w_0)} + C_\psi^2\sqrt{n}\|Dm_\ell - Dm\|_{L^2} \longrightarrow 0.$$

Boundedness of T and Cauchy–Schwarz also give

$$\begin{aligned}
&\mathbb{E}\left|\operatorname{tr}(TDm_\ell\, TDm_\ell) - \operatorname{tr}(TDm\, TDm)\right| \\
&\qquad \le C\left(\|Dm_\ell\|_{L^2} + \|Dm\|_{L^2}\right)\|Dm_\ell - Dm\|_{L^2} \longrightarrow 0, \\
&\mathbb{E}[m_\ell' T m_\ell] \to \mathbb{E}[m' T m], \quad \mathbb{E}[\mathfrak{D}(m_\ell) g_\ell] \to \mathbb{E}[\mathfrak{D}(m) g], \quad \mathbb{E}[m_\ell' T \nabla g_\ell] \to \mathbb{E}[m' T \nabla g].
\end{aligned}$$

The weighted identities follow from these limits.

Finally, for the offset fit, Lemma D.1 and (S15) imply

$$\|m\|_2 \le \|\Pi + v\|_2 + \|b\|_2, \qquad Dm = P \quad \text{almost everywhere}, \qquad \|Dm\|_{\mathrm{op}} \le 1.$$

The product density assigns zero probability to the exceptional Lebesgue-null set in Lemma D.1. We then obtain (S26) by substituting into (S24) with $g \equiv 1$ and into (S25). □

S5 Sparse Fits, Uniform Projections, and Effective Dimension

Let

$$\mathcal{G} = \{\|Z'v/n\|_\infty \le \lambda/c_\lambda\}. \tag{S28}$$

Lemma S4 (LASSO Rates under Non-Gaussian Errors)**.** *Under Assumptions S1 and S2,* $\mathbb{P}(\mathcal{G}^c) \le n^{-20}$*. On* $\mathcal{G}$*, eventually,*

$$\begin{aligned} \|Z(\widehat{\pi} - \pi^0)\|_2/\sqrt{n} &\le 2C_a\sqrt{s/n} + 2(1 + c_\lambda^{-1})\lambda\sqrt{s}/\kappa_0, \\ d < \lceil \vartheta s \rceil, \qquad \|\widehat{\pi} - \pi^0\|_2 &\le \underline{b}_n/2. \end{aligned} \tag{S29}$$

The coefficient bound uses the Borel solution in Lemma D.2. Under Assumption 3, $A \subseteq \operatorname{supp}(\widehat{\pi})$ *with correct signs, and* $k \le d$ *on this event. Furthermore, with* $a_n = s\lambda^2 + s/n$*,*

$$\begin{aligned} \mathbb{E}[\|\widehat{\Pi} - \Pi\|_2^2]/n \le Ca_n, \quad |h - H| \le C\sqrt{a_n}, \\ 0 \le \mathcal{A} \le Ca_n = o(1), \quad \mathbb{E}[d^r] \le Cs^r \quad (1 \le r \le 8). \end{aligned} \tag{S30}$$

In particular, h *is bounded away from zero eventually and* $\mathbb{E}[d] \asymp s$ *under Assumption 3.*

Proof. By normalization, (S18), and (S5),

$$\mathbb{P}(\mathcal{G}^c) \le 2p\exp\{-n\lambda^2/(2c_\lambda^2 C_\psi^2)\} \le n^{-20}.$$

From Assumption S2(ii) and (S5) we also have

$$\frac{\|\xi\|_2}{\sqrt{n}\,\lambda\sqrt{s}} \le \frac{C_a}{c_\lambda C_\psi\sqrt{2\{\log(2p) + 20\log n\}}} \to 0. \tag{S31}$$

On $\mathcal{G}$, (S29) follows from the deterministic arguments in the proof of Lemma F.1, since the approximation, restricted eigenvalue, and sparse eigenvalue bounds used there hold here by Assumption S2(ii),(iv) and (S31). Under Assumption 3, the same proof yields the asserted inclusion, signs, and $k \le d$.

For the moments, let $U(v) = \|\mu(\Pi + v) - \Pi\|_2/\sqrt{n}$. By Lemma D.1, U is $n^{-1/2}$-

Lipschitz, and therefore $\mathrm{Var}(U) \le C_\psi^2/n$ by (S19). On $\mathcal{G}$, (S29) and $\|\xi\|_2/\sqrt{n} \le C_a\sqrt{s/n}$ give $U \le \overline{U} := 3C_a\sqrt{s/n} + 2(1+c_\lambda^{-1})\lambda\sqrt{s}/\kappa_0$, so that

$$\mathbb{P}(\mathcal{G})(\mathbb{E}U - \overline{U})_+^2 \le \mathrm{Var}(U), \qquad \mathbb{E}[U^2] \le 2\overline{U}^2 + 5C_\psi^2/n \le Ca_n,$$

where we use that $\mathbb{P}(\mathcal{G}) \ge 1/2$ eventually and that $s \ge 1$. By the definitions of $h, \mathcal{A}$ and Cauchy–Schwarz,

$$|h - H|^2 \le H\mathbb{E}[U^2] \le Ca_n, \qquad 0 \le \mathcal{A} \le \mathbb{E}[U^2] \le Ca_n = o(1),$$

where the limit follows from (S6).

We will also use the following bound in the next section. For $1 \le r \le 4$, we have $\mathbb{E}[(n + \|v\|_2^2)^{2r}] \le C_r n^{2r}$ by (S18), and if $\mathbb{P}(\mathcal{B}) \le Cn^{-20}$, Cauchy–Schwarz then gives

$$\mathbb{E}[(n + \|v\|_2^2)^r \mathbb{1}_{\mathcal{B}}] \le C_r n^{r-10}. \tag{S32}$$

Finally, $d = \mathrm{rank}(Z_E) \le n$, so for $1 \le r \le 8$,

$$\mathbb{E}[d^r] \le (\vartheta s + 1)^r + n^r \mathbb{P}(\mathcal{G}^c) \le Cs^r.$$

Under Assumption 3,

$$k(1 - n^{-20}) \le \mathbb{E}d \le Cs, \qquad k \le s \le C_s k,$$

so $\mathbb{E}d \asymp s$. Also, h is bounded away from zero eventually since $h \ge H - C\sqrt{a_n}$. □

For a deterministic subset $B \subseteq [p]$, let P_B be the projection onto $\mathrm{col}(Z_B)$, even if the columns of Z_B are dependent.

Lemma S5 (Projections onto Sparse Sets of Columns)**.** *Under Assumptions S1 and S2, set* $m^* = \lceil(\vartheta + 1)s\rceil \wedge p$. *For* $\zeta = v$ *and for* $\zeta = W\Pi$, *and every fixed* $r \in [1, 8]$,

$$\mathbb{E}\left[\max_{|B| \le m^*} \|P_B\zeta\|_2^r\right] \le C_r\{s\log(pn)\}^{r/2}. \tag{S33}$$

There are a fixed $C_\eta < \infty$ *and an event* $\mathcal{W}$, *with* $\mathbb{P}(\mathcal{W}^c) \le n^{-20}$, *on which, for every*

$|B| \le m^*$,

$$\|P_B W P_B\|_{\mathrm{op}} \le \eta, \qquad \eta := C_\eta s\sqrt{\log(pn)/n} = o(1). \tag{S34}$$

Consequently, for the random projection P,

$$\mathbb{E}[\|Pv\|_2^r] + \mathbb{E}[\|PW\Pi\|_2^r] \le C_r\{s\log(pn)\}^{r/2}, \qquad 1 \le r \le 8, \tag{S35}$$

$$\mathbb{E}[(D_\tau - \sigma_v^2 d)^2] \le C\eta^2 s^2 + Cn^{-18}. \tag{S36}$$

Proof. Step 1: Uniform projection moments. For $\zeta = v$, (S18) gives a centered sub-Gaussian bound on the moment generating function. The coordinates of $\zeta = W\Pi$ are independent, centered by (S17), and bounded by (S17) and Assumption S1(iii), so by Hoeffding's lemma the same bound holds with a possibly larger fixed constant. In both cases, for every $b \in \mathbb{R}^n$,

$$\mathbb{E}e^{b'\zeta} \le \exp\{K_{\mathrm{sg}}^2\|b\|_2^2/2\}, \qquad K_{\mathrm{sg}} < \infty.$$

For deterministic B with $r_B := \mathrm{rank}(P_B) > 0$, apply Hsu et al. (2012, Theorem 2.1) with P_B in the role of their matrix A, so that $\Sigma = P_B$, and with zero mean:

$$\mathbb{P}\Big\{\|P_B\zeta\|_2^2 > K_{\mathrm{sg}}^2\{r_B + 2\sqrt{r_B t} + 2t\}\Big\} \le e^{-t}, \qquad t > 0.$$

The rank-zero projection is identically zero. Since $r_B \le |B|$ and $\sum_{j=0}^{\ell}\binom{p}{j} \le (p+1)^\ell$, a union bound with $t = \ell\log(p+1) + u$ gives, for $1 \le \ell \le p$,

$$\mathbb{P}\left\{\max_{|B|\le\ell}\|P_B\zeta\|_2 > C\sqrt{\ell\log(2p) + u}\right\} \le e^{-u}, \qquad u \ge 0.$$

Integrating this tail for fixed $r \in [1,8]$ and using $\ell = m^* \le Cs$ and $\log(2p) \le C\log(pn)$, we obtain (S33).

Step 2: Restricted weighted Gram matrix. For fixed j and k, the variables $z_{ij}z_{ik}W_{ii}$, $i \in [n]$, are independent, centered, and uniformly bounded by Assumption S2(i) and (S17). Hoeffding's inequality and a union bound over the pairs (j,k) give an event $\mathcal{W}$, with $\mathbb{P}(\mathcal{W}^c) \le n^{-20}$, on which

$$\max_{j,k}|(Z'WZ/n)_{jk}| \le C\sqrt{\log(pn)/n}.$$

For $g = Z_B b$ with $|B| \le m^*$, it then follows from the lower sparse-eigenvalue bound in Assumption S2(iv) that

$$|g'Wg| \le n \max_{j,k} |(Z'WZ/n)_{jk}| \|b\|_1^2 \le C|B|\sqrt{\log(pn)/n}\, \|g\|_2^2.$$

Symmetry of W implies $\|P_B W P_B\|_{\mathrm{op}} \le \eta$, which also holds for $B = \varnothing$. As $\eta \to 0$ by (S6), this proves (S34).

Step 3: Selected projections. On $\mathcal{G}$, an independent basis $B \subseteq E(x)$ has $|B| = d < \lceil \vartheta s \rceil$ and $|B| \le m^*$, so $P = P_B$. On $\mathcal{G}^c$ we use $\|Pv\|_2 \le \|v\|_2$ and $\|PW\Pi\|_2 \le C\sqrt{n}$ together with Cauchy–Schwarz and (S18), so that for $\zeta = v$ and for $\zeta = W\Pi$,

$$\mathbb{E}[\|P\zeta\|_2^r] \le \mathbb{E} \max_{|B| \le m^*} \|P_B \zeta\|_2^r + O(n^{r/2-10}) \le C_r \{s \log(pn)\}^{r/2}, \quad 1 \le r \le 8.$$

On $\mathcal{G} \cap \mathcal{W}$, $|D_\tau - \sigma_v^2 d| = |\operatorname{tr}(PWP)| \le \eta d \le Cs\eta$. Since $D_\tau + d \le Cn$ everywhere,

$$\mathbb{E}[(D_\tau - \sigma_v^2 d)^2] \le C\eta^2 s^2 + Cn^2\{\mathbb{P}(\mathcal{G}^c) + \mathbb{P}(\mathcal{W}^c)\} \le C\eta^2 s^2 + Cn^{-18}.$$

These are (S35)–(S36). □

Lemma S6 (Concentration of the Effective Dimension)**.** *Under the hypotheses of Theorem S1,*

$$\begin{aligned} \operatorname{Var}(d) &\le C\{s \log(pn) + \eta^2 s^2 + n^{-18}\} = o(s^2), \\ \mathbb{E}[d^2 + d] &= \mathbb{E}[d]^2\{1 + o(1)\} \asymp s^2. \end{aligned} \tag{S37}$$

Consequently,

$$\frac{d}{\mathbb{E}[d]} \to_p 1, \quad \frac{d^2 + d}{\mathbb{E}[d^2 + d]} \to_p 1. \tag{S38}$$

Proof. Let $m_0(v) = \mu(\Pi + v) - \mathbb{E}[\mu(\Pi + v)]$. We apply (S19) to each coordinate and use (S15) and Lemma S4 to obtain

$$\mathbb{E}\|m_0\|_2^2 \le \mathbb{E}[\operatorname{tr}(PTP)] = \mathbb{E}[D_\tau] \le C\mathbb{E}d \le Cs.$$

The function $F_0 = v'm_0$ has gradient $\nabla F_0 = m_0 + Pv$ almost everywhere, so (S19), (S35), and (S26) give

$$\operatorname{Var}(F_0) \le Cs \log(pn), \qquad \mathbb{E}[(F_0 - D_\tau)^2] = \mathbb{E}[m_0' T m_0 + \operatorname{tr}(TPTP)] \le Cs.$$

Combining these bounds shows that $\mathrm{Var}(D_\tau) \le Cs\log(pn)$, and so by (S36),

$$\sigma_v^4 \mathrm{Var}(d) \le 2\,\mathrm{Var}(D_\tau) + 2\mathbb{E}[(D_\tau - \sigma_v^2 d)^2] \le C\{s\log(pn) + \eta^2 s^2 + n^{-18}\}.$$

Since $\mathbb{E}d \asymp s \ge k \to \infty$ by Lemma S4 and Assumption 3, $\log(pn)/s \to 0$, and $\eta \to 0$, this is (S37). By Chebyshev's inequality, $d/\mathbb{E}d \to_p 1$. Moreover,

$$\frac{\mathbb{E}[d^2 + d]}{(\mathbb{E}d)^2} = 1 + \frac{\mathrm{Var}(d)}{(\mathbb{E}d)^2} + \frac{1}{\mathbb{E}d} \to 1,$$

and $(d^2 + d)/(\mathbb{E}d)^2 \to_p 1$, so (S38) follows. □

S6 The Second Moment of the Numerator

Lemma S7 (Second Moment of the Numerator)**.** *Under the hypotheses of Theorem S1,*

$$\mathbb{E}[(\varepsilon'\widehat{\Pi})^2] = \sigma_\varepsilon^2 \mathbb{E}[\|\widehat{\Pi}\|_2^2] + \sigma_{\varepsilon v}^2 \mathbb{E}[d^2 + d] + o(n\mathcal{A} + s^2), \tag{S39}$$

$$\mathbb{E}[(\Gamma - L_n)^2] \le CS. \tag{S40}$$

Also, $S \asymp \mathcal{A} + s^2/n = o(1)$ *and* $nS \to \infty$.

Proof. By Lemmas S4 and S6, (3.7) in the main text, and $a \neq 0$,

$$\mathcal{R} := n\mathcal{A} + s^2 \asymp nS, \qquad S = o(1), \qquad s^2 \le CnS.$$

Since $\log(pn) \ge \log n$, (S6) forces $s \to \infty$, so that $nS \ge s^2/C \to \infty$. With m as in (S11), we replace Π in the centering term $(h/H)\Pi$ by its sparse approximation $Z\pi^0$ to define the auxiliary fit $\tilde{m} = m + (h/H)\xi = \mu(\Pi + v) - (h/H)Z\pi^0$. By Assumption S2(ii) and (S12),

$$\mathbb{E}\|\tilde{m}\|_2^2 \le 2\mathbb{E}\|m\|_2^2 + 2\frac{h^2}{H^2}\|\xi\|_2^2 \le C(n\mathcal{A} + s). \tag{S41}$$

Step 1: the decomposition (S42). Define $F = v'\mu$ and $\delta = F - D_\tau$. The weighted covariance and second-moment identities (S24) and (S25), together with $\nabla F = \mu +$

Pv, give

$$\mathbb{E}[\delta F] = \mathbb{E}[\mu' T\mu + \mu' TPv], \qquad \mathbb{E}[\delta^2] = \mathbb{E}[\mu' T\mu + \operatorname{tr}(TPTP)].$$

Using $T = W + \sigma_v^2 I$, $P\mu = \mu$, and $\mathbb{E}[\mu' v] = \mathbb{E}[D_\tau]$ by (S26), we can write

$$\begin{aligned}\mathbb{E}[2\delta F] &= 2\sigma_v^2 \mathbb{E}[\|\mu\|_2^2] + 2\mathbb{E}[\mu' W\mu] + 2\sigma_v^2 \mathbb{E}[D_\tau] + 2\mathbb{E}[\mu' WPv],\\ \mathbb{E}[\delta^2] &= \sigma_v^2 \mathbb{E}[\|\mu\|_2^2] + \mathbb{E}[\mu' W\mu] + \mathbb{E}[\operatorname{tr}(TPTP)].\end{aligned}$$

Since $D_\tau = F - \delta$, squaring and rearranging gives $F^2 = D_\tau^2 + 2\delta F - \delta^2$. We substitute the expressions above, subtract $\sigma_v^2 \mathbb{E}[\|\mu\|_2^2] + \sigma_v^4 \mathbb{E}[d^2 + d]$, and collect terms to obtain

$$\begin{aligned}\mathbb{E}[F^2] - \sigma_v^2 \mathbb{E}[\|\mu\|_2^2] - \sigma_v^4 \mathbb{E}[d^2 + d] = \mathbb{E}[\mu' W\mu] + \mathbb{E}[D_\tau^2 - \sigma_v^4 d^2] \\ - \mathbb{E}[\operatorname{tr}(TPTP) - \sigma_v^4 d] + 2\sigma_v^2 \mathbb{E}[D_\tau - \sigma_v^2 d] \\ + 2\mathbb{E}[\mu' WPv]. \qquad \text{(S42)}\end{aligned}$$

Step 2: the bounds (S43). Because Π, ξ, and h/H are nonrandom and $\mathbb{E}W = 0$,

$$\mathbb{E}[\mu' W\mu] = 2\frac{h}{H}\mathbb{E}[\Pi' W\tilde{m}] + \mathbb{E}[m' Wm].$$

For the Borel solution in Lemma D.2, whose support lies in E by (D.1), set $B = \operatorname{supp}(\widehat{\pi}) \cup T^0$. On $\mathcal{G}$ we have $|B| \le m^*$ and $\tilde{m} = P_B \tilde{m}$ by (S29). We can bound $|\Pi' W\tilde{m}| \le \|P_B W\Pi\|_2 \|\tilde{m}\|_2$, so that (S33), (S41), and Cauchy–Schwarz control the first term on that event. For the second, we write $m = \tilde{m} - (h/H)\xi$ and use $\|P_B W P_B\|_{\mathrm{op}} \le \eta$ on $\mathcal{G} \cap \mathcal{W}$ together with $\|\xi\|_2^2 \le Cs$. On the complements of these events we use (S32), since $\|\mu\|_2 \le \|\Pi + v\|_2$, $\|\Pi\|_2 = O(\sqrt{n})$, and $\|\xi\|_2 = O(\sqrt{s})$ give $\|m\|_2 + \|\tilde{m}\|_2 \le C(\sqrt{n} + \|v\|_2)$, so that each quadratic product is bounded by $C(n + \|v\|_2^2)$. These arguments show that

$$\begin{aligned}|\mathbb{E}[\Pi' W\tilde{m}]| &\le C\sqrt{s\log(pn)(n\mathcal{A} + s)} + Cn^{-8},\\ |\mathbb{E}[m' Wm]| &\le C\{\eta(n\mathcal{A} + s) + \sqrt{s(n\mathcal{A} + s)} + s + n^{-8}\}. \qquad \text{(S43)}\end{aligned}$$

Step 3: the bounds (S44) *and* $|\mathbb{E}[\mu' WPv]| \le C\{s\log(pn) + \sqrt{n\mathcal{A}\, s\log(pn)}\}$. As in Step 3 of the proof of Lemma S5, $|D_\tau - \sigma_v^2 d| \le \eta d$ on $\mathcal{G} \cap \mathcal{W}$. Moreover, $0 \le D_\tau \le Cd$

and $0 \le \operatorname{tr}(TPTP) \le Cd$ by (S17) and $P^2 = P$. Since $\mathbb{E}[d^j] \le Cs^j$ for $j = 1, 2$, these bounds control the expectations over $\mathcal{G} \cap \mathcal{W}$. On its complement, $D_\tau + d \le Cn$ and $\mathbb{P}((\mathcal{G} \cap \mathcal{W})^c) \le 2n^{-20}$, so that the quadratic and linear terms contribute $O(n^{-18})$ and $O(n^{-19})$, respectively. We therefore have

$$\begin{aligned} |\mathbb{E}[D_\tau^2 - \sigma_v^4 d^2]| &\le C\eta s^2 + Cn^{-18}, \\ |\mathbb{E}[D_\tau - \sigma_v^2 d]| &\le C\eta s + Cn^{-19}, \\ |\mathbb{E}[\operatorname{tr}(TPTP) - \sigma_v^4 d]| &\le Cs. \end{aligned} \tag{S44}$$

Finally, write $\mu = (h/H)\Pi + m$. Since P is an orthogonal projection and $\|W\|_{\mathrm{op}} \le C$, the moment bounds in (S35) give

$$|\mathbb{E}[\mu' W P v]| \le C\mathbb{E}[(\|PW\Pi\|_2 + \|m\|_2)\|Pv\|_2] \le C\{s\log(pn) + \sqrt{n\mathcal{A}\, s\log(pn)}\}.$$

Step 4: (S39) *and* (S40). By the bounds above, the five terms in (S42) have total absolute value at most

$$C\big\{\eta\mathcal{R} + \sqrt{s\log(pn)(n\mathcal{A}+s)} + \sqrt{s(n\mathcal{A}+s)} + s\log(pn) + s + n^{-8}\big\}.$$

Since $s \ge 1$, we have $n\mathcal{A} + s \le \mathcal{R}$, and $\mathcal{R} \ge s^2$ by definition. Relative to $\mathcal{R}$, this total is at most

$$C\left\{\eta + \sqrt{\log(pn)/s} + s^{-1/2} + \log(pn)/s + s^{-1} + n^{-8}/\mathcal{R}\right\} \longrightarrow 0$$

by (S6) and $s \to \infty$. The conditional structural moments (S14) now imply

$$\mathbb{E}[(\varepsilon'\widehat{\Pi})^2] = a^2\mathbb{E}[F^2] + \sigma_u^2\mathbb{E}[\|\mu\|_2^2] = \sigma_\varepsilon^2\mathbb{E}[\|\mu\|_2^2] + \sigma_{\varepsilon v}^2\mathbb{E}[d^2 + d] + o(\mathcal{R}),$$

which is (S39).

Lastly, (S13) and $D_\tau \le Cd$ yield

$$\mathbb{E}[(v'm)^2] \le 2\mathbb{E}[(v'm - D_\tau)^2] + 2\mathbb{E}[D_\tau^2] \le C\{n\mathcal{A} + s + \mathbb{E}d^2\} \le C\mathcal{R}.$$

By (S14) and $\Gamma - L_n = \varepsilon' m/(\sqrt{n}\, h)$, we then have

$$\mathbb{E}[(\Gamma - L_n)^2] = \frac{a^2\mathbb{E}[(v'm)^2] + \sigma_u^2\mathbb{E}\|m\|_2^2}{nh^2} \leq C\mathcal{R}/n \leq CS.$$

□

S7 The Denominator and the Optimal Instrument

Lemma S8 (The Sample Denominator)**.** *Under the hypotheses of Theorem S1,*

$$\mathbb{E}[\widehat{h}] = h + \mathbb{E}[D_\tau]/n, \qquad \mathbb{E}[(\widehat{h} - h)^2] \leq C/n, \tag{S45}$$

$$\widehat{h} - h = O_p(n^{-1/2}), \qquad \mathbb{P}(\mathcal{H}_n) \to 1. \tag{S46}$$

Consequently, $\Delta = O_p(n^{-1/2})$.

Proof. By the weighted first-moment identity (S26) with $b = 0$ we have $\mathbb{E}[v'\mu] = \mathbb{E}[D_\tau]$, so that $\mathbb{E}[\widehat{h}] - h = \mathbb{E}[D_\tau]/n$, and $\mathbb{E}[D_\tau] \leq C\mathbb{E}[d] \leq Cs$ by (S17) and (S30). For the variance, the gradient of $x'\mu$ with respect to v is $\mu + Px$ almost everywhere. Since $\|\mu\|_2 \leq \|x\|_2$ by Lemma D.1 and P is a projection, the weighted Poincaré inequality (S19) yields

$$\mathrm{Var}(\widehat{h}) \leq \frac{C}{n^2}\mathbb{E}\|\mu + Px\|_2^2 \leq \frac{C}{n^2}\mathbb{E}\|x\|_2^2 \leq \frac{C}{n}.$$

By (S6), $s = o(\sqrt{n})$, so that $\mathbb{E}[(\widehat{h}-h)^2] = \mathrm{Var}(\widehat{h}) + \mathbb{E}[D_\tau]^2/n^2 \leq C/n + Cs^2/n^2 \leq C/n$, which proves (S45). Since $h \geq \underline{H}/2$, the first bound in (S46), the bound on Δ, and $\mathbb{P}(\mathcal{H}_n^c) \leq \mathbb{P}(|\widehat{h} - h| > \underline{H}/4) \leq C/n \to 0$ then follow from Markov's inequality. □

Lemma S9 (Moments of the Denominator Correction)**.** *Under the hypotheses of Theorem S1, with* Δ *as in* (B.1)*,*

$$\mathbb{E}[\Delta^2 L_n^2] \leq C/n, \tag{S47}$$

$$|\mathbb{E}[\Delta L_n^2]| \leq C\{\sqrt{S/n} + 1/n\}. \tag{S48}$$

Proof. We follow the proof of Lemma B.3 in the main appendix, replacing (B.11) by the weighted Stein identity

$$\mathbb{E}[(\Pi' v) f(v)] = \mathbb{E}[(T\Pi)'\nabla f(v)]. \tag{S49}$$

This identity is (S24) applied with the vector Π in place of the field there, for which the weighted divergence in (S23) is $\Pi' v$.

As in the Gaussian case, we use three facts. First, $\nabla\Delta = (\mu + Px)/(nh)$, so that $\mathbb{E}\|\nabla\Delta\|_2^2 \le 4\mathbb{E}\|x\|_2^2/(nh)^2 \le C/n$. Second, Lemma S8 gives $\mathbb{E}[\Delta] = \mathbb{E}[D_\tau]/(nh) = O(s/n)$ and $\mathbb{E}[\Delta^2] \le C/n$. Third, by the conditional structural moments (S14),

$$\mathbb{E}[L_n^2 \mid v] = \frac{\sigma_u^2}{H} + \frac{a^2}{nH^2}(\Pi' v)^2. \tag{S50}$$

Since Δ depends only on v, we may multiply (S50) by functions of Δ and take expectations.

Step 1: $\mathbb{E}[\Delta^2 L_n^2] \le C/n$. By (S50), it suffices to show that $\mathbb{E}[(\Pi' v)^2\Delta^2] \le C$. Apply (S49) with $f = (\Pi' v)\Delta^2$. Because $\nabla(\Pi' v) = \Pi$, the product rule gives

$$\begin{aligned}\mathbb{E}[(\Pi' v)^2\Delta^2] &= \mathbb{E}[(\Pi' T\Pi)\Delta^2] + 2\mathbb{E}[(\Pi' v)\Delta \cdot (T\Pi)'\nabla\Delta] \\ &\le Cn\mathbb{E}[\Delta^2] + \tfrac{1}{2}\mathbb{E}[(\Pi' v)^2\Delta^2] + 2\mathbb{E}[((T\Pi)'\nabla\Delta)^2],\end{aligned}$$

where the inequality uses $\|T\|_{\mathrm{op}} \le C$ and Young's inequality, as in Step 1 of the proof of Lemma B.3. The last term is at most $Cn\mathbb{E}\|\nabla\Delta\|_2^2 \le C$ because $\|T\Pi\|_2^2 \le Cn$. Moving $\frac{1}{2}\mathbb{E}[(\Pi' v)^2\Delta^2]$ to the left-hand side, we obtain $\mathbb{E}[(\Pi' v)^2\Delta^2] \le C$.

Step 2: $|\mathbb{E}[\Delta L_n^2]| \le C\{\sqrt{S/n} + 1/n\}$. Multiply (S50) by Δ and take expectations. Applying (S49) with $f = (\Pi' v)\Delta$, we have $\mathbb{E}[(\Pi' v)^2\Delta] = \mathbb{E}[(\Pi' T\Pi)\Delta] + \mathbb{E}[(\Pi' v)(T\Pi)'\nabla\Delta]$, where $\Pi' T\Pi = \sigma_v^2 nH + \Pi' W\Pi$ because $T = \sigma_v^2 I + W$. Since $\sigma_\varepsilon^2 = \sigma_u^2 + a^2\sigma_v^2$, the two multiples of $\mathbb{E}[\Delta]$ again add up to σ_ε^2/H, so that

$$\mathbb{E}[\Delta L_n^2] = \frac{\sigma_\varepsilon^2}{H}\mathbb{E}[\Delta] + \frac{a^2}{nH^2}\left\{\mathbb{E}[(\Pi' W\Pi)\Delta] + \mathbb{E}[(\Pi' v)(T\Pi)'\nabla\Delta]\right\}. \tag{S51}$$

The first term is $O(s/n)$. Since W is diagonal with independent, mean-zero entries and $|\Pi_i| \le K_\Pi$ by (S4),

$$\mathbb{E}[(\Pi' W\Pi)^2] = \mathrm{Var}(\tau(v_i))\sum_{i=1}^n \Pi_i^4 \le Cn,$$

so that $|\mathbb{E}[(\Pi' W\Pi)\Delta]| \le C$ by Cauchy–Schwarz.

For the last term in (S51), split $(T\Pi)'\nabla\Delta = \sigma_v^2\Pi'\nabla\Delta + (W\Pi)'\nabla\Delta$. The first part

satisfies $|\mathbb{E}[(\Pi' v)\Pi'\nabla\Delta]| \le C(1+\sqrt{n\mathcal{A}})$ by the argument in Step 2 of the proof of Lemma B.3. The only differences are that (S49) gives $\mathbb{E}[(\Pi' v)\Pi'\mu] = \mathbb{E}[(T\Pi)'P\Pi]$, which is at most Cn in absolute value, and that the bound $\mathbb{E}[(\Pi' v)^4] + \mathbb{E}\|x\|_2^4 \le Cn^2$ now follows from (S18). For the second part, $nh\,(W\Pi)'\nabla\Delta = (W\Pi)'(\mu + Px) = (PW\Pi)'(\mu + x)$, since $P\mu = \mu$. We apply Cauchy–Schwarz twice and then use $\|\mu + x\|_2 \le 2\|x\|_2$ and (S35) to bound

$$\begin{aligned}|\mathbb{E}[(\Pi' v)(PW\Pi)'(\mu+x)]| &\le \left\{\mathbb{E}[(\Pi' v)^4]\right\}^{1/4}\left\{\mathbb{E}\|\mu+x\|_2^4\right\}^{1/4}\left\{\mathbb{E}\|PW\Pi\|_2^2\right\}^{1/2}\\ &\le Cn\sqrt{s\log(pn)}.\end{aligned}$$

Dividing the second part by nh and combining it with the first, we have

$$|\mathbb{E}[(\Pi' v)(T\Pi)'\nabla\Delta]| \le C\left\{1 + \sqrt{n\mathcal{A}} + \sqrt{s\log(pn)}\right\},$$

so that by (S51),

$$|\mathbb{E}[\Delta L_n^2]| \le C\left\{\frac{s}{n} + \sqrt{\frac{\mathcal{A}}{n}} + \frac{\sqrt{s\log(pn)}}{n} + \frac{1}{n}\right\}.$$

Finally, $\mathcal{A} \le CS$ and $s/n \le C\sqrt{S/n}$ by Lemma S7, while

$$\frac{\sqrt{s\log(pn)}}{n} = \frac{s}{n}\sqrt{\frac{\log(pn)}{s}} = o\left(\frac{s}{n}\right)$$

by (S6). Combining these bounds gives (S48). □

Lemma S10 (Moments for the Optimal Instrument)**.** *Under Assumption S1,*

$$\mathbb{E}[L_n^2] = \sigma_\varepsilon^2/H, \quad |\mathbb{E}[qL_n^2]| + \mathbb{E}[q^2 L_n^2] \le C/n. \tag{S52}$$

Consequently, $C_0 = (1-q)^2 L_n^2$ *satisfies*

$$\mathbb{E}[C_0] = \sigma_\varepsilon^2/H + O(n^{-1}). \tag{S53}$$

Proof. By independence and centering, $\mathbb{E}[L_n^2] = \sigma_\varepsilon^2/H$. Since (S50) uses only (S14), it holds under Assumption S1. Multiplying it by $q = \Pi' v/(nH)$, which depends only

on v, and using independence and $\mathbb{E}v_i = 0$, we have

$$\mathbb{E}[qL_n^2] = \frac{a^2\mathbb{E}[(\Pi' v)^3]}{n^2H^3} = \frac{a^2\mathbb{E}[v_i^3]\sum_i \Pi_i^3}{n^2H^3} = O(n^{-1}),$$

where $\sum_i |\Pi_i|^3 \le K_\Pi nH$ by (S4) and $\mathbb{E}|v_i|^3 \le C$ by (S18). By the same tail bound, $\mathbb{E}[(\Pi' v)^4] \le Cn^2$, and by independence $\mathbb{E}[q^2] = \sigma_v^2/(nH)$. Therefore

$$\mathbb{E}[q^2L_n^2] = \frac{\sigma_u^2}{H}\mathbb{E}[q^2] + \frac{a^2\mathbb{E}[(\Pi' v)^4]}{n^3H^4} = O(n^{-1}).$$

Expanding $C_0 = (1 - 2q + q^2)L_n^2$ proves (S53). □

S8 Supporting Results for Theorem S2

Lemma S11 (Sample Counterparts of $\mathcal{A}$ and q)**.** *Under the hypotheses of Theorem S1, let* $\widehat{\mathcal{A}} = \min_{b\in\mathbb{R}} \mathbb{E}_n[(\widehat{\Pi}_i - b\Pi_i)^2]$*, and let* q *and* $\widehat{q}$ *be as in the proofs of the main-text Theorems 1 and 2. Then*

$$\widehat{\mathcal{A}} - \mathcal{A} = O_p\Big(\sqrt{\mathcal{A}/n} + n^{-1}\Big) = o_p(S), \tag{S54}$$

$$\widehat{q} - q = O_p\Big(\sqrt{\mathcal{A}/n} + s/n + n^{-1}\Big) = o_p(S), \tag{S55}$$

where the second assertion holds on $\mathcal{H}_n$*.*

Proof. Step 1: $\widehat{\mathcal{A}} - \mathcal{A} = O_p(\sqrt{\mathcal{A}/n} + 1/n)$. In Step 1 of the proof of Lemma C.1, we use Gaussian errors only through the Poincaré inequality, which we apply to $\mathbb{E}_n[\Pi_i\widehat{\Pi}_i]$ and $\widehat{\mathcal{A}}$. By (S15), their gradients are again $P\Pi/n$ and $(2/n)P\{I - \Pi\Pi'/(nH)\}\widehat{\Pi}$. With the weighted Poincaré inequality (S19) in place of (E.1), the same argument shows that $0 \le \mathcal{A} - \mathbb{E}[\widehat{\mathcal{A}}] \le C/n$ and $\mathrm{Var}(\widehat{\mathcal{A}}) \le C\mathcal{A}/n$, and the claim follows from Chebyshev's inequality.

Step 2: $\widehat{q} - q = O_p(\sqrt{\mathcal{A}/n} + s/n + 1/n)$ *on* $\mathcal{H}_n$. Since $\widehat{\Pi} = (h/H)\Pi + m$ by (S11), we have $\mathbb{E}_n[v_i\widehat{\Pi}_i] = v'm/n + hq$, and $\mathbb{E}[(v'm - D_\tau)^2] \le C(n\mathcal{A} + s)$ by (S13). In addition, $0 \le D_\tau \le Cd = O_p(s)$ by Lemmas S2 and S4, and since $s \ge 1$ it follows that $v'm/n = O_p(\sqrt{\mathcal{A}/n} + s/n)$. By independence, $\mathbb{E}[q^2] = \sigma_v^2/(nH)$, while $\widehat{h} - h = O_p(n^{-1/2})$ by

Lemma S8. On $\mathcal{H}_n$, where $\widehat{h} \geq \underline{H}/4$,

$$\widehat{q} - q = \frac{v'm/n - q(\widehat{h} - h)}{\widehat{h}} = O_p\Big(\sqrt{\mathcal{A}/n} + s/n + n^{-1}\Big).$$

Step 3: Both rates are $o_p(S)$. By Lemma S7, $S \asymp \mathcal{A} + s^2/n$. Using the inequality $2s\sqrt{\mathcal{A}/n} \leq \mathcal{A} + s^2/n$ and $s^2/n \leq CS$, we can bound

$$\frac{\sqrt{\mathcal{A}/n} + s/n + n^{-1}}{S} \leq C(s^{-1} + s^{-2}) \longrightarrow 0,$$

since $s \to \infty$ by (S6). □

Lemma S12 (Estimating the Error Moments)**.** *Suppose the hypotheses of Theorem S1 hold. For any (possibly data-dependent) $[J]$-valued index $\check{c}$, $\widehat{\beta}_{\check{c}} - \beta = O_p(n^{-1/2})$. If the pilot satisfies $\check{\beta} - \beta = O_p(n^{-1/2})$, the moment estimators in the main-text (3.11) satisfy*

$$\widehat{\sigma}^2_\varepsilon - \sigma^2_\varepsilon = O_p(n^{-1/2}), \quad \widehat{\sigma}_{\varepsilon v} - \sigma_{\varepsilon v} = O_p(n^{-1/2}). \tag{S56}$$

Proof. Step 1 of the proof of Theorem S1 shows that $\sqrt{n}(\widehat{\beta} - \beta) = O_p(1)$, and the first claim follows since $\check{c} \in [J]$ and J is fixed. By the moment conditions (S1), (S18), and (S4), $\sup_i \mathbb{E}[\varepsilon_i^4 + x_i^4] < \infty$. Independence and Chebyshev's inequality then give

$$\mathbb{E}_n[\varepsilon_i^2] = \sigma^2_\varepsilon + O_p(n^{-1/2}), \qquad \mathbb{E}_n[\varepsilon_i x_i] = \sigma_{\varepsilon v} + O_p(n^{-1/2}),$$

while $\mathbb{E}_n[x_i^2] = O_p(1)$ by Markov's inequality. Substituting these bounds and $\check{\beta} - \beta = O_p(n^{-1/2})$ into the algebraic identities (C.11) proves (S56). □

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